Why Underwater Segmentation Fails: Benchmark Contamination, Missing Annotations, and a Teacher-Assisted Masking Method (TAM-Net)

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A thesis submitted to the Department of Computer Science and Engineering
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B.Sc. in Computer Science and Engineering

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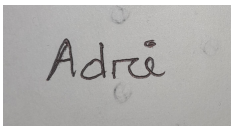
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Declaration

It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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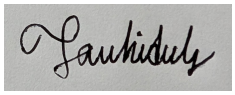
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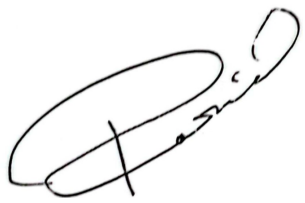
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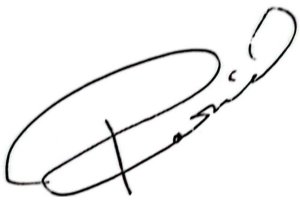
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Ethics Statement

This work uses images that were publicly available already. They were gathered from several public underwater datasets from the web and annotated by the authors themselves;. No human or animal subjects were used for the images and no personal data was collected either. Some of the images depict divers, but they are only used as examples of 'Human Activity' class and no attempt was made to identify the individuals. The image dataset is not redistributed with this research work.

Multiple members of the research team performed the blind review of candidate unlabelled objects described in Chapter 5. The reviewers had no prior knowledge about which model proposed the region, and background regions were added in randomly to measure the reviewers' inclination to say "yes". No external participants were recruited for this study.

The research also reports negative results and retracted measurements, in addition to positive ones (Section 5.2.8). Whenever a claim made in an earlier report was found to be wrong, it is corrected in the text instead of being excluded.

Abstract

Underwater images require semantic segmentation to support marine debris monitoring and the inspection of submerged infrastructure. However, the progress is hard to evaluate when the benchmark itself is unreliable. This study began with a survey and built a four-class underwater image dataset (Aquatic Animal, Artifacts and Metals, Human Activity, Marine Trash and Plastic). The initial benchmark reported a YOLO–U-Net hybrid as the best segmentation model (74.12% mIoU, 1.75 points above DeepLabv3+). Then the focus of the study shifted to whether this result could be trusted and what actually improves segmentation performance on this dataset. Later benchmarking did not support the previous claim. Upon further inspection, out of the 347 original test images, 143 (41.2%) contained a near-duplicate image in the training split, implying data leakage. After re-scoring the models on uncontaminated dataset, mean IoU dropped between 3.6 to 4.5 points across different models. Following that, we rebuilt the data splits using two independent methods, defined a single evaluation protocol for each of the models, and reported bootstrap confidence intervals. Furthermore, two-seed replication was performed on each model with matched controls. Using this protocol, the initial hybrid model achieved (68.80%) mIoU, a (5.2%) drop in performance. Additionally, a blind human review with random decoys estimated about 770 unlabelled objects in the training split and confirmed 158 unlabelled objects in the test split. We then moved on to evaluating six different modeling ideas. Replacing ResNet backbones with SegFromer-B2 resulted in the largest reliable gain (+0.08 to 0.11 mIoU). Frozen CLIP features (CoGS) improved a weak backbone by +0.071 (95% CI +0.053 to +0.093) but a stronger backbone such as Segformer saw a much smaller gain (+0.004 mIoU), which is well within seed-to-seed variation. Our proposed TAM-net, which combines a stride-2 detail path, flip and zoom-in augmentation, and the masking of background pixels that an exponential-moving average teacher confidently labels as foreground during training period, is the best performing model overall (0.7468 over two seeds, against 0.7361 for the tuned baseline). Pooled over two backbones (Segformer-B2 and DeepLabv3+), it improved the original model by 0.0126 mIoU (95% CI +0.0025 to +0.0225), with the same sign in all four seed pairs and an interval that stays above zero (+0.0007) when a third SegFormer-B2 seed is added; the pooling was chosen after the results, and on SegFormer-B2 alone the margin (+0.0108, CI −0.0013 to +0.0247) does not exclude zero. The background masking by itself adds +0.0077 mIoU (CI −0.0032 to +0.0192), and the detail path and augmentation contributes to gains in external SUIM images. Physics-based restoration (PADS-Net), scale-aware design (Aquascale), and missing-annotation-aware-training failed to beat their matched controls, and therefore, are retracted from the results. The primary contributions from this thesis are a corrected and audited benchmark, a reproducible evaluation protocol, a decoy-calibrated label audit, and proposed design that is the best-performing model in this comparison.

Keywords: Underwater Semantic Segmentation; Benchmark Contamination; Near-Duplicate Detection; Missing Annotations; SegFormer; CLIP; Evaluation Reliability; Bootstrap Confidence Intervals; Marine Debris; Deep Learning.

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We are also thankful to the Department of Computer Science and Engineering, BRAC University, for providing us with the academic environments and computational resources, which made all of this possible. We would also like to acknowledge the creators of the public underwater datasets and open-source tools (Ultralytics, PyTorch, Roboflow, and segmentation-models-pytorch) that we built upon. Finally, we would like to thank our families and friends for their continuous support and encouragement. With their kind assistance, we have finally reached the end of our undergraduate thesis journey.

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Chapter 1

Introduction

1.1 Background

The ocean covers more than 70 percent of the planet's surface and plays a vital role in the world's ecosystems. Human activities underwater have been rapidly increasing such as the building of oil pipelines, communication cables, etc. Meanwhile, the problem of marine pollution has grown into a serious environmental problem. Today, the sea bed is littered with plastic waste and metal garbage. To control these issues, underwater environments need to be monitored on a regular basis to enable identification of infrastructure damage and to locate and remove debris. Traditionally these tasks have been performed by human divers, but manual inspection is dangerous, slow, and expensive. This has steered the field to Remotely Operated Vehicles. Submersibles, also known as ROVs and Autonomous Underwater Vehicles (AUVs). To be useful, these vehicles must be able to perceive the environment autonomously, meaning they have to know what is happening around them without constant human supervision. This need has spurred the adoption of computer vision. Underwater imaging, however, is quite difficult. As mentioned in the work on the URPC dataset [35], underwater images are heavily degraded: light attenuation causes color distortion and strong blue/green hues, and particles suspended in the water cause noise and blur [31]. If the detectors are trained on clear terrestrial images, it results in lower accuracy when the detectors are deployed in underwater environments. Therefore, deep learning is widely used, and specialized datasets like TrashCan [32] and SUIM [33] exist because specialised training data are needed to build systems that can recognise marine debris, marine life, divers, and wrecks.

This thesis reports a study that proceeded in three phases (Figure 1.1). We first conducted a survey of underwater enhancement and detection, and made a proposal for a pipeline, with a longer term objective of open long-tailed recognition. We then proceeded from survey to implementation. We curated a four-class underwater dataset that has polygon masks,designed an enhancement and pre-processing pipeline and compared six families of models for detection, instance segmentation, semantic segmentation and classification. The original benchmark's best segmentation was achieved by a YOLO–U-Net hybrid It achieved a mean IoU of 74.12%, which is 1.75 points higher than DeepLabv3+.

The third stage began with an attempt to beat that record. However, it quickly became a different piece of work, as checking the initial numbers showed that they could not be trusted. The test set contained near-duplicates of training images, which inflated the results. The compared models afterwards had not been trained or scored under the same protocol, and the reported margin was smaller than the variation produced by changing only the random seed. Chapters 4 and 5 explain how each of the above was measured, the process to rebuild the benchmark, and re-evaluating which modeling ideas actually help. Figure 1.1 summarises the three phases.

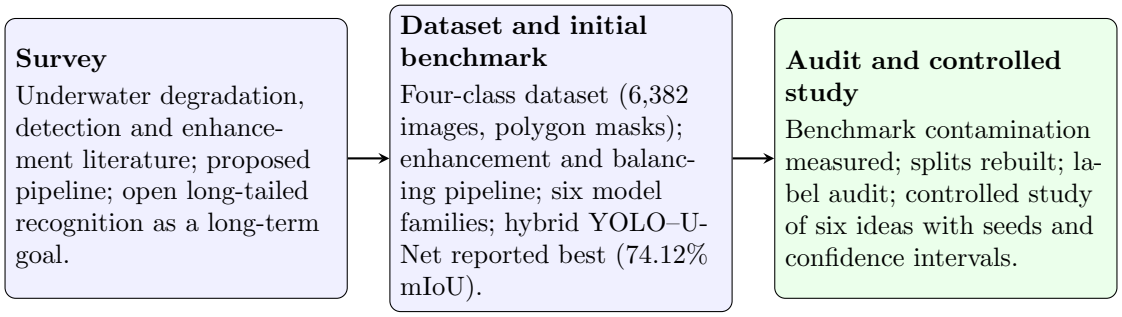


Figure 1.1: The three phases of the study. Most of this thesis reports the third phase and re-examines the results of the second in Chapter 5.

1.2 Rational of the Study or Motivation

The rationale for this work has four strands.

Environmental and safety needs. The buildup of human-made waste poses a threat to marine biodiversity and studies built on datasets like TrashCan [32] demonstrate that current robots are not able to precisely find debris in underwater environments. The process of automated clean-up and inspection can only be started once a dependable segmentation system is available, which is what our work is aimed at.

A technical gap between underwater and terrestrial vision. Object Recognition is well-developed on land, but it remains challenging in underwater environments, as shown with attention-based underwater detection [26], the main reason for the failure of the conventional algorithms is the clutter and visual noise. If we can segregate the model failure and data failure, we can develop another research question by understanding why the model fails.

Relevance to Bangladesh. Being a riverine country, with increasing maritime interests, Bangladesh is reliant on the underwater infrastructure for energy and connectivity. The risk of carrying out inspections of gas pipelines and bridge pillars is still significant as they require human divers.. An automated alternative that is reliable would be safer and would also be able to reveal the faults early on.

Trust in reported results. The distortion in benchmark results are mainly caused by data leakage, near-duplicate images, incomplete labels, and unreplicated single-run comparisons [24], [28], [37], [40]. Underwater datasets are particularly vulnerable

as they are typically made up from a number of repurposed public resources, as well as videos. In our own instance, the data had been collected in a similar manner; so we had to check whether our initial conclusions survived a careful audit afterwards. A negative favourable opinion or null finding that is measured properly is more useful to the field compared to a favourable opinion which is not measured rigorously.

1.3 Problem Statement

Underwater perception faces three familiar difficulties: image degradation (colour distortion, low visibility, changing illumination) [31], [70], the difficulty of separating and classifying objects against a blue-green background, and long-tailed class distributions in which a few classes dominate [50]. The initial benchmark addressed these by building a dataset, an enhancement pipeline, and a model benchmark.

The problem addressed in this thesis is different and comes first: *given a four-class underwater segmentation benchmark that was assembled from heterogeneous sources, can the model comparisons reported in the initial benchmark be trusted, and once the benchmark is repaired, which modelling choices improve pixel-level segmentation by more than run-to-run noise?*

We break this into five research questions:

- **RQ1 (benchmark integrity).** How much of the test set overlaps with the training set, where does the overlap come from, and how much did it inflate the initial scores?
- **RQ2 (evaluation reliability).** How large is the score variation caused by the random seed and by evaluation conventions such as test-time augmentation, compared with the differences that the initial benchmark reported?
- **RQ3 (label quality).** How many objects are missing from the annotations, and how much does that limit the achievable score?
- **RQ4 (modelling).** Under one protocol with replication and controls, which of the following help: a stronger backbone, frozen CLIP features, teacher-guided masking of suspected missing labels, physics-based restoration, and scale-aware design?
- **RQ5 (generalisation).** Do any gains persist on images from two external public datasets that were not used for training?

1.4 Objective

The primary objective is to establish a trustworthy benchmark for four-class underwater semantic segmentation and to determine, with proper statistical care, which modelling ideas improve on it. The specific objectives are:

- **O1.** To measure train–test overlap in the original split, trace its source, and quantify the resulting score inflation on the initial models.
- **O2.** To rebuild the data splits so that near-duplicates and frames from the same video cannot straddle a split, and to check the result independently.
- **O3.** To define one training and evaluation protocol for every compared model, with dataset-global IoU, bootstrap confidence intervals, replication over seeds, and a rule for when a difference counts as a result.
- **O4.** To audit the annotations for missing objects using a blind human review with random decoys, and to estimate the effect on scores.
- **O5.** To evaluate, against matched controls, a stronger backbone (SegFormer-B2), CLIP-feature guidance (CoGS), teacher-guided annotation masking (TAM-Net and MAAS), physics-based restoration (PADS-Net), and scale-aware design (AquaScale).
- **O6.** To test generalisation on images from two public datasets after removing everything that overlaps our data.

Contributions in brief. (i) A measured account of benchmark contamination in our own dataset (41% of the images in test splits had a near duplicate in the training split) and how it inflated initial scores (3.6 to 4.5 mIoU). (ii) An evaluation protocol that is reproducible and finding that seed-to-seed variation (0.02-0.03 mIoU) is greater than initial estimates (0.0175 mIoU). (iii) Estimated 770 unlabelled objects in training split and confirming 158 unlabelled objects in testing split through a decoy-calibrated audit. (iv) A controlled comparison confirming that SegFormer-B2 provides the largest reliable gain (+0.08 to +0.11 mIoU over ResNet-based baselines) and that CLIP guidance helps weak backbones (+0.071 on Att-UNet) but not stronger backbones such as SegFormer. (v) A proposed design, TAM-Net, which combines a stride-2 detail path, flip and zoom-in augmentation, and the masking of background pixels that an exponential-moving average teacher confidently labels as foreground during training period, is the best performing model overall (0.7468 over two seeds, against 0.7361 for the tuned baseline). Pooled over two backbones (Segformer-B2 and DeepLabv3+), it improved the original model by 0.0126 mIoU (95% CI +0.0025 to +0.0225), with the same sign in all four seed pairs and an interval that stays above zero (+0.0007) when a third SegFormer-B2 seed is added; the pooling was chosen after the results, and on SegFormer-B2 alone the margin (+0.0108, CI −0.0013 to +0.0247) does not exclude zero. The background masking by itself adds +0.0077 mIoU (CI −0.0032 to +0.0192). (vi) Retracted results of physics-based restoration, scale-aware design, and missing-annotation-aware for failing to beat their controls. We do not claim state-of-the-art performance on other benchmarks, which we did not run; within our controlled comparison TAM-Net is the best model, and the audited benchmark, the protocol, and the label audit are contributions in their own right.

1.5 Methodology in Brief

The study proceeds in five stages, shown in Figure 1.2. First, we *audit* the initial benchmark: near-duplicate detection between the original splits, overlap with the public SUIM and TrashCan datasets, and re-scoring of the initial checkpoints on the uncontaminated subset. Second, we *rebuild* the splits by clustering near-duplicates, mirrored copies, and video frames and assigning whole clusters to one split, stratified so that each class keeps a 70/15/15 share. Third, we *define a protocol* shared by every model: identical optimiser, schedule, loss, and input size within each comparison (and identical epochs, except for the retrained hybrid, which the platform's session limit restricted to 10 epochs), dataset-global IoU, bootstrap intervals, replication over seeds, and model selection on validation only. Fourth, we *evaluate ideas*: baselines from four architecture families, CLIP-guided variants, teacher-guided masking, and three designs that did not help, each against a matched control. Fifth, we *interpret* the results with a blind human review with random decoys, which measures how many objects are missing from the labels, and with two external datasets.

The work was carried out in two independently constructed pipelines (Pipeline A and Pipeline B, described in Chapter 4) on different machines. Both reproduced the central contamination finding with separately written code, which we treat as a form of replication.

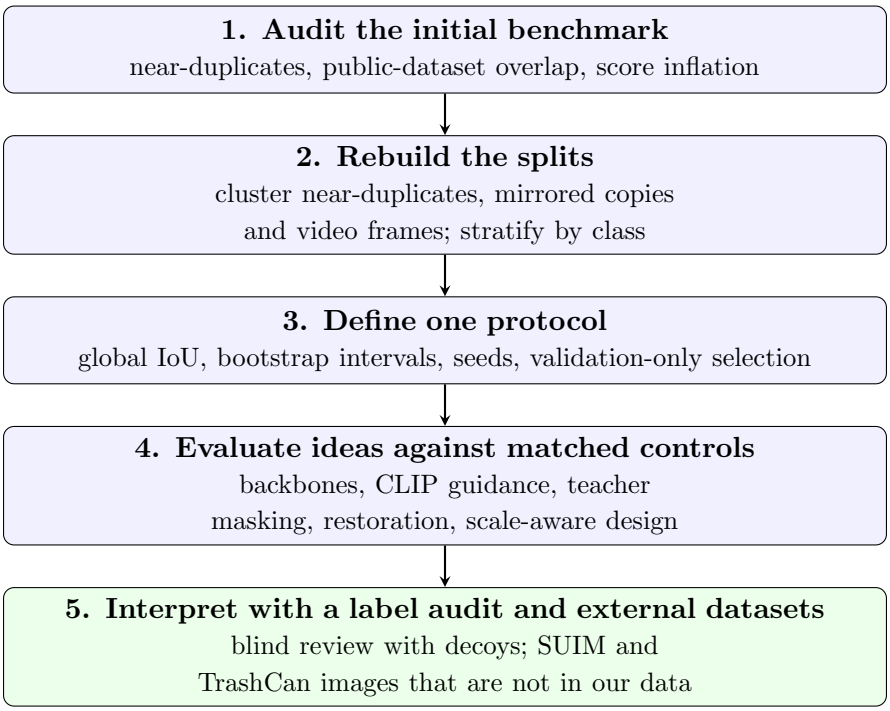


Figure 1.2: Overall methodology of the study.

1.6 Scopes and Challenges

Scope. This thesis is about semantic segmentation of still underwater images to four classes of objects + background. It provides the initial detection, instance

segmentation and classification results for completeness, but the new work is ongoing on segmentation. Only the ablation (Section 5.2.1) is discussed when it comes to image enhancement. Set the limit to follow the recommendations of the auditor, since the audit revealed the lack of colour degradation was not the constraint on this data. The proposed recognition direction mentioned at the beginning of the study, which was the open long-tailed recognition, was not continued: When it was determined that the benchmark was unreliable, the study was discontinued and then repaired. It was our priority to measure it and this change of scope is admitted.

Challenges. Difficulty with underwater perception: light gets absorbed by water, vehicle lights can create harsh shadows, floating particles can create noise, and the light can lead to a loss of contrast, with the result that the seafloor can appear like something man-made. Data challenges include expensive collection, expert labelling, domain shift between locations and class imbalance. This work poses challenges as well. The elimination of the near *Near-duplicates* is difficult because the same scene may be shown in a cropped, recompressed, mirrored, or as adjacent video frames. *Incomplete labels* can be a correct prediction and still count as an error. There are many *Small objects* that have high levels of dominance over the animal class, but relatively few pixels. *Run-to-run variance* is large in comparison to a lot of published margins, so any conclusion must be replicated. The cost of replication would have been limited if that had been the case, as just about that cost was 162 GPU-hours consumed by the controlled study itself.

1.7 Organisation of the Thesis

Chapter 2 reviews the literature, including the works added for the audit and the controlled study on benchmark reliability, missing annotations, and language-guided dense prediction. Chapter 3 states the requirements, impacts, and constraints of the work. Chapter 4 describes the data, the audit methods, the models, and the evaluation protocol. Chapter 5 presents and analyses the results, and Chapter 6 concludes with the findings, contributions, and future work. Appendix A lists every run and Appendix B documents how each number in this thesis was checked against the raw result files.

Chapter 2

Literature Review

2.1 Preliminaries

Light absorption, low visibility, and complex environmental conditions make underwater imaging and object understanding uniquely challenging. This thesis studies four-class semantic segmentation of underwater images (man-made objects and marine organisms): it audits the benchmark on which the work started, and then tests which modelling ideas improve segmentation, ending with the proposed TAM-Net design. Traditional underwater detection relied on hand-crafted features and classical computer vision — colour-correction algorithms, histogram equalisation, and template matching with support-vector machines or sliding windows — which require extensive manual feature engineering and perform poorly in real underwater scenes. Modern deep learning has changed this picture by learning end-to-end from raw pixels: convolutional neural networks (CNNs), two-stage detectors such as Faster R-CNN, and single-stage detectors such as YOLO consistently outperform classical methods, and recent variants add attention and multi-scale feature aggregation for further gains. This chapter reviews the key works that address these challenges and positions our own contribution against them.

Problems with underwater photography. Capturing images underwater is difficult. Water rapidly attenuates light and distorts colour: red components are lost first, so scenes appear blue or green. The images are blurry, noisy, and low in contrast, which makes object recognition hard.

Object detection. Object detection answers two questions about an image: *what* objects are present and *where* they are. It is widely used in autonomous vehicles, surveillance, and environmental monitoring.

Semantic vs. instance segmentation. Segmentation pushes beyond bounding boxes to pixel-level understanding. *Semantic* segmentation assigns a class label to every pixel (e.g. U-Net, DeepLabv3+), whereas *instance* segmentation additionally separates individual objects of the same class (e.g. Mask R-CNN, YOLO-seg). For underwater tasks such as debris quantification or robotic grasping, masks are more informative than boxes, which is why a large part of this work targets segmentation.

Deep learning and its limitations. CNN-based detectors such as YOLO and

Faster R-CNN learn rich features automatically and perform well on clean, balanced data. Their accuracy degrades, however, on degraded underwater imagery and on imbalanced datasets.

Long-tail problem. In underwater datasets a few classes appear very frequently while many appear rarely; this is the long-tail problem. Standard models detect common classes well but fail on rare ones.

Unknown-object (open-set) problem. In the real world not all objects are known in advance. Objects absent from the training data may appear at test time, and conventional closed-set detectors cannot handle them. This matters for practical deployment.

OLTR concept. Open Long-Tailed Recognition (OLTR) unifies three tasks: recognising common (head) classes, recognising rare (tail) classes, and rejecting unknown classes. Because all three co-exist in underwater environments, OLTR is a natural long-term direction for underwater recognition; we did not pursue it in this thesis (Chapter 6).

Evaluating segmentation. Fully convolutional networks [5] made dense pixel-wise prediction the standard approach to semantic segmentation. Quality is normally reported as intersection over union (IoU), $\mathrm{IoU}_c = \mathrm{TP}_c/(\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c)$ for class c, averaged over classes to give the mean IoU (mIoU). Two conventions exist. *Dataset-global* IoU accumulates true positives, false positives, and false negatives over every pixel of the test set and divides once. *Per-image* IoU computes one value per image and averages, which weights a rare class in a single small image as heavily as a common class in many large ones. The two can differ enormously on sparse classes: in the initial benchmark the same U-Net scored 0.699 by the first convention and 0.250 by the second (Section 5.1.2). We use the dataset-global convention throughout and average over the four object classes only, so that the large background class cannot inflate the score.

Benchmark reliability. A benchmark score is only meaningful if the test set is independent of the training set, the labels are correct, and the comparison is replicated. Each of these has been shown to fail on well-known datasets. Recht et al. [24] built new test sets for CIFAR-10 and ImageNet and found that accuracy dropped for a wide range of models. Barz and Denzler [28] found near-duplicate images shared between the CIFAR training and test sets and released a duplicate-free test set. Northcutt et al. [40] estimated label errors in the test sets of widely used benchmarks and showed that such errors can change the ranking of models. Bouthillier et al. [37] showed that several sources of variance, including weight initialisation, data order, and data splits, must be accounted for when comparing methods, because differences smaller than that variance are not reliable. We use the bootstrap [1] to attach confidence intervals to test-set scores.

Incomplete annotations and the background class. In segmentation, every pixel not covered by a label is treated as background and the model is penalised for predicting anything else there. If the annotator missed an object, the model is punished for being right. This is the "missing annotation" problem. Zhang et al. [36] address it for object detection with a loss that re-calibrates the loss signal of background regions. Cermelli et al. [29] study a related problem in incremental

semantic segmentation, where pixels labelled background at one step may belong to other classes. A frequently used stabiliser in semi-supervised learning is the exponential-moving-average (EMA) teacher of Tarvainen and Valpola [12], whose weights are a running average of the student's and whose predictions are therefore more stable than the student's at any single step.

Transformer segmentation and language-guided features. SegFormer [44] pairs a hierarchical Transformer encoder (Mix Transformer, MiT) that produces multi-scale features with a light all-MLP decoder. CLIP [41] learns image and text embeddings jointly from a very large set of image–text pairs, and several works turn its features into dense predictions: MaskCLIP [54] extracts dense labels from CLIP without fine-tuning, CLIPSeg [51] adds a decoder that segments from text or image prompts, LSeg [47] trains pixel embeddings to align with text embeddings of class names, and DenseCLIP [52] uses context-aware text prompting for dense prediction.

Physics of underwater imaging. Jaffe [2] modelled underwater image formation as an attenuated direct signal plus backscatter, and Akkaynak and Treibitz [18] showed that the attenuation coefficient of the direct signal depends on range and reflectance and differs from the backscatter coefficient. Restoration methods invert such a model to recover the colours of the scene; whether that helps a downstream segmenter is an empirical question (Section 5.2.1).

2.2 Review of Existing Research

Underwater enhancement, detection, and datasets

Han et al. [31] proposed methods for image preprocessing and object recognition underwater. For preprocessing they combined the Max-RGB and Shades-of-Gray methods to estimate the illuminant colour, and alternatively used a CNN to predict an illumination map for weakly-illuminated inputs, enhancing the image accordingly. For detection they used a CNN that takes the whole image as input to predict bounding boxes and classify objects in one pass, exploring two structural schemes; the second (down-sample then convolve) performed slightly better after prolonged training. Both schemes outperformed YOLOv3, Fast R-CNN, and Faster R-CNN on underwater data, although the authors acknowledged the methods are tuned specifically for underwater scenes and do not generalise to other datasets.

Yeh et al. [68] proposed a lightweight deep network that jointly learns colour conversion and object detection, together with a method to synthesise underwater images for training. A colour-conversion module (FPN-style) transforms the input to grayscale to lower computational cost, followed by a detection module with feature-extraction and FPN-based feature-aggregation sub-modules. The method matches or outperforms Faster R-CNN, YOLOv2, and YOLOv3 on the Brackish dataset in mAP while being far more efficient in GFLOPs and inference time. Its robustness is not fully established, however, as it is trained on synthetic data without a cross-domain test.

Zhang et al. [63] proposed MetaUE, a multi-variable CNN with meta-learning

for underwater image enhancement. Three U-Net-like sub-networks (with skip and shortcut connections) translate a degraded image to a clean one, supervised by a composite loss over clean image, background light, transmission map, and the underwater image. After pre-training on a large synthetic set (27 distortion types, 100k samples), the model is fine-tuned on UIEB and EUVP. MetaUE matches or beats strong baselines on PSNR/SSIM, but fine-tuning on UIEB reduces generalisation to EUVP, highlighting data sensitivity and over-fitting.

Zhuang et al. [**69**] proposed UWNet, a YOLOv8-based detector enhanced with a Structured Space Model (SSM) and multi-scale fusion for small-object detection underwater. Down-sampling convolutions are replaced by SPD-Conv, an MDSBlock provides global feature extraction, and multi-scale implicit fusion improves small-target discrimination. UWNet improves mAP50-95 by 1.1–3.4% over YOLOv9/v10/v11 and Mamba-YOLO-B with 40–390% fewer parameters, though its GFLOPs are slightly higher due to SSM scanning overhead, and colour distortion is not directly addressed.

Li et al. [**65**] proposed a three-part deep CNN for small underwater objects. The backbone uses lightweight CSPSL and VKConv modules for multi-scale extraction while preserving detail; an SPPFMS module adds further multi-scale extraction at low cost; a dual-path PAN-FPN neck fuses features at different resolutions; and an anchor-free head performs detection. The model reaches 64.1% mAP, 0.3–3% ahead of YOLOv5s, YOLOv8n, and YOLOv5-Lite-g with the lowest FLOPs, but is evaluated only on UDD and DUO without a cross-domain test or explicit enhancement.

Dai et al. [**55**] proposed ERL-Net, an edge-directed representation-learning network for underwater detection. Observing that object edges remain discriminative in low-contrast scenes, they introduce an edge-directed attention module, a feature-aggregation module that fuses global and local information across three levels, and a wide, asymmetric receptive-field block for small objects. ERL-Net achieves superior performance on the UOD task; its key innovation is the explicit use of edge cues for discrimination.

Peng et al. [**60**] address salient object detection (SOD) in underwater scenes, producing pixel-level prominence labels. Their key insight is to use blurriness as a depth cue, since distant objects appear more blurred. A blurriness map (multi-scale Gaussian filtering refined by max filtering, morphological reconstruction, and matting) feeds a dual-stream U-Net with ResNet-50 backbones. Cross-Channel Attention Fusion (CCAF) outperforms simple addition fusion, and a FocusAugment data-augmentation strategy further improves four SOD baselines. The method struggles on camouflaged or smooth objects.

Reddy et al. [**42**] review image and signal processing in the underwater environment. They survey physical-model methods that invert the degradation process, enhancement methods (histogram equalisation, contrast and colour correction), and learning-based methods using GANs and CNNs. They note remaining challenges in generalising across water types, real-time motion processing, and the trade-off between quality and computational efficiency.

Jiji and Dhanalakshmi [**19**] propose a hybrid technique for enhancing blurry underwater images. They distinguish motion, defocus, and scatter blur and com-

bine colour correction, dehazing, and sharpening in sequence (or jointly), adaptively weighting each step. The hybrid method is more robust than any single step, yielding better colour fidelity, contrast, and detail, with measurable PSNR/SSIM gains; its modularity allows adaptation to specific environments.

Wang and Liu [71] propose FEFM-YOLO11 for underwater detection in low-light, turbid water. A Feature Enhancement Module replaces two C3K2 blocks to increase feature diversity and receptive field; a Context-Aware Fusion Module models global and local features; and a dedicated small-target layer with the Wise-IoU loss improves dense, small-object detection. On URPC2020 the model gains 2.0 points mAP@50 and 1.9 points mAP@50:95 over baseline YOLO11.

Chen et al. [30] tackle small/blurry objects and label noise. They pair SWIPENET — a small-object detector that integrates dilated convolutions into a deconvolutional single-shot backbone to keep high-resolution, semantically rich feature maps — with Invert Multi-Class AdaBoost (IMA), which *lowers* the weights of hard-to-classify (likely noisy) samples. Combined with a sample-weighted loss, the framework is robust to label noise and beats several state-of-the-art detectors on two underwater datasets.

Wang et al. [27] address colour-accurate restoration from degraded images using a two-stage approach: UWGAN, an improved physical underwater imaging model embedded in a GAN, synthesises realistic degraded images from clear ones to create paired training data; a U-Net then learns end-to-end restoration. The restoration network, trained on realistic synthetic data, generalises well to real scenes and outperforms several prior methods qualitatively and quantitatively.

Xu et al. [67] propose YOLOv5-MH for poor visibility, colour distortion, and multi-scale variability. Improvements include adaptive contrast enhancement with linear mapping, replacing C3 with the more efficient C2f, a multi-head bottleneck transformer with coordinate attention, and a bi-directional FPN. The model reaches 83.1% mAP on URPC and 81.6% on DeepPlastic, beating YOLOv5/v8, Faster R-CNN, and RetinaNet, with ablations confirming each component.

Guo et al. [46] improve recognition by training a detector jointly with an image-reconstruction task. A shared backbone and FPN serve both detection and reconstruction; the reconstruction loss adapts the feature extractor to underwater characteristics. Their UNet-based FPN fuses multi-level features through top-down, bottom-up, and skip connections, improving RetinaNet by 1.4% and Faster R-CNN by 1% with negligible overhead.

Ferreira et al. [56] present a machine-learning pipeline for navigation in visually degraded water. A YOLOv11-nano detector performs detection and coarse classification, HSV filtering identifies buoys, and PaddleOCR reads numbered markers, evaluated on the RAMI 2023 data. Detection and coarse classification are good, but identification of unseen markers and manipulation objects is poor, exposing the weaknesses of closed-set, supervised pipelines on long-tailed underwater data and motivating open-set, long-tailed approaches.

Li et al. [48] integrate a self-supervised image-deblurring sub-network with a Faster R-CNN detector, enabling shared feature learning through multi-task optimisation rather than treating enhancement as a separate step. An improved spatial trans-

former network based on perspective transformation enriches features and adds robustness to viewpoint changes. On URPC2017/2018 the framework improves mAP over baselines, though it focuses on closed-set, balanced detection.

Saoud et al. [**61**] introduce ADOD (Adaptive Domain-Aware Object Detection) to handle domain shift. Built on YOLOv3 with Residual Attention Blocks, it highlights informative spatial/channel features and suppresses background noise. An attention-based domain-classification module learns domain-invariant features during training. ADOD outperforms baseline YOLOv3 and domain-aware variants on URPC2019, especially under simulated domain shifts, but is evaluated on few classes and ignores long-tailed/open-set issues.

Ritzau et al. [**43**] present real-time detection and tracking on forward-looking sonar to overcome optical limitations in low-visibility water. A sliding-window preprocessing strategy integrates vehicle motion data to improve accuracy and efficiency, enabling continuous localisation even when objects are temporarily undetectable. It improves precision, recall, F1, and speed over detection-only baselines, but is evaluated mainly on single, stationary objects and depends on accurate inertial navigation.

Liu et al. [**50**] address long-tailed and open-ended distributions with OLTR++, a unified framework for class imbalance and open-set recognition. A dynamic meta-embedding combines direct features with memory-induced representations from class centroids, enabling head-to-tail knowledge transfer; embeddings are calibrated by reachability to visual memory to reduce over-confidence on unknowns; a modulated attention enhances head/tail discrimination; and an energy-based uncertainty measure selects informative open samples. OLTR++ improves consistently across open-set scenarios and is a key inspiration for our long-term direction.

Chen et al. [**45**] propose SWIPENET with a Curriculum Multi-class AdaBoost (CMA) strategy for noisy underwater imagery. A Sample-Weighted Hyper Network produces high-resolution, semantically rich feature maps for small objects; a sample-weighted loss down-weights noisy samples; and CMA progressively learns from clean to noisy samples, ensembling diverse detectors. The method achieves state-of-the-art results on URPC2017/2018/2019 and ChinaMM, at the cost of higher inference complexity.

Liu et al. [**35**] introduce the URPC dataset for marine-life detection, focusing on holothurian, echinus, scallop, and starfish. Its images suffer from colour casts, low contrast, and blur, providing a realistic benchmark that exposes the failure of models trained on clear images and supporting research on enhancement and detection for robot picking.

Hong et al. [**32**] introduce TrashCan, a semantically-segmented dataset of more than 7,000 images built from JAMSTEC footage, containing many forms of debris (plastic, metal, trash) and marine creatures. It provides both bounding-box and instance-segmentation versions. Standard models perform poorly on it, demonstrating the need for specialised data to build smarter clean-up robots. TrashCan directly motivates our debris-focused class design.

Wang et al. [**26**] review and analyse attention and multi-scale fusion for detection in murky water. An attention mechanism acts like a spotlight on important regions

while suppressing water-induced noise, and multi-scale fusion lets the model see large and small objects simultaneously. Together they substantially reduce false alarms, which is valuable for safe AUV operation.

Kim and Yu [39] target perception in dark, cloudy water where cameras fail, using forward-looking/side-scan sonar. Sonar images are noisy and grayscale, so the authors clean them and analyse shapes with a CNN trained to recognise the sharp shadows and highlights of hard man-made materials versus muffled seabed reflections, enabling automatic classification useful for mapping, wreck location, and infrastructure inspection.

Li et al. [34] provide a full enhancement solution. They first build the UIEB benchmark of 890 raw/reference image pairs, then propose Water-Net, which learns to fuse white balance, gamma correction, and histogram equalisation into a single clear image, adapting automatically to water conditions. Water-Net produces superior colour and contrast, aiding both humans and robots. UIEB and Water-Net underpin much of the enhancement literature we build on.

Fulton et al. [20] present the CADDY dataset for diver–robot collaboration, providing stereo footage of divers performing gestures and using tools. By focusing on interaction, it helps train "buddy" robots to follow divers, carry tools, and interpret gestures despite lighting and bubble variation, improving safety and efficiency for professional divers.

Underwater semantic and instance segmentation

Islam et al. [33] introduced SUIM. Its a dataset of underwater images with pixel-level annotations. It has categories such as fish, reefs, plants, wrecks, divers, robots, and sea floor, together with a benchmark of segmentation models. It is the closest public counterpart to our dataset and, as Chapter 5 shows, about half of its images also appear in ours.

Lian et al. [59] introduced the UIIS dataset (4,628 images, seven categories, pixel-level instance annotations) and WaterMask, an instance-segmentation model whose Difference Similarity Graph Attention module recovers detail lost through image degradation and down-sampling. Another paper which was proposed by the same group [66] built USIS10K (10,632 images, seven categories). The purpose of this model is to segment underwater salient instances and it adapted the Segment Anything Model [58] with underwater-specific visual prompts (USIS-SAM). Zuo et al. [72] proposed UWSegFormer. It is a SegFormer-based model which has an image-quality attention, a multi-scale aggregation attention, and an edge-learning loss, and reports 82.12 mIoU on SUIM and 71.41 on DUT-USEG. These are not comparable to our numbers as the label sets, splits, and metrics are all different; here, SUIM is only used as an external test set. We refer to them because they serve as examples of how the concepts of transformer backbones, foundation-model prompts and attention modules are already employed in the marine environment. While these reported gains may or may not be larger than run-to-run variance, as was not mentioned in the abstracts of the studies reviewed, this is the type of question the protocol of this thesis is meant to address.

Benchmark reliability, incomplete annotations, and language guidance

The below publications involve aspects of reliability of benchmarks, incomplete annotations, language-guided dense prediction, and the design ideas explored in this thesis.

Xie et al. [44] proposed SegFormer, which consists of a Mix Transformer encoder that needs no positional encoding and generates multi-scale features, and an all-MLP decoder. In our study its B2 variant is the backbone that works best.

Radford et al. [41] proposed that, CLIP is contrastively trained on an extremely large dataset of image–text pairs. It has visual similarity that carries over to many tasks, without any training for them, so it is a good inspiration to use it as a frozen source of general visual knowledge for a small dataset like ours.

Language-guided dense prediction. Zhou et al. [54] (MaskCLIP), Lüddecke and Ecker [51] (CLIPSeg), Li et al. [47] (LSeg), and Rao et al. [52] (DenseCLIP) adapts CLIP to segmentation or dense prediction. They generally show improvement over previous norms on general purpose tests. If the baseline for underwater data is already a strong transformer, as we don't know, is such guidance still beneficial? This is an issue we will address with our controlled comparison.

Tarvainen and Valpola [12] proposed for semi-supervised classification mean teachers whose weights are an exponential moving average of the student's weights. We have made a new use of the EMA teacher as an arbiter in suspected missing labels within fully labelled data, in our TAM-Net and MAAS.

Zhang et al. [36] presented a Background Recalibration Loss method for detecting objects without annotations. TAM-Net shares a spirit with the MAAS, where the concept of detection is extended to semantic segmentation using an EMA-teacher selection rule, but it is not considered as an entirely new concept.

In incremental semantic segmentation, **Cermelli et al. [29]** model the change in the background class that contains the unlabelled pixels, demonstrating that assuming a fixed background class for all unlabelled pixels is a distortion of learning.

Reliability of benchmarks. Recht et al. [24], Barz and Denzler [28], Northcutt et al. [40], and Bouthillier et al. [37] report on generalisation errors on new test sets, near-duplicates of training and test images, label errors in test sets, and the size of the run-to-run variance compared to reported gains. To our knowledge, there is no similar audit of an underwater segmentation benchmark, and Chapter 5 provides one for our data.

Physics-based restoration. Jaffe [2] and Akkaynak and Treibitz [18] give the image-formation models that are the basis of physics-based restoration. We propose a model that is inverted in a segmentation network and trained with the segmentation loss, which we call PADS-Net(Section 5.2.1).

Architectural ingredients. Oktay et al. [14] introduced attention gates for U-Net decoders, used in our Att-UNet baseline and in the initial hybrid. For the scale-aware design, Chen et al. [7] learn per-pixel weights over multi-scale features, Yu et al. [16] keep a shallow high-resolution "spatial path" beside a deep context path,

and Singh et al. [15] train on context regions around objects instead of whole image pyramids. Zhang et al. [62] attach new branches to a pretrained network through zero-initialised convolutions so that the initial output is unchanged, a device we use in CoGS and TAM-Net.

2.3 Summary of Key Findings

The reviewed literature shows a clear shift in underwater vision towards deep learning. Early solutions revolve around colour correction, dehazing, and deblurring to make images clearer before detection, while more recent works fuse enhancement and detection into a single end-to-end model to reduce error propagation. A second trend is improved detection of small objects in low-contrast, noisy water: many models keep the YOLO backbone but add new necks or specialised blocks, or exploit edge, blur, and saliency cues to emphasise object shape. A third trend is better feature representation through attention modules and multi-scale fusion, sometimes combined with meta-learning, reconstruction, or hyper-networks. Datasets are central to all of this: marine-life benchmarks (URPC), debris datasets (TrashCan), enhancement datasets (UIEB), and diver–robot datasets (CADDY) each expose real problems such as colour cast, blur, low contrast, long-tailed class distributions, and label scarcity. Finally, robustness and generalisation are recurring concerns, addressed through domain adaptation and joint auxiliary training; yet many methods are still tested on small or synthetic data, leaving cross-domain robustness, open-set detection, and long-tailed imbalance largely open. Table 2.1 summarises the reviewed works and their relevance to this thesis.

Table 2.1: Summary of key findings from related research works: underwater enhancement, detection, segmentation, and datasets, followed by the works on benchmark reliability, incomplete annotations, and language-guided prediction that motivate this thesis.

Work	**Task and setting**	**Method**	**Data**	**Result, or relevance here**
Underwater enhancement, detection, and recognition				
Han et al. [31]	Detection, closed-set	Max-RGB + Shades-of-Gray, CNN illumination estimation, custom detector	Proprietary underwater set	Beat YOLOv3, Fast R-CNN and Faster R-CNN underwater; limited generalisation
Yeh et al. [68]	Detection, closed-set	Lightweight CNN, FPN colour-to-grayscale	Brackish	Comparable mAP, far lower GFLOPs
Zhang et al. [63]	Enhancement, closed-set	Meta-learning U-Net (MetaUE), multi-loss	UIEB, EUVP	High PSNR/SSIM; weaker cross-domain after fine-tuning
Zhuang et al. [69]	Detection, closed-set	YOLOv8 + SSM + multi-scale fusion	URPC	+1.1 to +3.4% mAP, fewer parameters
Li et al. [65]	Detection, closed-set	CSPSL, VKConv, PAN-FPN, anchor-free head	UDD, DUO	64.1% mAP; limited diversity

Work	**Task and setting**	**Method**	**Data**	**Result, or relevance here**
Dai et al. [55]	Detection, closed-set	Edge-directed attention, multi-scale aggregation	URPC	Better small-object detection via edges
Peng et al. [60]	Salient object detection, closed-set	Dual-stream U-Net, blurriness-guided (CCAF)	U-SOD	Large gain; fails on camouflaged objects
Wang and Liu [71]	Detection, closed-set	FEFM-YOLO11, CAFM, Wise-IoU	URPC2020	+2.0% mAP@50, better small objects
Ferreira et al. [56]	Detection and identification, closed-set	YOLOv11-nano, HSV, OCR	RAMI 2023	Good detection; poor on unseen or rare objects
Saoud et al. [61]	Detection, domain-aware	YOLOv3 + residual attention + domain classifier	URPC2019	Robust under domain shift
Ritzau et al. [43]	Sonar detection and tracking, closed-set	CNN sonar detection + motion-aware tracking	Sonar basin	Better recall, stable tracking
Liu et al. [50]	Recognition, open-set and long-tailed	OLTR++ dynamic meta-embedding	ImageNet-LT	Strong on rare and unseen classes
Hong et al. [32]	Dataset (segmentation)	TrashCan debris dataset	TrashCan	Standard models detect debris poorly
Li et al. [34]	Enhancement, closed-set	Water-Net + UIEB benchmark	UIEB	Superior colour and contrast restoration
Segmentation, foundation models, benchmark reliability, and missing annotations				
Islam et al. [33]	Semantic segmentation, dataset	SUIM dataset and benchmark	SUIM	External test source; about half of its images are also in our data
Lian et al. [59]	Instance segmentation	WaterMask with a graph-attention detail module	UIIS (4,628 images)	Closest underwater work on masks; instance-level, so not comparable
Lian et al. [66]	Salient instance segmentation	SAM with underwater visual prompts (USIS-SAM)	USIS10K (10,632 images)	Foundation-model guidance underwater; not comparable with ours
Zuo et al. [72]	Semantic segmentation	UWSegFormer: SegFormer with quality and multi-scale attention, edge loss	SUIM, DUT-USEG	82.12 mIoU on SUIM as reported; not comparable with ours
Xie et al. [44]	Semantic segmentation	Mix Transformer encoder, all-MLP decoder	Natural images	Strongest backbone in our study
Radford et al. [41]	Image–text representation	Contrastive image–text pre-training	Large image–text collection	Frozen features in CoGS
Zhou et al. [54]; Lüddecke and Ecker [51]; Li et al. [47]; Rao et al. [52]	Language-guided segmentation	CLIP-based dense prediction	Natural images	Precedent for CoGS; not tested against strong baselines underwater

Work	Task and setting	Method	Data	Result, or relevance here
Tarvainen and Valpola [12]	Semi-supervised learning	EMA teacher	—	Teacher in TAM-Net and MAAS
Zhang et al. [36]	Missing annotations, detection	Background recalibration loss	—	Closest precedent for TAM-Net and MAAS
Cermelli et al. [29]	Incremental segmentation	Modelled background shift	—	Background is not always background
Recht et al. [24]; Barz and Denzler [28]; Northcutt et al. [40]; Bouthillier et al. [37]	Benchmark reliability	Generalisation gaps, near-duplicates, label errors, variance	Public datasets	Motivate the audit and the protocol
Jaffe [2]; Akkaynak and Treibitz [18]	Restoration models	Underwater image-formation models	—	Basis of PADS-Net

Research gap and positioning of this work. Four gaps stand out from this body of work and shape the present study. *First*, most underwater studies focus on bounding-box detection of a small set of marine-life classes; pixel-level segmentation of *man-made* categories such as artifacts, metals, and plastics is comparatively rare, even though masks are what downstream robots need. *Second*, the underwater segmentation literature we reviewed rarely reports whether test images overlap the training set, and rarely reports run-to-run variance or confidence intervals; papers typically compare one proposed model with weaker baselines on different datasets and metrics, which makes it hard to know which model family is genuinely better. Benchmarks assembled from redistributed public sources, as ours was, are especially exposed. *Third*, missing annotations have been studied for detection [36] and for incremental segmentation [29], but we did not find a measurement of how many objects are missing from an underwater segmentation dataset, or of how much that limits scores. *Fourth*, CLIP-guided dense prediction is usually compared with weaker baselines on general-purpose data, so whether it helps on top of a strong transformer backbone underwater is unknown.

This thesis addresses these gaps. It builds a four-class dataset that explicitly includes man-made objects, with polygon masks; it audits the benchmark (about 41% of the original test images have a near-duplicate in the training set), rebuilds the splits, and fixes one protocol with seeds, confidence intervals, and controls (Chapter 4); it estimates the number of unlabelled objects with a decoy-calibrated blind review (about 770 in the training set); and it tests a stronger backbone, CLIP guidance, and teacher-guided masking against matched controls, ending with TAM-Net as the best-scoring model (Chapter 5). The open long-tailed recognition direction highlighted by Liu et al. [50] was our longer-term goal at the start of the study; it is left to future work because the benchmark had to be repaired before any new model could be judged.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

This section states what the system and the study had to satisfy: what the software must do, the data and compute it needs, and the conditions under which a result is accepted.

3.1.1 Functional requirements

- **FR1.** Given an RGB underwater photograph, classify each pixel into one of five labels which are: Background, Aquatic Animal, Artifacts and Metals, Human Activity, Marine Trash and Plastic. Images are letter-boxed to 512×512 pixels for all segmentation models.
- **FR2.** Provide an evaluation harness that computes dataset-global IoU, Dice coefficient per class, bootstrap confidence intervals, and an option for flip augmentation during testing phase.
- **FR3.** Provide audit tools that (a) detect near-duplicate and mirrored images between the 3 data splits, (b) mine candidate unlabelled objects in the dataset, and (c) serve a blind review page with background as random decoys.
- **FR4.** Provide a packaged, runnable model that generates a class mask, an overlay, and a per-class coverage table for a folder of photographs (Pipeline B's deployable SegFormer-B2 package).

3.1.2 Non-functional requirements and acceptance rules

- **Comparability.** All models share the number of epochs, optimiser, schedule, loss, input size, data loader, and rule in every comparison. Difference in any of these stated parameters is stated explicitly.

- **Replication.** A result is only accepted if the difference between its paired bootstrap interval excludes zero and it bears the same same sign in every seed pair. Otherwise, the result is reported as "not established". Chapter 5 goes into detail about why the first condition alone is not enough for this dataset.
- **Selection without leakage.** Checkpoints and hyperparameters are selected based on the validation split only; the test split is scored only once and not used to tune any models or hyperparameters.
- **Traceability.** Every number in this thesis can be traced back to the stored result file in (Appendix B).

3.1.3 Hardware and software

Table 3.1 lists the environments actually used. Where a value could not be recovered from the stored logs, it is left unstated instead of estimated.

For orientation, one epoch of SegFormer-B2 on the 10,446-image training set of Pipeline A took a median of 149 seconds at batch size 8. A TAM-Net epoch took a median of 252 seconds; its no-masking control, which has the same detail path and augmentation, took 185 seconds, so the teacher's forward pass and the masking add about 67 seconds per epoch (medians over the two main-analysis runs of each arm). A trained SegFormer-B2 checkpoint is about 99 MB.

3.1.4 Data requirements

The study requires the Roboflow export of the dataset in YOLO polygon format (6,382 images and 19,114 annotation rows in four classes), from which both pipelines derive their masks. CLIP-guided models additionally need the frozen OpenCLIP ConvNeXt-Large tower (`convnext_large_d_320`, LAION-2B weights [38], [49], [53]) or a cache of its features. The external evaluation requires the public SUIM [33] and TrashCan [32] images.

3.1.5 Constraints

Table 3.2 lists the constraints under which the requirements above had to be met and what each one changed in the study.

3.2 Societal Impact

Safety and inspection. Reliable segmentation of debris, wrecks can minimize the number of dives required by human divers to inspect underwater pipelines, cables, bridge foundations, which is directly relevant to safety in a riverine and maritime country such as Bangladesh. This thesis contributes to this indirectly, its main

Table 3.1: Compute environments used in this study.

Environment	**Hardware**	**Used for**
Pipeline B workstation	NVIDIA GeForce RTX 4070 Ti SUPER, 16 GB (15.99 GiB); Windows 11; Python 3.12.3; PyTorch 2.11.0+cu128; torchvision 0.26.0; segmentation-models-pytorch 0.5.0; timm 1.0.29; NumPy 2.5.2; OpenCV 5.0.0; SciPy 1.18.1	19 trained runs (SegFormer-B2 baselines and seeds, U-Net, audit models, MAAS, AquaScale); 62 GPU-hours
Pipeline A workstation	NVIDIA GeForce RTX 4080 SUPER, 16 GB; segmentation-models-pytorch 0.5; OpenCLIP	33 runs (baselines, PADS-Net, CoGS, TAM-Net on SegFormer-B2 and on DeepLabv3+, controls); 96 GPU-hours
Kaggle notebook	NVIDIA T4, 12-hour session limit	Initial hybrid retrain (10 epochs, 4 GPU-hours) and TAM-Net seed 2 (time not logged)
Initial benchmark	Notebook logs: NVIDIA GeForce RTX 4050 Laptop GPU (6 GB) for YOLOv8m-seg, U-Net and Mask R-CNN; RTX 4080 SUPER for YOLOv11m-seg and DeepLabv3+; RTX 4070 Ti SUPER for the hybrid and the two classifiers. Ultralytics 8.4.57 and PyTorch 2.5.1+cu121 for the YOLO models	Initial benchmark (six model families)

insight being about how much a reported score can be trusted, which determines whether a system can be relied upon.

Environmental monitoring. Pixel-level segmentation masks allow debris to be counted and mapped, which aids clean-up planning. The results also highlights a limit: on TrashCan images from a deep-sea domain, the best models only achieved about 0.34 to 0.37 mean IoU on the classes scored compared to 0.70 to 0.75 for the same models tested on our own dataset. This implies that a system trained on shallow, well-lit images should not be trusted in deep-sea footage without significant re-validation.

The research community. We discovered that a dataset assembled from publicly available sources might carry significant overlap between its different splits without anyone intending it. Reporting this finding, the detection method, and the corrected splits lets others avoid the same error.

Table 3.2: Constraints on the study and their consequences.

Constraint	**Limit**	**Consequence for the study**
GPU memory	About 16 GB on each GPU of the controlled study (RTX 4080 SUPER, RTX 4070 Ti SUPER, Kaggle T4)	512×512 inputs, mixed precision, batch size 8 (Pipeline A) or 16 (Pipeline B); the CLIP tower kept frozen, with its features cached once per image
Session limit	Kaggle sessions last at most 12 hours	The initial hybrid was retrained for 10 epochs (Chapter 5)
Compute	About 162 GPU-hours in total, no paid compute	62 trained networks; two seeds per arm in the main analysis, with a third seed for five arms
Time	The thesis had to be completed by September 2026	Fixed schedules of 40 to 60 epochs; the learning rate was tuned only for the SegFormer-B2 baseline, so the MAAS and AquaScale comparisons use an untuned baseline
Data	6,382 images, 19,114 annotation rows, four classes; 7.0% of the rows carry only a bounding box; labels are incomplete	Scores are limited by label quality; the label audit (Chapter 5) measures the gap
Data rights	Images come from public datasets and the web and are not redistributed	Only scripts, splits, per-image counts, and stored results are shared; the external evaluation uses public SUIM and TrashCan images with no counterpart in our data

Legal and cultural aspects. Images of human divers appear across the dataset. However, we do not attempt to identify the individuals and the images are not redistributed either. The models are not designed for any kind of surveillance activity, and we oppose using them to identify the individuals.

3.3 Environmental Impact

The primary environmental cost of this work comes in the form of electricity used to train the models. The controlled study used about 162 GPU-Hours: 96 hours on the Pipeline A workstation, 4 hours on the hybrid retrain in Kaggle, and 62 hours on the Pipeline B workstation. One further run (TAM-Net seed 2) and the initial benchmark are not included as they were not recorded. Assuming each GPU drew its full rated board power throughout (285 W for the RTX 4070 Ti SUPER of Pipeline B, 320 W for the RTX 4080 SUPER of Pipeline A, and 70 W for the

Kaggle T4), the upper bound for consumed power is about 27.4 kWh for Pipeline A, 19.9 kWh for Pipeline B, 0.3 kWh for the hybrid retrain in Kaggle, with the total being around 47.6 kWh. Although the average power consumption during training is lower, the figures are calculated using upper bounds so that it represents the maximum possible impact.

There are two features of the work that help to minimize the waste of computation resources. First, the audit (near duplicate detection over all 6,382 images) takes only a few seconds on a CPU and it determined which conclusions were safe to make, as opposed to training the models on an invalid benchmark, which is far more expensive computationally. Second, TAM-Net's teacher and masking are only used during training, meaning there are no additional parameters at inference. Although the small stride-2 detail path is kept, it adds only 0.33% additional parameter at inference, making it much lighter than CoGS, which requires a large frozen CLIP tower at inference. In contrast to this, the study replicated its comparisons over multiple seeds and used matched controls, which added additional compute times. However, we judge this as a defensible use of compute as replication is what makes the results credible.

3.4 Ethical Issues

Data provenance. Our dataset combines images from several public datasets and from the web. The audit revealed that 773 of the 1,598 SUIM images (48.4%) and 1,645 of the 7,212 TrashCan images (22.8%) are also in our dataset, and that 70 of SUIM's official test images are in our dataset, 53 of which are in our own training split. As a result, our models cannot be evaluated on SUIM's official test set, and our external evaluation is performed only on the images that do not have a counterpart in our data. We describe this as a warning, not an accusation, about assembling datasets from redistributed sources.

Image provenance. All images used in this work are real underwater photographs or video frames sourced from the public datasets and web sources, and no synthetic or AI-generated images were used anywhere in this work. The only altered images are colour-enhanced copies of real photographs (red-channel restoration and CLAHE, used in the initial benchmark and in the enhancement comparison), augmented copies of real images (the initial class-aware augmentation and the on-the-fly flips and zoom of the controlled study), which were used for training only. The 100 pre-augmented images of the export are described in Chapter 4.

Transparent reporting. We withdrew one of our own measurements (Section 5.2.8), corrected an earlier measurement of a diagnostic (Section 5.2.3), and withdrew a stability claim when a third seed arrived (Section 5.2.3). We view this as our responsibility to not present any report or measurements as contributions if it does not pass the control tests.

Reviewer bias. There are two sources of human bias in this work. A total of 5 reviewers and annotators worked on the dataset, each labelling with their own judgement of what counts as an object, and its boundary, which led to inconsistencies in the annotations: the same type of object (e.g. small animal) was annotated in

some images and not in others, and 7.0% of the annotation rows contained only a bounding box. This inconsistency is a probable reason for the lack of labels measured in Chapter 5 and a restriction on the accuracy that any model can achieve on this dataset. To mitigate this, we have conducted our own blind review of candidate regions, which was done by two team members. To limit the effect of a reviewer who tends to answer "yes", random decoys were mixed in and all rates are reported next to the decoy rate.

Dual use. Theoretically, segmentation of divers and vehicles could aid in surveillance. We restrict our claims to environmental monitoring and infrastructure inspection and oppose its use for any surveillance purposes.

3.5 Standards

The study follows these conventions and standards:

- **Annotation formats.** COCO segmentation JSON and the YOLO polygon text format, both exported from the same annotations.
- **Metrics.** The standard IoU and Dice definitions, computed dataset-globally and averaged over the four object classes; confidence intervals by the percentile bootstrap [1].
- **Reporting.** Comparisons are made at a single test-time-augmentation setting (flip-TTA is reported where it was run), with seed replication and a stated selection rule (Chapter 4).
- **Documentation.** The IEEE citation style and the Brac University thesis template.
- **Software licences.** The open-source tools (PyTorch [23], segmentation-models-pytorch [21], OpenCLIP, Ultralytics) are used under their own licences.

3.6 Project Management Plan

The study was carried out by the five authors under the supervision indicated in the front matter, and in the phases described in Table 3.4. The controlled study was set up as two separate pipelines running concurrently on two different machines, and a Kaggle notebook for the two runs that required additional capacity.

Resources and budget. The GPUs were provided by the authors and the department, with the rest provided by Kaggle's free tier. The experiments were not conducted with any external funding and no paid service was bought for the experiments. The actual expenses were compute time (approximately 162 GPU-hours, Section 3.3), the time spent by the team members to annotate the dataset, the two members to review the annotations, and the authors' time. We do not assign a monetary value to them because the varying electricity pricing and the value of student

time relies on assumptions which we cannot verify. The two pipelines were executed on separate machines to allow them to run in parallel, and interrupted runs were restarted from the last checkpoint. Table 3.3 summarises the budget.

Table 3.3: Budget and resources of the project.

Item	**Source**	**Cost to the project**
GPUs (RTX 4080 SUPER, RTX 4070 Ti SUPER)	University workstations	Not charged; equipment not costed
Kaggle T4 sessions	Free tier	None
Electricity (47.6 kWh upper bound)	University	About 470 taka at an illustrative 10 taka per kWh (assumed price)
Software (PyTorch, segmentation-models-pytorch, OpenCLIP, Ultralytics)	Open source	None
Images	Public datasets and web sources	None
People (five authors; two of them for the blind review)	Students	Not valued

Table 3.4: Project plan by stage: activities, outputs, and period.

No.	**Stage**	**Main activities**	**Output**	**Period**
1	Survey and planning	Review of underwater enhancement, detection and segmentation literature; problem definition; proposed pipeline	Literature review; research direction	Up to June 2026
2	Dataset construction	Collection from public and web sources; polygon labelling in Roboflow; cleaning, enhancement and class balancing	Four-class dataset of 6,382 images in COCO and YOLO formats	Up to June 2026
3	Initial benchmark	Training and evaluation of six model families and a hybrid model on the original split	Initial results; the initial hybrid model	Up to June 2026
4	Audit	Near-duplicate detection, overlap with public datasets, re-scoring of the initial models	Contamination and score-inflation measurements	June–September 2026
5	Clean splits and protocol	Cluster-based rebuilding of the splits in two pipelines; one evaluation protocol	Splits A and B; evaluation protocol	June–September 2026
6	Controlled study	Baselines, PADS-Net, CoGS, TAM-Net (Pipeline A); seed noise, MAAS, AquaScale (Pipeline B); external validation	Scores with intervals, seeds, and controls	June–September 2026

No.	Stage	Main activities	Output	Period
7	Label audit	Candidate mining; blind review with decoys in two rounds; corrected scoring	Estimates of unlabelled objects	June–September 2026
8	Analysis and writing	Statistical analysis; verification of every number against the stored files; thesis writing	This thesis	To September 2026

3.7 Risk Management

Table 3.5 lists the risks we identified. Most are not hypothetical; they occurred, and the mitigation is what we did about them.

Table 3.5: Risks, what happened, and the mitigation.

Risk	What happened	Mitigation
Train–test leakage	41% of original test images had a near-duplicate in training	Cluster-based splits; independent re-implementation of the detector
Seed noise larger than the effect	Seed-only differences reached 0.020–0.028 mIoU	Seed replication; decision rule; controls
Over-claiming from few seeds	A two-seed TAM-Net result (0.7468) fell to 0.7433 with a third seed	Report two seeds per arm as the main analysis, disclose the third seed as a sensitivity analysis, and grade each claim: the SegFormer-B2 margin alone is not established, the pooled effect of the whole design over two backbones is (the pooling was chosen after the results)
Flawed measuring instrument	A first video-consistency check flagged 3,546 Marine Trash regions, more than the 3,395 large Marine Trash instances in the training set, before a local-registration test removed 97% of all flags; a filename-based noise measurement was invalid	Verify by pixel identity; retract what fails (Section 5.2.8)

Risk	What happened	Mitigation
Hardware failure	A crash on 20 September truncated a checkpoint, costing about 80 minutes of retraining	Write checkpoints to a temporary file, flush, verify by loading, then rename
Compute limits	The 12-hour Kaggle limit forced the hybrid retrain to 10 epochs	State the deviation; bound its effect using the initial benchmark's own validation curve
Reviewer yes-bias	The first review round had a 14% decoy yes-rate	Decoys in every round; second round reached 2%
Loss of reproducibility	Large caches and optimiser states cannot be shipped	Keep scripts, splits, per-image counts, stored results, and best checkpoints; document rebuild steps

3.8 Economic Analysis

The economic question with a research project of this kind is whether the cost of measuring reliability is small compared with the cost of acting on an unreliable result.

Costs. About 162 GPU-hours in the controlled study (Section 3.3), the annotation time spent by the authors, and the review time of the 2 team members (385 crops in round 1 and 429 in round 2). The audit itself is inexpensive: hashing and comparing all 6,382 images only took a few seconds on a CPU in our independent re-implementation, and needs no GPU.

Benefits. The audit ensured that no wrong conclusion was built. Without it, the first hybrid would have been deployed as it was the best model. Under a matched protocol, it gets 0.6435 compared to 0.7361 for a tuned SegFormer-B2. If a model is selected that is approximately 0.09 mIoU worse, then it would result in more missed debris or require additional manual review for a real inspection program. In conclusion, plain SegFormer-B2 is the least expensive, while TAM-Net only introduces a small amount of additional cost (0.33% more parameters). On the contrary, CoGS adds a large frozen CLIP tower at inference which contributes to significant additional cost.

Sensitivity. he conclusion that replication is worth its cost even considering the price of compute, because the alternative, an unreplicated comparison, produced a claimed margin (0.0175) that is smaller than the seed noise (0.0198 to 0.0276). It would only reverse if a single training run were so expensive that two seeds were unaffordable, which is not the case for models of this size.

Relevance to Bangladesh. Automated inspection can reduce the cost of recurring inspection, and risk of diver based pipeline and bridge pillar surveys. This thesis does not quantify that saving, and we make no claim about it; it gives the validated

segmentation baseline and the process required before such a system is trusted for real world deployment.

Chapter 4

Proposed Methodology

This chapter presents the methodology. It describes the overall design process, the specification of the candidate models, the construction and preprocessing of the dataset (collection, cleaning, transformation, integration, and reduction), the audit and the rebuilt splits, the evaluation protocol, and the implementation. The data pipeline of the initial benchmark is retained because the final dataset is built from it;

Figure 4.1 is a detailed flow diagram of the full pipeline, from the collection of the data to the final conclusions, including the two pipelines of the controlled study (Table 4.1). The header of each stage names the section or chapter that describes it.

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4.1 Design Process or Methodology Overview

Our initial design process was *data-centric*: we treated underwater performance as limited as much by data quality and class imbalance as by architecture. So we worked on enhancement, cleaning, and balancing before training. We also compared architectures on the same data. The later stage keeps that view but adds an *audit-centric* step in front of it: before any new model is trained, the benchmark is examined for leakage, label incompleteness, and evaluation noise, because those determine what a score can mean.

The design has five steps. (i) Audit the original benchmark (Section 4.4). (ii) Rebuild the splits so that near-duplicates, mirrored copies, and video frames stay together (Section 4.4.2). (iii) Fix one evaluation protocol (Section 4.5). (iv) Evaluate ideas: specify candidate designs and controls (Section 4.2), then train and evaluate them under that protocol with replication (Section 4.6). (v) Interpret the results

1. Dataset — Section 4.3

Public underwater datasets and web images, labelled with polygon masks in Roboflow (SAM-assisted). Six classes reduced to four: 6,382 images and 19,114 annotation rows. Exported as YOLO polygons, COCO masks and classification crops.

↓

2. Preparation — Section 4.3

Cleaning removes zero-area, tiny and degenerate polygons. Colour correction and CLAHE are tested only as an ablation. Images are letter-boxed to 512 (YOLO 640) and masks rasterised. Class-aware augmentation grows the training set from 5,402 to 12,543 images.

↓

3. Initial benchmark — Section 4.2

Six model families trained on the original split: YOLO-seg, Mask R-CNN, U-Net, DeepLabv3+, crop classifiers and the YOLO+U-Net hybrid. Scored with mIoU, Dice, pixel accuracy, mAP@50 and macro-F1. Best score: the hybrid, 74.12% mIoU.

↓

4. Audit of the benchmark — Section 4.4

Near-duplicate detector (dHash distance ≤ 5 and correlation ≥ 0.95): 143 of 347 test images (41.2%) have a near-duplicate in training. 48.4% of SUIM and 22.8% of TrashCan are in our data. On the clean subset, U-Net loses 0.0445 and DeepLabv3+ 0.0357 mIoU.

↓

5. Rebuilt splits — Section 4.4.2

Images are linked by name stem, near-duplicate, video id and 250-frame block (Pipeline B also links flipped and 180°-rotated copies). Union-find clusters go whole to train, valid or test, rarest class first. Split A: 4,448 / 980 / 954 images; Split B: 4,467 / 957 / 958 images.

↓

6. One evaluation protocol — Section 4.5

Dataset-global IoU per class, mIoU over the four classes; paired bootstrap with 95% intervals. Two seeds: a result counts only if the interval excludes 0 and all seed pairs agree. Selection on validation only; flip-TTA is reported apart.

↓

7A. Pipeline A: model comparison

Split A: 4,448 / 980 / 954 images; 60 epochs.

- Baselines: Att-UNet, U-Net, DeepLabv3+, SegFormer-B2.
- PADS-Net and CoGS, with matched controls.
- TAM-Net on SegFormer-B2 and DeepLabv3+, with no-masking controls.
- Retrained hybrid; external test on SUIM and TrashCan.

7B. Pipeline B: missing-label audit

Split B: 4,467 / 957 / 958 images; 40 epochs.

- SegFormer-B2 (three seeds) and U-Net.
- Label audit: 3-fold models, video check, blind review with decoys.
- About 770 unlabelled training objects; 158 confirmed in 150 test images.
- MAAS, AquaScale, deployable SegFormer-B2.

↓

8. Results and conclusions — Chapters 5 and 6

Results (Chapter 5): designs compared under one protocol; each claim graded as established or not established. **Conclusions** (Chapter 6): contaminated benchmark, incomplete labels, TAM-Net on SegFormer-B2 best-scoring (mIoU 0.7468); contributions and future work.

Figure 4.1: Detailed flow diagram of the full pipeline, from data collection to the conclusions. Each box is one stage, with its heading at the top and a short description below; the stages follow one another from top to bottom, and the heading of each stage names the section or chapter that describes it. Stage 7 shows the two independently built pipelines, A and B, side by side; scores are compared only within a pipeline.

Figure 4.1: Detailed flow diagram of the full pipeline, from data collection to the conclusions. Each row is one stage, and its steps are read from left to right; the stages follow one another from top to bottom. The header of each stage names the section or chapter that describes it. Stage 7 shows the two independently built pipelines, A and B, side by side; scores are compared only within a pipeline.

with a blind human review with random decoys (Section 4.4.5) and with two external datasets (Chapter 5). The sections of this chapter follow the template, so the audit and the protocol are described after the models and the data.

The work was carried out in two independently constructed pipelines, described in Table 4.1. They share the raw data and the goal, but differ in how the splits were built, how the few box-only labels are treated, how augmentation is applied, and which loss and schedule are used. Because they use different test sets, scores from one pipeline are *never* compared with scores from the other. Their value is different: Pipeline A hosts the model comparison, and Pipeline B hosts the label audit and the missing-annotation experiments. That both independently reproduced the central contamination finding is our replication; Section 5.1.7 lists what they reproduce.

Table 4.1: The two pipelines. Scores are comparable only within a pipeline.

Aspect	**Pipeline A**	**Pipeline B**
Images (train / valid / test)	4,448 / 980 / 954	4,467 / 957 / 958
Duplicate relations used to build clusters	Same base filename stem; dHash-and-correlation-confirmed pairs; same video id; same 250-frame block	The same, plus filename stems with mirror markers removed, and every image matched against the horizontal flip, vertical flip, and 180° rotation of every other
Label rows with no polygon (box-only)	Dropped, so they become background	Painted as IGNORE and excluded from loss and metrics
Training data	4,448 originals expanded offline to 10,446 images by class-balancing copies; TAM-Net variants add flip and zoom-in augmentation on the fly	4,467 originals; every augmentation applied on the fly, none pre-baked
Loss	Focal cross-entropy with class weights, plus focal Tversky	Cross-entropy plus soft Dice, no class weights
Schedule	60 epochs, batch 8, AdamW, OneCycle; learning rate 3×10^{-4}, or 6×10^{-5} for SegFormer-B2	40 epochs, batch 16, AdamW 3×10^{-4}, 2-epoch warm-up then cosine
Models studied	Att-UNet, U-Net, DeepLabv3+, SegFormer-B2, retrained initial hybrid, PADS-Net, CoGS, TAM-Net	U-Net, SegFormer-B2, 3-fold audit models, MAAS, AquaScale
Main outputs	Model comparison; external validation	Label audit; missing-annotation and scale studies; deployable package

4.2 Preliminary Design or Design (Model) Specification

In the spirit of presenting and verifying multiple alternative solutions, we do not commit to one architecture; we specify several families and let controlled experiments decide.

4.2.1 Initial benchmark model families

All initial models use backbones pretrained on ImageNet [4] or COCO and fine-tuned on our data. Table 4.2 maps each to its task, backbone, data format, and input size.

Table 4.2: Initial candidate models, tasks, and data formats.

Model	Task	Backbone	Data format	Input
YOLOv8m-seg	Detection + instance seg.	CSPDarknet-m	YOLO polygons	640
YOLOv11m-seg	Detection + instance seg.	C3k2-m	YOLO polygons	640
Mask R-CNN	Detection + instance seg.	ResNet-50-FPN	COCO JSON	800 (multi-scale)
U-Net	Semantic seg.	ResNet-50	COCO masks	512
DeepLabv3+	Semantic seg.	ResNet-101	COCO masks	512
EfficientNet-B4 / ResNet-50	Classification	—	Crops	224
Hybrid (YOLO+U-Net)	Seg. (+ det. aux)	ResNet-50	COCO masks (+det.)	512

YOLOv8m-seg and YOLOv11m-seg. YOLO is a single-stage detector that predicts boxes, class scores, and, in the segmentation variant, prototype mask coefficients in one pass. We use the medium variants (`yolov8m-seg` [57] and `yolo11m-seg` [64]) initialised from COCO-pretrained weights.

Mask R-CNN. Mask R-CNN [9] is the canonical two-stage instance-segmentation model: a ResNet-50-FPN backbone feeds a Region Proposal Network, and each proposal receives a class, a box, and a mask from a small FCN head. It serves as a two-stage reference.

U-Net and DeepLabv3+. U-Net [6] is an encoder–decoder with skip connections; DeepLabv3+ [13] adds atrous spatial pyramid pooling and a light decoder to a deeper (ResNet-101) encoder. Both use the `segmentation-models-pytorch` implementation [21] with ImageNet-pretrained encoders and five output classes.

Classification. To isolate classification from localisation we train EfficientNet-B4 [25] and ResNet-50 [8] on 224×224 object crops.

Preliminary hybrid. The hybrid shares one ResNet-50 encoder between an attention-gated U-Net decoder [14] and a YOLO-style detection head, with a detection-guided fusion gate. In the initial benchmark it was presented as groundwork; here we retrain it under the common protocol (Section 5.2.7).

4.2.2 Baselines used in the controlled study

Pipeline A retrains four architecture families under one protocol. **Att-UNet** is a ResNet-50 encoder with an attention-gated decoder (256/128/64/32 channels) and an auxiliary head at stride 8 (34.4 M parameters); it is the initial hybrid's decoder without the detection head and is our initial baseline. **U-Net** (32.5 M) and **DeepLabv3+** with a ResNet-101 encoder (45.7 M) are the standard reference models. **SegFormer-B2** [44] (24.7 M) has a hierarchical Mix Transformer encoder that yields features at strides 4, 8, 16, and 32 and an all-MLP decoder that fuses them at stride 4. We include it because ResNet-era baselines are 2016–2018 designs, and a modern baseline is needed before any claim of improvement can be judged.

4.2.3 PADS-Net: physics-based restoration

PADS-Net (Physics-Adaptive Dense-Small Network) tests whether restoring underwater colour inside the network helps segmentation. Its Differentiable Underwater

Restoration Module (DURM) predicts per-image physical parameters, namely attenuation coefficients β_c^D and β_c^B and veiling light B_c^∞ for each colour channel c, together with a dense range map $z(x, y)$. It then inverts the image-formation model [2], [18]

$$I_c = J_c\, e^{-\beta_c^D z} + B_c^\infty \left(1 - e^{-\beta_c^B z}\right) \implies J_c = \frac{I_c - B_c^\infty \left(1 - e^{-\beta_c^B z}\right)}{e^{-\beta_c^D z}}, \tag{4.1}$$

where I is the observed image and J the restored one. DURM is trained with the segmentation loss only, with no restoration ground truth. The nine physical parameters also condition every encoder stage through feature-wise linear modulation (FiLM), and a stride-4 density head with an area-inverse pixel weighting is added to help small objects. The network has 34.6 M parameters against 34.4 M for Att-UNet. Figure 4.2 shows the data flow.

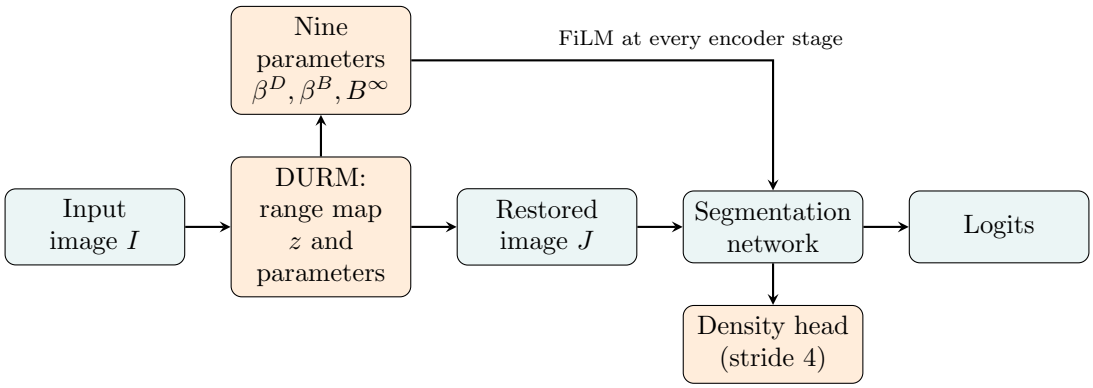


Figure 4.2: PADS-Net. DURM predicts nine physical parameters and a range map from the input image and restores it by inverting the image-formation model of Equation (4.1). The same parameters modulate every encoder stage through FiLM, and a stride-4 density head with an area-inverse pixel weighting supports small objects. Orange: new trainable parts.

4.2.4 CoGS: concept-guided segmentation

Each coarse class is a union of very different visual concepts; "Aquatic Animal" covers fish, octopus, jellyfish, turtles, and manatees, and a jellyfish can resemble a plastic bag from a different class. CoGS decomposes each class into named sub-concepts and lets a frozen language–image model supply a dense prior over them. Figure 4.3 shows the design.

Vocabulary. 57 concepts frozen on 14 September 2026, before any CoGS training: 21 for Aquatic Animal (fish, school of fish, shark, ray, eel, jellyfish, octopus, squid, sea turtle, dolphin, whale, seal, manatee, crab, lobster, shrimp, starfish, sea urchin, sea anemone, nudibranch, sea slug), 10 for Artifacts and Metals, 8 for Human Activity, 10 for Marine Trash and Plastic, and 8 background concepts (sand, rocks, coral reef, open water, seaweed, sea floor, algae, sunlight in water). Each is embedded through four templates such as "an underwater photo of {}". The concept list was written from the class definitions alone, with no test image consulted.

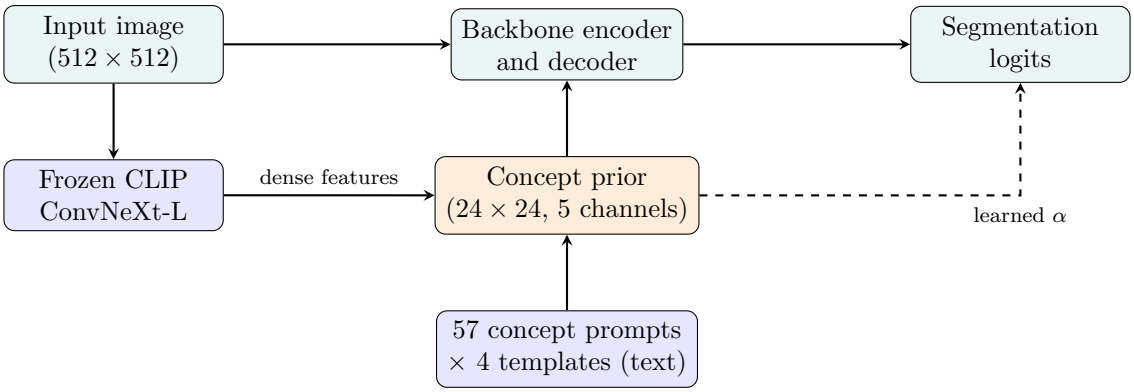


Figure 4.3: CoGS. A frozen CLIP tower produces dense features that are scored against text prototypes of 57 named concepts; the resulting class prior is fused into the decoder (solid arrow up, through zero-initialised convolutions) and added to the final logits (dashed arrow, scaled by a learned scalar α). All fusion weights start at zero, so at initialisation the network equals its baseline. Blue: frozen; orange: new trainable parts.

Image features. We use the OpenCLIP ConvNeXt-Large tower [38], [49] with LAION-2B weights [41], [53]. Each 512×512 letter-boxed image is resized to 768×768 and yields a 24×24 grid of 768-dimensional embeddings, cached once per image.

Prior and fusion. A residual 1×1 adapter (zero-initialised) is applied to the embeddings; cosine similarity to every concept prototype is scaled by a learned temperature; the maximum over a class's concepts gives that class's score, and a per-class bias (zero at start) calibrates them. The softmaxed prior and a 64-channel projection of the embeddings are fused into the decoder through zero-initialised convolutions [62], at strides 16 and 8 for Att-UNet, at the ASPP output and the stride-4 decoder for DeepLabv3+, and once at stride 4 for SegFormer-B2, and the log-prior is added to the final logits with a learned scalar that starts at zero. An extra loss (weight 0.4) supervises the prior directly. The trainable parameters added are 0.97 M for Att-UNet (35.3 M in total), 1.1 M for DeepLabv3+ (46.8 M), and 0.9 M for SegFormer-B2 (25.6 M); the CLIP tower is frozen and must be run at inference.

The prior in equations. Let $f(x)$ be the adapted CLIP embedding at position x and t_k the text prototype of concept k. The class score, the class prior, and the corrected logits are

$$S_c(x) = b_c + \max_{k \in \mathcal{K}_c} \lambda \cos\big(f(x),\, t_k\big), \tag{4.2}$$

$$P_c(x) = \frac{\exp S_c(x)}{\sum_{c'} \exp S_{c'}(x)}, \qquad \tilde{z}_c(x) = z_c(x) + \alpha \log P_c(x), \tag{4.3}$$

where $\mathcal{K}_c$ is the set of concepts of class c (the eight background concepts form the background class), b_c is a per-class bias that starts at zero, λ is a learned scale (initialised at 50, the inverse of a temperature), z are the logits of the backbone, and α is a learned scalar that starts at zero.

Ablations. Three modes change one thing each: *sub-concepts* (57 prompts); *class names* (only the four class names); and *visual only* (no text: a learned 1×1 classi-

fier on the CLIP features). A fourth control replaces CLIP with an ImageNet-22k ConvNeXt-L (`convnext_large.fb_in22k`), whose features pass through a fixed random orthogonal map from 1536 to 768 dimensions, to test whether language-aligned pre-training is needed or any strong frozen backbone will do.

4.2.5 TAM-Net: teacher-guided annotation masking

TAM-Net (Teacher-guided Annotation-Masking Network) targets the missing-annotation problem: a background-labelled pixel that in fact belongs to an unlabelled object penalises the network for being right. Figure 4.4 shows the design; it has four parts.

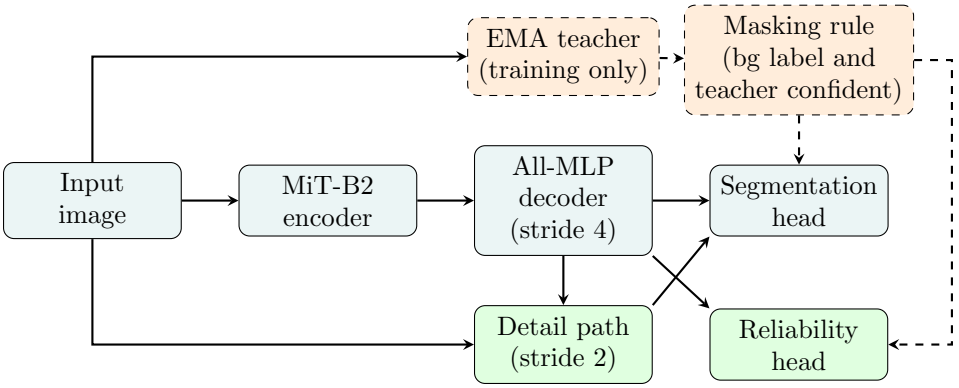


Figure 4.4: TAM-Net. The student is a SegFormer-B2 with a stride-2 detail path and a reliability head. An EMA teacher, used only during training, marks background-labelled pixels it confidently calls foreground; those pixels are removed from the loss (dashed arrow from the masking rule to the head) and the reliability head is trained to predict the resulting mask. The stride-2 detail path adds a zero-initialised residual to the logits. Dashed parts do not exist at inference.

(1) Student. SegFormer-B2 with an ImageNet-pretrained encoder. **(2) Detail path.** A shallow CNN on the raw image at stride 2, whose features are concatenated with a projection of the stride-4 decoder features and reduced by convolutions to a residual logit map that is added to the ordinary logits. The last convolution is zero-initialised, so at step 0 the network is numerically identical to a plain SegFormer-B2. It exists because the median Aquatic Animal covers only about 600 pixels (Section 5.1.9), so a stride-4 decoder represents it in a few cells. **(3) EMA teacher.** A copy of the student whose weights are an exponential moving average (momentum 0.999) of the student's; it is never back-propagated through. After a 15-epoch warm-up (of 60), a pixel labelled background whose maximum foreground softmax probability under the teacher exceeds a per-class threshold is *excluded* from both loss terms; foreground labels are never softened, because the audit indicated that annotations are missing rather than spurious. **(4) Reliability head.** A 1×1 convolution on the decoder features, trained with binary cross-entropy (weight 0.2) to predict the keep mask. It does not feed back into the main loss, to avoid a learned weight collapsing to "trust nothing".

The teacher and the mask in equations. With student weights θ^S and teacher

weights θ^T, the teacher update and the masking rule are

$$\theta^T \leftarrow m\,\theta^T + (1-m)\,\theta^S, \qquad m = 0.999, \tag{4.4}$$

$$\hat{c}_i = \arg\max_{c\in\{1,\dots,4\}} p^T_{i,c}, \qquad d_i = \mathbf{1}\big[\, y_i = 0 \;\wedge\; p^T_{i,\hat{c}_i} > \tau_{\hat{c}_i} \big], \qquad k_i = 1 - d_i, \tag{4.5}$$

where p^T is the softmax output of the teacher, y_i the label of pixel i (0 is background), and τ_c the per-class threshold. Only background labels can be masked. A masked pixel ($k_i = 0$) is removed from both loss terms: its cross-entropy is multiplied by k_i, and both its prediction and its target are set to zero in the soft counts of the Tversky term.

The masked share is 0 during warm-up, first appears at epoch 16, peaks at 0.82% of pixels at epoch 20, and is 0.62% at epoch 60. The parameters added over SegFormer-B2 are 82,598 (the detail path 82,341 and the reliability head 257), giving 24.81 M against 24.72 M. The teacher and the masking exist only during training; the detail path remains at inference.

Thresholds. The per-class thresholds (Aquatic Animal 0.846, Artifacts and Metals 0.833, Human Activity 0.795, Marine Trash 0.923) were originally computed from a per-class annotation-inconsistency measurement that we later found to be invalid (Section 5.2.8). They must therefore be regarded as hand-set hyper-parameters, not as calibrated from measured annotation reliability.

Controls. The *control* run uses exactly the same code, detail path, reliability head, augmentation, and schedule but a threshold above 1, so the mask can never fire. It separates the effect of masking from the effect of everything else added to the baseline. Note that both differ from the plain SegFormer-B2 baseline in three ways beyond masking: the detail path, the on-the-fly flip and zoom-in augmentation, and the absence of the area-inverse pixel weighting that the baseline used.

Second backbone. The design does not depend on SegFormer: the masking rule uses only the teacher's logits and the label mask, and the detail path and the reliability head attach to any decoder that emits 256 channels. After the SegFormer-B2 results we therefore attached the same design to DeepLabv3+ (ResNet-101 encoder, 45.8 M parameters against 45.7 M), with the same thresholds, EMA momentum, warm-up, augmentation and loss, and with the learning rate of the DeepLabv3+ baseline (3×10^{-4}, 60 epochs, batch 8). We trained TAM-Net on DeepLabv3+ with seeds 42 and 1 and a no-masking control (threshold 1.01) with seed 42; a control with seed 1 was not trained. The thresholds were not re-tuned for this backbone. Five of the 10,446 training images (copies of one source photograph) could not be read on the machine that trained these three runs, and whenever one of them was drawn the data loader used a random other training image, which affects at most 0.05% of the draws. Section 5.2.4 reports the results.

4.2.6 MAAS and the random and threshold controls

MAAS (Missing-Annotation-Aware Segmentation, Pipeline B) uses the same idea with a softer rule. An EMA teacher (0.999) runs once per epoch over the training images after a warm-up of 20% of the epochs. A background pixel is *suspected* if

the teacher's foreground probability exceeds τ for the last $K = 3$ evaluations, its connected region has at least a minimum area, and the region does not touch the boundary band of any labelled instance. For suspected pixels the background cross-entropy is scaled by $(1 - p_{fg})$; they are never relabelled as foreground, and at most 5% of an image's background may be down-weighted. The values τ, minimum area, and boundary width are chosen on validation only, by deleting a known 30% of validation instances and scoring how well the rule recovers them, as precision (the share of down-weighted pixels that lie inside a deleted object), recall (the share of deleted pixels reached), and the chance rate (the precision of a random choice of background pixels; its computation is given in Section 5.2.5). For the real-label run they are $\tau = 0.8$, minimum area 200, boundary width 0, $K = 3$.

In equations, with ℓ_i the cross-entropy of pixel i, $p^T_{\mathrm{fg},i}$ the foreground probability of the teacher, and $\mathcal{S}$ the set of suspected pixels defined above,

$$\ell_i^{\mathrm{MAAS}} = s_i\,\ell_i, \qquad s_i = \begin{cases} 1 - p^T_{\mathrm{fg},i} & i \in \mathcal{S}, \\ 1 & \text{otherwise.} \end{cases} \tag{4.6}$$

Figure 4.5 shows the mechanism and the two controls described next.

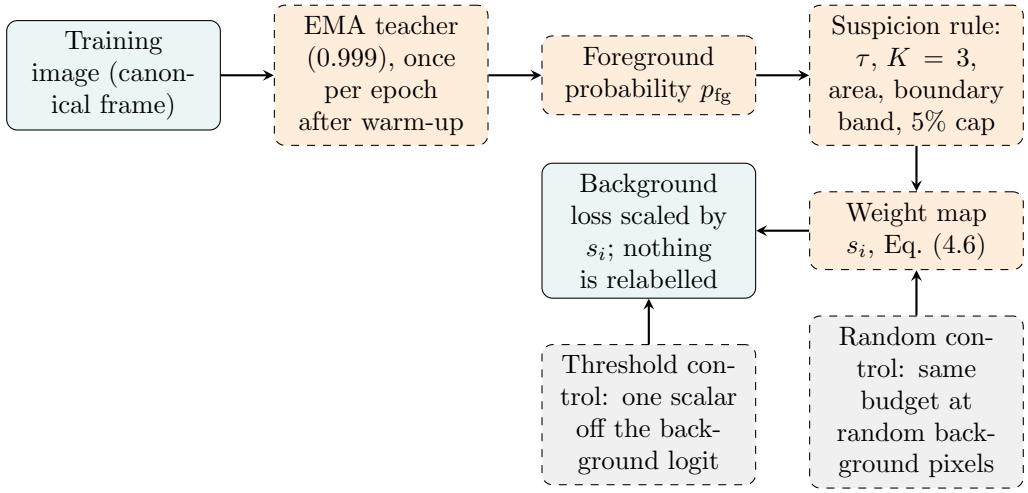


Figure 4.5: MAAS. An EMA teacher marks background pixels that it confidently calls foreground as suspected missing annotations, and their background loss is scaled by s_i (Equation (4.6)). Orange dashed: parts that exist only during training. Grey dashed: the two controls, which replace the suspicion rule by random locations with the same pixel budget (random control) or by a matched shift of the background logit (threshold control).

Two controls isolate what MAAS adds. The *random control* down-weights the same number of pixels, in the same region-size distribution and with the same mean weight, but at random background locations, so it separates selecting the right pixels from merely having less background loss. The *threshold control* subtracts one scalar from the base model's background logit, fitted on validation to match MAAS's foreground recall, so it separates a genuine improvement from a shifted operating point.

4.2.7 AquaScale: scale-aware design

AquaScale (Pipeline B) targets the finding that animal detection is U-shaped in object size (Section 5.1.9). It keeps the SegFormer-B2 encoder and adds (i) per-pixel scale attention over the four encoder levels, after Chen et al. [7]; (ii) a three-convolution stride-2 detail branch on the raw image, in the spirit of BiSeNet's spatial path [16]; and (iii) instance-centred zoom augmentation from a 1024-pixel cache, in the spirit of scale-normalised chips [15]. It has 24.95 M parameters against 24.72 M, so a gain cannot be attributed to size. Two zoom variants were tried: version 1 zooms toward all size classes and version 2 only toward large and huge animals. Figure 4.6 shows the design.

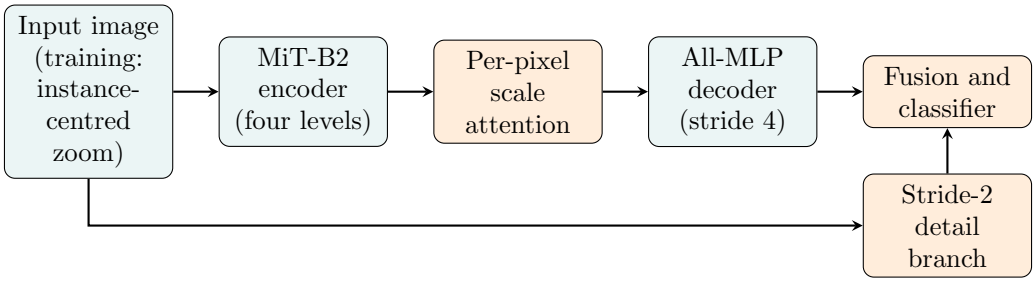


Figure 4.6: AquaScale. A SegFormer-B2 encoder–decoder with (i) per-pixel attention over the four encoder levels, (ii) a three-convolution stride-2 detail branch on the raw image that is fused before the classifier, and (iii) instance-centred zoom augmentation from a 1024-pixel cache during training. Orange: new parts.

4.2.8 Summary of the candidate designs

Table 4.3 summarises what each design targets and what it adds.

Table 4.3: Designs evaluated in the controlled study.

Design	**Targets**	**Added trainable parameters**	**Extra at inference**
SegFormer-B2	Stronger backbone than ResNet	(baseline, 24.7 M)	none
PADS-Net	Colour degradation, small objects	about 0.2 M over Att-UNet	restoration module
CoGS	Weak semantic knowledge on small data	0.9–1.1 M	large frozen CLIP tower
TAM-Net	Missing annotations; small objects	82,598 (0.33%)	stride-2 detail path
MAAS	Missing annotations	none	none
AquaScale	Objects from a few pixels to whole frame	about 0.23 M	scale-attention and detail branch

4.3 Data Collection

No single public dataset covers the four categories we target, so we built our own. We combined several public underwater datasets, manually selected suitable images, and added web images to improve diversity. All images were uploaded to Roboflow, where each object was labelled with a polygon mask using a SAM-assisted (Segment Anything [58]) semi-automatic workflow: the annotator clicks an object, SAM proposes a mask, and the annotator refines it and assigns the class. The audit in Section 4.4 later found that 1,332 of the 19,114 annotation rows (7.0%) carry only a bounding box and no polygon, and that near-identical images can appear under different filenames.

Dataset evolution. Our first iteration had roughly 4,000 images in six classes (Aquatic Animal, Artifacts and Metals, Human Activity, Marine Trash, Natural Formation, Plastic). Baseline models performed poorly on it: "Natural Formation" was ill-defined and overlapped the background, and "Marine Trash" and "Plastic" were confused. We removed "Natural Formation" and merged the two debris classes into "Marine Trash and Plastic", and expanded the data. Version 3 has **6,382 images** (the COCO export lists one additional `.jpeg` image and 19,115 annotations; the label files that both pipelines use contain 6,382 images and 19,114 annotation rows) in **four classes**, exported in COCO segmentation and YOLOv8 polygon formats from the same annotations.

The classes and their raw counts are in Table 4.4. The distribution is long-tailed: aquatic animals and marine trash dominate and artifacts and metals are rare. Source and class are correlated (Figure 4.7); we did not stratify by source, and the cluster-based splits of Section 4.4.2 keep near-duplicates together whatever their source.

Table 4.4: Target classes and raw annotation counts (dataset v3), recounted from the label files.

ID	Class Name	Annotations	Share
0	Aquatic Animal	9,077	47.5%
1	Artifacts and Metals	1,275	6.7%
2	Human Activity	2,038	10.7%
3	Marine Trash and Plastic	6,724	35.2%
	Total	**19,114**	**100%**

4.3.1 Data Cleaning

A detailed inspection of the raw export revealed quality issues that would corrupt mask rasterisation or destabilise training (Table 4.5). On the COCO branch, all 1,298 run-length-encoded annotations were decoded (719 were converted to polygons; 579 were discarded: 211 tiny, 188 degenerate, and 180 for other reasons such as a failed conversion or an empty mask). Zero-area annotations, tiny annotations (<100 px), and degenerate polygons (<4 vertices) were removed. The 167 unannotated training images were kept as background negatives.

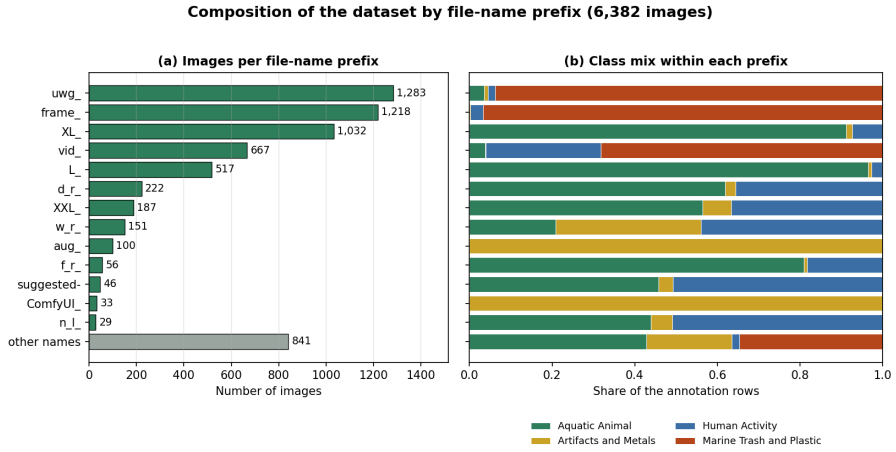


Figure 4.7: Composition of the dataset by file-name prefix, the only source information stored with the images (all 6,382 images). (a) Images per prefix. (b) Share of each class among the annotation rows of each prefix: source and class are strongly correlated.

Table 4.5: Raw-data quality issues identified and resolved during cleaning (initial benchmark).

Issue	**Count**	**Action**
Zero-area annotations	17	Removed (cause NaN in mask loss)
Tiny annotations (<100 px)	1,935	Removed (label noise)
Degenerate polygons (<4 vertices)	106	Removed (corrupt rasterisation)
RLE-encoded annotations	1,298	Converted to polygon (719) / removed (579)
Unannotated train images	167	Kept as background negatives
Mixed formats (.jpg/.jpeg/.webp)	—	Unified to JPEG

4.3.2 Data Transformation

Two transformations prepare the data. **Underwater enhancement** corrects wavelength-dependent attenuation through (i) *adaptive red-channel restoration*, applied only when the red mean is below 88% of the green mean, (ii) *blue-cast suppression*, applied when the blue mean exceeds 115% of the green mean, and (iii) *CLAHE* [3] (clip limit 2.0, 8×8 tiles) on the luminance channel in LAB colour space. Measured over the 6,382 images, the red rule fires on 67.5% of images (mean multiplier 1.63 when it fires) and the blue rule on 22.9%. Figure 4.8 shows examples. **Geometric and encoding transformation** then letter-boxes every image to a fixed size (512 for segmentation, 640 for YOLO) and rasterises a single-channel semantic mask (0 background, 1–4 class), filling polygons largest first so that small objects survive overlap. In the controlled study, enhancement is applied only in an ablation; the main models use the un-enhanced letter-boxed image, and inputs are normalised with ImageNet statistics inside each network.

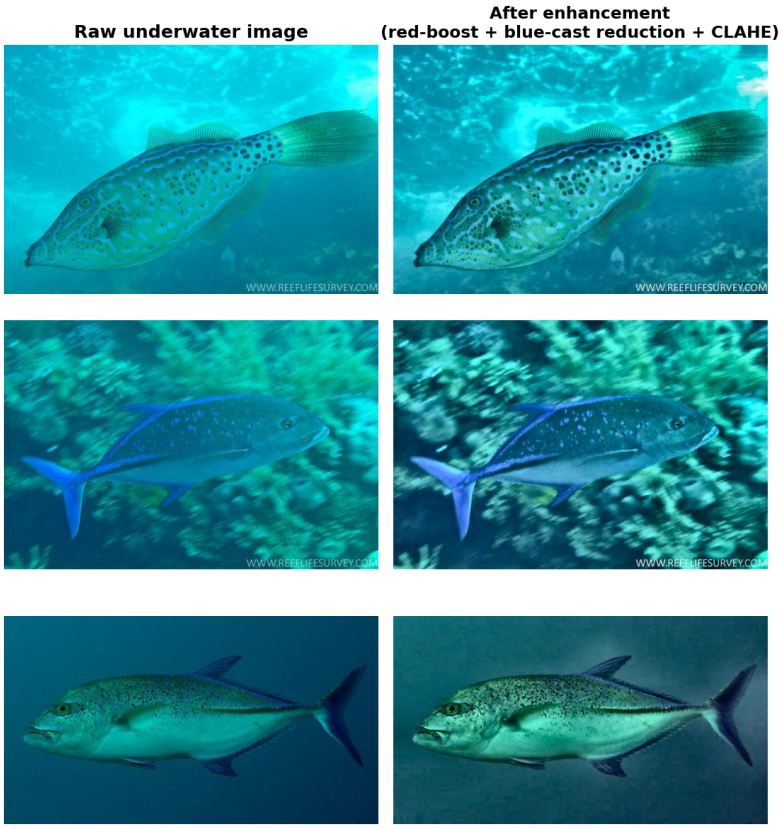


Figure 4.8: Underwater enhancement results (initial benchmark). Left: raw images with a blue-green cast. Right: after red-channel restoration, blue-cast suppression, and CLAHE.

4.3.3 Data Integration

The dataset is assembled from heterogeneous origins, so integration was a deliberate step. All sources were unified into one Roboflow project with one four-class schema and one annotation standard, and from this single source we export synchronised representations that share identical pixels and labels: YOLO polygon labels, a COCO JSON with semantic masks, and classification crops. This guarantees that every model family is trained and tested on the same underlying data. It does not, as Section 4.4 shows, guarantee that the training and test images are independent. The export also contains 100 images whose file names start with `aug_`: pre-augmented versions of photographs, none of which has a counterpart with the same source name elsewhere in the data. They were split like any other image (16 of Pipeline A's 954 test images and 44 of Pipeline B's 958). Thirty-three images have file names beginning with `ComfyUI_`; we could not confirm their origin (Chapter 3). These 133 images contain only Artifacts and Metals annotations, and they account for 10.4% of that class's annotation rows in Pipeline A's test set and 23.0% in Pipeline B's (Figure 4.7).

4.3.4 Data Reduction

At the *class* level, six classes were reduced to four. At the *annotation* level, cleaning removed thousands of noisy, tiny, and degenerate annotations. At the *image* level, all images are letter-boxed to a fixed working resolution, which bounds the input size despite an original spread of 357 distinct resolutions up to 6512×4366. The initial benchmark assumed that a held-out validation and test set that is never augmented would give a fair evaluation; Section 4.4 shows that assumption was false for the original split.

4.3.5 Summary of Preprocessed Data

The initial preprocessing expanded the training set from 5,402 to 12,543 images through class-aware augmentation and roughly doubled the share of the rare Artifacts and Metals class (6.9% to 13.9%), as shown in Figure 4.9; Figure 4.10 shows the growth of the training set. Table 4.6 summarises the initial export.

Table 4.6: Summary of the preprocessed dataset of the initial benchmark across the three export formats (original split).

Format	**Train**	**Valid**	**Test**	**Notes**
YOLO (polygons)	12,537	634	347	640×640 input
COCO (masks)	12,543	634	347	512×512 semantic masks
Classification crops	29,930	2,803	1,673	224×224 crops

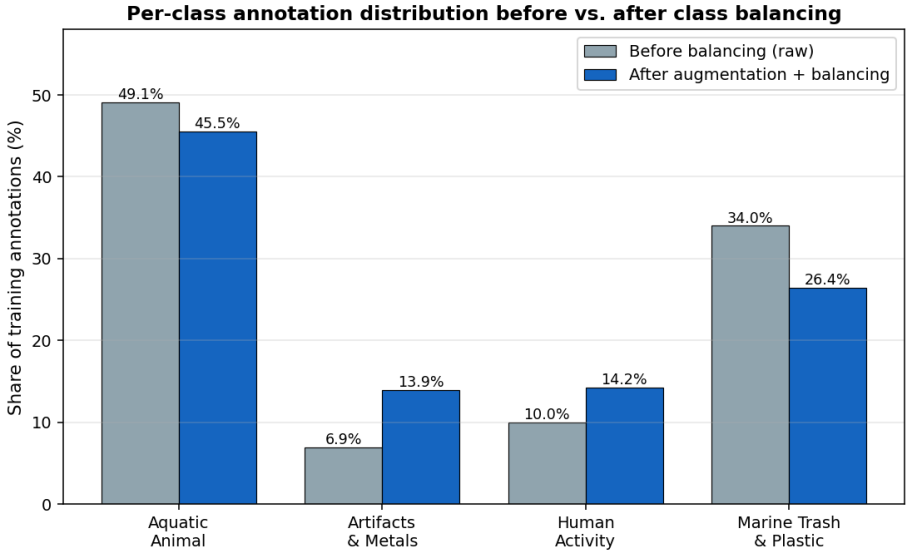


Figure 4.9: Per-class annotation share in the initial training set before and after class-aware augmentation and balancing.

Instance sizes. We recomputed instance sizes from the polygon labels of all 6,382 images (Table 4.7). The median Aquatic Animal covers 609 pixels in the 512×512 frame, about 25×25, and 75.9% of animals cover less than 1% of the frame; a

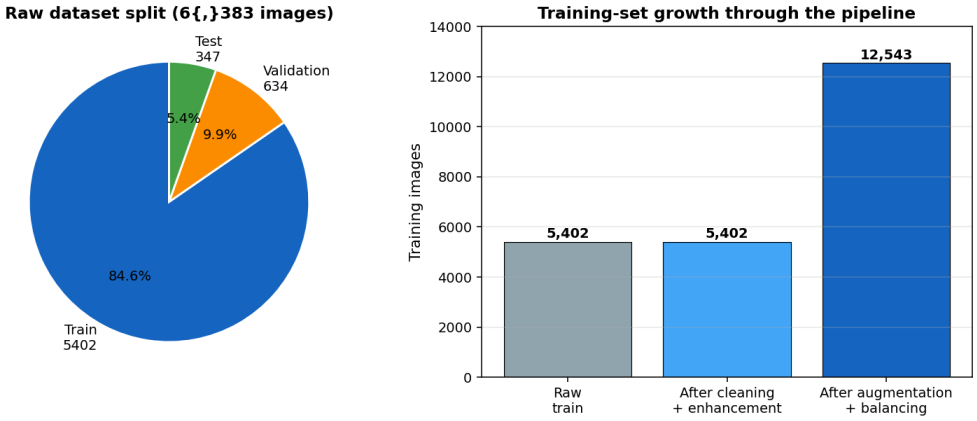


Figure 4.10: Left: raw dataset split. Right: growth of the initial training set through cleaning, enhancement, and class-aware augmentation.

median Artifacts and Metals object is about 27 times larger. Animals also make up 47% of all instances but a small share of pixels. This drives several design choices and the small-object analysis in Chapter 5.

Table 4.7: Instance size by class, computed from the polygon rows (at least four vertices) of all 6,382 images, in the letter-boxed 512 × 512 frame. Box-only and degenerate rows are excluded, so the counts are lower than the raw annotation counts of Table 4.4.

Class	**Instances**	**Median area (px)**	**Under 1% of frame**	**Under 32×32 px**	**Pixel share of those**
Aquatic Animal	9,017	609	75.9%	60.5%	3.4%
Artifacts and Metals	1,271	16,615	22.0%	10.2%	0.2%
Human Activity	2,036	6,495	28.5%	10.1%	0.4%
Marine Trash and Plastic	5,331	12,443	19.6%	10.2%	0.2%

4.4 Benchmark Audit and Clean Splits

Figure 4.11 summarises the audit of the original split and the construction of the two clean splits; the subsections below describe each step.

4.4.1 Near-duplicate detection

Two images are treated as near-duplicates if (i) the Hamming distance between their 64-bit difference hashes (dHash, computed on a 9 × 8 grey-scale reduction) is at most 5, *and* (ii) the normalised cross-correlation of their mean-centred 32 × 32 grey-scale thumbnails is at least 0.95. The hash is a cheap candidate generator; the correlation is the confirmation. The second condition is necessary: dHash alone

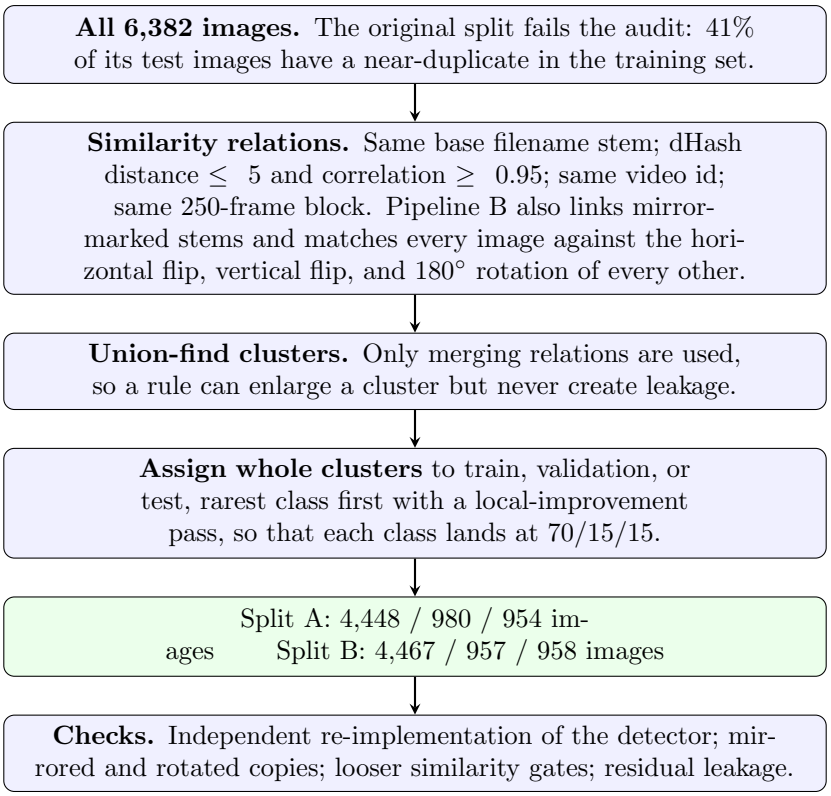


Figure 4.11: Audit of the original split and construction of the clean splits.

links unrelated low-texture underwater scenes. In our re-implementation, 619 of the 1,050 hash candidates between the original test and training images (59%) were rejected by the correlation check, and Pipeline B's construction rejected 11,784 hash candidates in the same way. Pipeline B additionally compares every image with the horizontal flip, vertical flip, and 180° rotation of every other, because the export contains pre-mirrored copies of images: 770 filenames carry a flip marker.

Independent verification. For this thesis we re-implemented the detector from its specification, using a different image library and interpolation, and applied it to the original splits (Chapter 5). We also applied it to both rebuilt splits, including the mirrored variants.

4.4.2 Clusters and splits

Images are merged into indivisible clusters by union-find over the relations in Table 4.1. Only merging relations are used, so a rule can never create leakage; it can only make a cluster larger. Whole clusters are then assigned to train, validation, or test, greedily in rare-class-first order with a local-improvement pass, so that each class lands at 70/15/15 of its annotations. The largest cluster in Pipeline B contains 987 images and is assigned to training. Video frames are near-duplicate-dense, so a valid split has to keep them on one side; the consequence is that the test sets contain few video frames and measure mainly the still-image domain, which applies equally to every model compared.

Table 4.8 gives the two resulting splits, recounted from the label files, next to the

original one. We call them "near-duplicate and same-video free", never "leakage-free": the criterion is a similarity threshold and not a proof of independence.

Table 4.8: The original export and the two rebuilt splits, recounted from the raw label files. Box-only rows carry a bounding box but no polygon.

	Original export			**Split A**			**Split B**		
Quantity	Train	Valid	Test	Train	Valid	Test	Train	Valid	Test
Images	5,401	634	347	4,448	980	954	4,467	957	958
Aquatic Animal	8,253	575	249	6,353	1,361	1,363	6,354	1,362	1,361
Artifacts and Metals	1,166	39	70	892	191	192	893	191	191
Human Activity	1,677	177	184	1,426	306	306	1,427	306	305
Marine Trash and Plastic	5,711	642	371	4,706	1,008	1,010	4,707	1,009	1,008
All annotation rows	16,807	1,433	874	13,377	2,866	2,871	13,381	2,868	2,865
of which box-only rows	1,331	0	1	975	68	289	1,168	92	72
Background-only images	167	0	0	83	0	84	167	0	0

Two design differences matter later. Pipeline A places 84 of the 167 background-only images in its test set (and 83 in training); Pipeline B puts all 167 in training. And Pipeline A's construction did not link mirrored copies, so, as Chapter 5 shows, about 11% of its test images have a mirrored or rotated copy in training.

4.4.3 Label handling and masks

Of the 19,114 annotation rows, 1,332 are box-only (1,320 of them Marine Trash). The initial pipeline dropped them, turning those objects into background. Pipeline B paints them as IGNORE (255) where the mask is still background, and excludes them from loss and metrics. Pipeline A still drops them, as the initial benchmark did: its 954 test masks therefore contain 706 of the 1,010 Marine Trash instances, the rest being box-only rows or degenerate polygons. Measured in the 512 frame these boxes are tiny (median about 16 pixels), so the effect on IoU is small, but it is a difference between the pipelines. In both pipelines polygons are rasterised in the 512 frame, largest first, so that small objects survive overlap.

4.4.4 Public-dataset overlap

The same detector was applied between our images and the public SUIM [33] and TrashCan [32] datasets, to trace where the contamination came from and to obtain external test images that are guaranteed not to be in our data.

4.4.5 Annotation audit

Missing annotations were measured with three streams, none of which trusts the labels. **Video consistency:** a region annotated in one frame and unannotated at the same coordinates in a near-identical frame means one label is wrong. **Model disagreement:** a 3-fold cross-validation on the training set (20 epochs per fold)

gives every training image a prediction from a model that never saw it; a region is a candidate if the label is background, the model is confident (probability at least 0.7) in a foreground class, the region has at least 100 pixels, and it does not touch a labelled boundary (5-pixel band). **Blind human review with decoys:** crops with the candidate outlined are shown one at a time, shuffled, with opaque identifiers; the reviewers answer yes, no, or unsure (round 1) or pick a class or background (round 2). Two team members carried out the review, and the stored answers hold one final answer per crop. Random background regions, drawn with the same box shapes as the candidates, are mixed in as decoys, so the rate at which the reviewers answer yes to nothing is measured and reported next to every rate.

Round 1 covered training candidates (385 crops, 60 of them decoys). Round 2 covered 150 test images fixed in advance by a random seed, with candidates pooled from the union of the predictions of the three SegFormer-B2 seeds so that no single seed is favoured (429 crops, 50 of them decoys). The number of missing objects per class is estimated as the number of mined candidates times the decoy-adjusted precision,

$$\hat{p}_c = \frac{r_c - d}{1 - d}, \qquad \hat{N}_c = n_c\,\hat{p}_c, \tag{4.7}$$

where n_c is the number of mined candidates of class c, r_c the yes-rate among decisive answers for that class, and d the decoy yes-rate. Because only objects that some model detected can be reviewed, the estimate is a lower bound.

4.5 Evaluation Protocol

Metric. The metric is dataset-global IoU per class, with the true-positive, false-positive, and false-negative pixel counts summed over every pixel of the test set, and the reported mIoU is the mean over the four object classes,

$$\mathrm{IoU}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \qquad \mathrm{mIoU} = \frac{1}{4}\sum_{c=1}^{4} \mathrm{IoU}_c, \tag{4.8}$$

so an all-background prediction scores 0. Pixel accuracy is about 96% for our models and is uninformative.

Other metrics. The initial benchmark also reported the Dice coefficient, pixel accuracy, detection average precision, and, for the classification of object crops, accuracy and macro-averaged F1,

$$\mathrm{Dice}_c = \frac{2\,\mathrm{TP}_c}{2\,\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \qquad \mathrm{mDice} = \frac{1}{C}\sum_c \mathrm{Dice}_c, \tag{4.9}$$

$$\mathrm{PA} = \frac{\sum_c \mathrm{TP}_c}{N}, \qquad \mathrm{AP}_{50} = \int_0^1 p(r)\,\mathrm{d}r, \qquad \mathrm{mAP}_{50} = \frac{1}{C}\sum_c \mathrm{AP}_{50,c}, \tag{4.10}$$

$$\mathrm{F1}_c = \frac{2\,P_c R_c}{P_c + R_c}, \qquad \text{macro-F1} = \frac{1}{C}\sum_c \mathrm{F1}_c, \tag{4.11}$$

where C is the number of classes, N the number of pixels, $p(r)$ the precision of a detector at recall r when a prediction counts as correct if its IoU with an object

is at least 0.5, and P_c and R_c the precision and recall of class c. Accuracy is the share of correctly classified crops. IGNORE pixels (Pipeline B) are excluded from all counts.

Uncertainty. The bootstrap [1] resamples test images with replacement (1,000 to 2,000 resamples) and gives 95% percentile intervals. Every difference between two models uses a *paired* bootstrap: the same resample of images is used for both, and image order is checked to be identical. For groups of seeds, the per-resample score of a group is the mean over its members.

Replication and the decision rule. Every arm of the Pipeline A comparison is trained with two seeds, 42 and 1, the two seeds that exist for all arms, so that the choice depends on availability and not on the score; the rule was applied after all runs had finished, which is why every third seed is disclosed in Section 5.4.1. A third seed (seed 2) was trained later for five arms only (Att-UNet, the two CoGS variants on Att-UNet, the ImageNet-feature control, and TAM-Net); to keep the number of seeds equal across arms it is not mixed into the group means, and is reported instead in a separate sensitivity analysis (Section 5.4.1). Pipeline B trained SegFormer-B2 with three seeds as a dedicated measurement of seed noise (Section 5.1.4); none of its other comparisons uses more than one run per arm. A difference counts as a result only if the paired interval excludes zero and the difference has the same sign in every seed pair; otherwise it is reported as "not established". For comparisons between groups with few seeds we additionally report the exact permutation p-value computed on the seed-level means, which does not depend on the image bootstrap; with two seeds per arm its smallest possible value is 0.33, so no claim rests on it. One further analysis pools the effect of the TAM-Net design over its two backbones (Section 5.2.4); it was decided after the DeepLabv3+ runs had finished, is labelled exploratory, and uses a control with a single seed on DeepLabv3+.

Test-time augmentation. Horizontal-flip TTA (the average of the prediction on the image and on its mirror) is reported where it was run (Table 5.6); the main tables are without TTA, and every comparison is made at the same setting, because TTA alone is worth up to 0.019 mIoU (Section 5.1.4).

Selection. Checkpoints and all design and hyper-parameter choices use the validation split only; the test split is used only to report the final scores and never to choose a checkpoint, a setting, or a model. For Pipeline B we also report the final-epoch checkpoint as an exploratory analysis, labelled as such.

Losses. Pipeline A combines a focal cross-entropy [10] and a focal Tversky loss [17],

$$c_i = -w_{y_i} \log p_{i,y_i}, \qquad \mathcal{L}_{\mathrm{FCE}} = \frac{1}{N} \sum_{i=1}^{N} \omega_i \left(1 - e^{-c_i}\right)^{\gamma} c_i, \tag{4.12}$$

$$\mathrm{TI}_c = \frac{\mathrm{TP}_c + 1}{\mathrm{TP}_c + \alpha\,\mathrm{FP}_c + \beta\,\mathrm{FN}_c + 1}, \qquad \mathcal{L}_{\mathrm{FT}} = \frac{1}{4} \sum_{c=1}^{4} \left(1 - \mathrm{TI}_c\right)^{\gamma_T}, \tag{4.13}$$

$$\mathcal{L} = \mathcal{L}_{\mathrm{FCE}} + \mathcal{L}_{\mathrm{FT}}, \tag{4.14}$$

where $p_{i,c}$ is the softmax probability of class c at pixel i, $\gamma = 2$, and w_c are class weights, $(0.595, 1.904, 1.567, 0.865)$ for the four object classes and 1.0 for background (inverse frequency in the training split). The pixel weight ω_i is 1, or for some

baselines $\sqrt{\text{reference area/instance area}}$ clipped to $[1, 8]$. TP_c, FP_c, and FN_c are the soft counts of class c over the batch with background excluded, and $\alpha = 0.3$, $\beta = 0.7$, $\gamma_T = 0.75$, so that false negatives are penalised more than false positives. Pipeline B uses cross-entropy plus a soft multiclass Dice loss, with no class weights,

$$\mathcal{L}_B = \frac{\sum_i v_i s_i \, \text{CE}_i}{\sum_i v_i s_i} + \left(1 - \frac{1}{5} \sum_{c=0}^{4} \frac{2 \sum_i v_i \, p_{i,c} \, t_{i,c} + 1}{\sum_i v_i \, p_{i,c} + \sum_i v_i \, t_{i,c} + 1}\right), \tag{4.15}$$

where $v_i = 0$ marks an IGNORE pixel, $t_{i,c}$ is the one-hot label, and s_i is the pixel weight of MAAS and its random control (1 for every other run).

4.6 Implementation of Selected Design

All models were implemented in PyTorch [23] with automatic mixed precision. Table 4.9 lists the initial hyper-parameters (retained for the initial results in Chapter 5), and Table 4.10 those of the controlled study.

Initial augmentation and balancing. Each training image is replicated by its dominant class (5× for Artifacts and Metals, 3× for Human Activity, 2× for Aquatic Animal and Marine Trash) and the extra copies are augmented. Geometric transforms (flips, ±15° rotation, scale and shift) are applied identically to image and mask; photometric transforms (brightness and contrast, HSV jitter, blur, noise) to the image only. In Pipeline A the same balancing produces the 10,446-image training set.

Table 4.9: Key training hyper-parameters per model in the initial benchmark.

Model	**Epochs**	**Batch**	**Init. LR**	**Optimiser**	**Scheduler**	**Input**
YOLOv8m-seg	100 (stopped at 88)	7 (auto)	0.01	auto: SGD type (m=0.9, wd=5e-4)	linear (lrf=0.01)	640
YOLOv11m-seg	150 (stopped at 82)	5 (auto)	0.01	auto: SGD type (m=0.9, wd=5e-4)	linear (lrf=0.01)	640
U-Net (R50)	120	32	1e-3	AdamW (wd=1e-4)	CosineAnnealing	512
DeepLabv3+ (R101)	50	32	1e-3	AdamW (wd=1e-4)	CosineAnnealing	512
Mask R-CNN (R50-FPN)	30 (stopped at 17)	8	5e-3	SGD (m=0.9, wd=5e-4)	MultiStep [15,22]	800
EfficientNet-B4	60 (stopped at 27)	64	1e-3	AdamW (wd=1e-4)	CosineAnnealing	224
ResNet-50 (cls)	60	64	1e-3	AdamW (wd=1e-4)	CosineAnnealing	224
Hybrid (YOLO+U-Net)	100 (stopped at 72)	24	1e-3	AdamW (warmup 3)	cosine	512

Which learning rate. The SegFormer-B2 baseline was first trained at the shared rate of 3×10^{-4} (test mIoU 0.6883, seed 42) and then at 6×10^{-5} (0.7395, same seed), a difference of 0.051 that is larger than any architectural effect except the backbone

Table 4.10: Training configuration of the controlled study. All runs use AdamW [22] with weight decay 10^{-4}, 512×512 input, and mixed precision; the best-validation checkpoint is evaluated.

Model (pipeline)	**Epochs, batch**	**Learning rate**	**Schedule and notes**
Att-UNet, U-Net, DeepLabv3+, PADS-Net, CoGS on Att-UNet / DeepLabv3+ (A)	60, 8	3×10^{-4}	OneCycle [11], 10% warm-up; gradient clipping 5.0
SegFormer-B2 baseline, CoGS on SegFormer, TAM-Net and control (A)	60, 8	6×10^{-5}	As above; TAM-Net variants: EMA 0.999, masking after epoch 15, flip and zoom-in augmentation
TAM-Net and its control on DeepLabv3+ (A)	60, 8	3×10^{-4}	As the TAM-Net variants above: EMA 0.999, masking after epoch 15, flip and zoom-in augmentation; the control has masking disabled
Retrained initial hybrid (A, Kaggle)	10, 8	3×10^{-4}	OneCycle sized to 10 epochs; loss adds two auxiliary and one detection term
SegFormer-B2, U-Net, MAAS, controls, AquaScale (B)	40, 16	3×10^{-4}	2-epoch warm-up, cosine; flips, scale 0.75–1.25, rotation $\pm 15°$, brightness/contrast $\pm 20\%$ on the fly
3-fold audit models (B)	20, 16	3×10^{-4}	Each trained on two folds of the training set

change itself. The lower rate was chosen on validation (0.7155 against 0.7516) and used for every SegFormer-family run in Pipeline A. Pipeline B's SegFormer runs stayed at 3×10^{-4}, so they are, by this evidence, less well tuned than Pipeline A's.

Reproducibility measures. Checkpoints are written to a temporary file, flushed to disk, verified by loading, and only then renamed, after a workstation crash on 20 September truncated one. Augmentation in Pipeline B is deterministic per seed, epoch, and index. A reproduction gate re-evaluated the stored TAM-Net checkpoint on the machine used for the last evaluations and returned the earlier score to within one millionth (0.747073 against 0.747072) before the new run was scored.

Selected design. The design carried forward as the final configuration is TAM-Net on SegFormer-B2 (second row of Table 4.10): learning rate 6×10^{-5}, 60 epochs, batch 8, EMA momentum 0.999, masking from epoch 16, flip and zoom-in augmentation, and flip test-time augmentation at inference. The teacher and the masking are needed only during training; the stride-2 detail path (82,598 parameters, 0.33%) stays at inference. Chapter 5 gives the evidence and its limits.

Chapter 5

Result Analysis

The findings of the audit and the controlled experiments are reported in this chapter. All numbers are re-calculated rather than copied from previous notes and Appendix B contains details of how. If any number on a working document was not the same as the recomputed number, the recomputed number is listed, and the discrepancy is noted.

5.1 Performance Evaluation

5.1.1 Evaluation criteria and testing methods

Models are evaluated using dataset-global IoU per class and by mIoU (Equation (4.8)) on the test split of their pipeline with 95% bootstrap intervals and using multiple replications over seeds and the decision rule of Section 4.5. For each result, the evaluation questions are: Is it accurate (mIoU and per class IoU)? Is it reliable (does it hold across seeds and under a paired bootstrap)? Is it efficient (parameters, what must run at inference)? Testing methods include: paired bootstrap over test images, replication over seed with exact permutation test on seed means, controls to isolate single components, and independent recomputation of the audit.

5.1.2 Initial benchmark results on the original split

Table 5.1 reproduces the initial benchmark, read from the stored result files. Detection is scored by mAP@50 and segmentation by dataset-global mIoU on the original 347-image test set.

Table 5.1: Initial-benchmark results on the original (contaminated) test split. YOLOv8m-seg test mAP@50-95 is 52.63%.

Model	**Seg. mIoU (%)**	**mDice (%)**	**Pixel acc. (%)**	**Det. mAP@50 (%)**
YOLO-UNet hybrid (with flip-TTA)	74.12	84.80	96.85	32.15
YOLOv8m-seg	72.40	82.47	97.34	72.87
DeepLabv3+ (ResNet-101)	72.37	83.57	96.62	—
U-Net (ResNet-50)	69.94	81.49	96.58	—
YOLOv11m-seg	68.89	79.74	96.70	72.81
Mask R-CNN (ResNet-50-FPN)	56.08	70.18	93.64	63.71

EfficientNet-B4 reached 93.48% accuracy (macro-F1 0.906) and ResNet-50 92.83% (macro-F1 0.903) in order to classify 224 × 224 object crops on 1,673 test crops. In this table, there were two problems that were present in the stored files, but not detected during the time of the scan. First, a hybrid's detection branch was only able to achieve 32.15% mAP@50, far below a standalone detector, despite the model being described as a hybrid detection and segmentation model. Second, the AP@50 of 0.00% for each class is listed next to an AP@[.50:.95] of 15–69%, but AP cannot improve when the IoU threshold is raised, so these per-class Mask R-CNN results should not be reused (most likely an indexing error in the per-class evaluation code). Finally, a per-image averaging of IoU was applied, resulting in 25.0% (U-Net), 24.4% (DeepLabv3+) and 25.4% (hybrid), where the pitfall mentioned in the previous Chapter 2 is observed, as the figures for the results of the datasets above are in the range of 70–74%.

5.1.3 Benchmark contamination and its effect

Contamination of the original split. Of the 347 original test images 143 were confirmed to have a near-duplicate in the original training set (41.2%) using the detector of Section 4.4.1. We re-implemented this algorithm independently, using a different image library, and obtained 162 (40.6%) of the original 143, and found that 132 (38.0%) had a base filename stem that matched a training image. More was found to be contaminated as well: 251 images (39.6%) in the original validation set were also a near duplicate in the training set, which had an impact on validation-based decisions in the initial benchmark. When the test images are mirrored, Pipeline B's detector also finds that 47.6% of the images in the test set have training images when flipping them. The results of the audit of the original and rebuilt splits are summarised in Table 5.2 and Figure 5.1 (a) and Figure 5.2 provides some randomly selected examples of the matches.

Table 5.2: Test images that have a copy in the training set, by split and by detector setting (our re-implementation unless stated).

Test images with a copy in the training set	**Original test (347)**	**Split A test (954)**	**Split B test (958)**
Near-duplicate, same orientation (dHash ≤ 5, NCC ≥ 0.95)	141 (40.6%)	1 (0.1%)	0 (0.0%)
Same detector as used to build the splits	143 (41.2%)	0 by construction	0 by construction
Mirrored or rotated copy (horizontal flip, vertical flip, 180°)	39 (11.2%)	109 (11.4%)	0 (0.0%)
Looser gate (dHash ≤ 6, NCC ≥ 0.90)	—	38 (4.0%)	9 (0.9%)
Looser gate (dHash ≤ 8, NCC ≥ 0.85)	—	76 (8.0%)	33 (3.4%)
Same base filename stem as a training image	132 (38.0%)	—	—

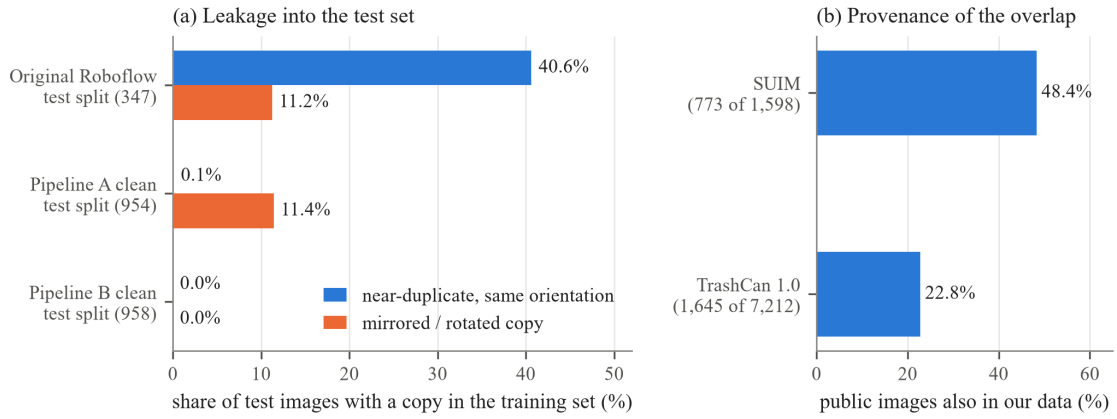


Figure 5.1: (a) Share of test images with a near-duplicate, mirrored or rotated copy in the training set, for the original split and the two rebuilt splits. (b) Share of two public datasets that is also in our data.

Where the overlap came from. Out of the 1,598 SUIM images and 7,212 TrashCan images (mostly from the public) examined 773 (48.4%) and 1,645 (22.8%) images, respectively, were found in our data(Figure 5.1(b)); with 660 of the SUIM matches having a thumbnail correlation of 1.0000, and 765 images being at least 0.999. The 70 images used for the official tests of SUIM are in our data, and 53 of them are in our training split A. Because our benchmark picked up images from those sources without doing a check for duplicate images, contamination was almost guaranteed. This is not an accusation, but a warning on the issue of using the public domain, which is redistributed, as a basis for the construction of benchmarks.

Score inflation on the initial models. We then re-scored the first U-Net and DeepLabv3+ checkpoints, without re-training, on the 204 original test images, none of which is close to being duplicated in the training set. The scoring harness was first

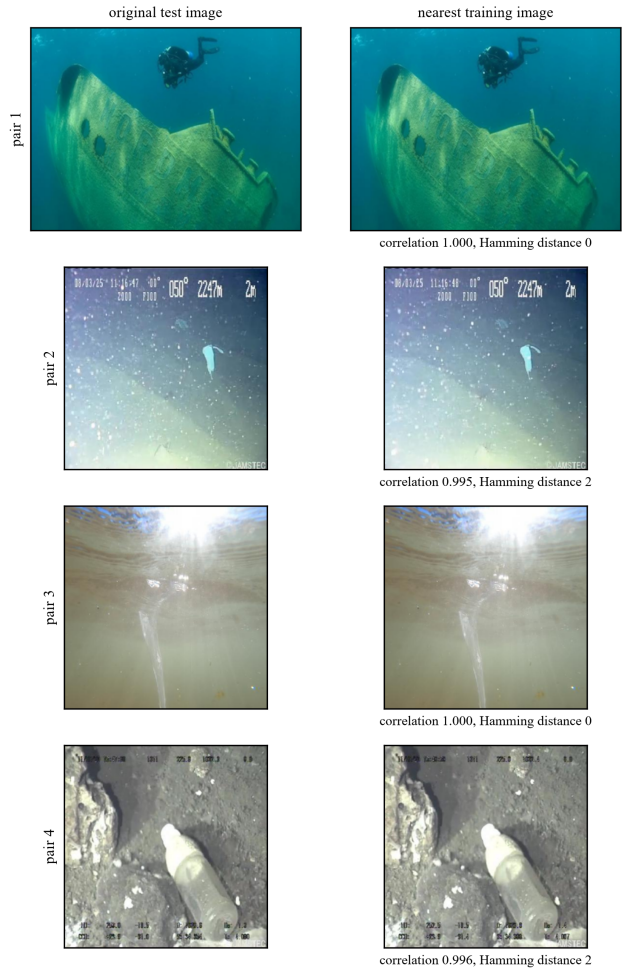


Figure 5.2: Four randomly selected near duplicate pairs of images from original test set (left) and training set (right) are shown, along with the correlation between the 32×32 thumbnails and Hamming distance of the different hashes. Pairs 1 and 3 are the same size at thumbnail resolution, pairs 2 and 4 are next frames of a video. A fixed random seed was used for the draw.

reproduced to reproduce the two published IoU values, retained up to four decimal places. As can be seen in Table 5.3 and Figure 5.3, both models suffer from low accuracy on the uncontaminated subset (0.0445 mIoU (6.8%) for U-Net and 0.0357 (5.2%) for DeepLabv3+).

The gap may be due to a change in class composition, as the uncontaminated subset is not a random sample. We were unable to distinguish between the two. The agreement on the largest per-class change (Human Activity: +0.078 and +0.079) supports the real effect. The Artifacts and Metals figure is not interpreted as only 10 of the 56 original test images with that class are contaminated, and the Aquatic Animal figure is treated with caution as only 30 test images contain that class at all (8 of which are contaminated). We do not interpret the Marine Trash change (+0.019 for both) either: 123 of the 262 test images with that class (47%) are in the contaminated set, which makes the two sets very different. The direction of the effect (reported scores too high) is determined; the magnitude of the effect is only approximate.

Table 5.3: Initial checkpoints scored on the full original test set and on its uncontaminated subset.

Model	Metric	Reported (347)	Clean subset (204)	Inflation
U-Net (ResNet-50)	mean IoU	0.6994	0.6549	+0.0445 (6.8%)
	Aquatic Animal IoU	0.4933	0.4407	+0.0526
	Human Activity IoU	0.7479	0.6695	+0.0783
	Marine Trash IoU	0.8799	0.8605	+0.0194
DeepLabv3+ (ResNet-101)	mean IoU	0.7237	0.6880	+0.0357 (5.2%)
	Aquatic Animal IoU	0.5828	0.5532	+0.0296
	Human Activity IoU	0.7187	0.6392	+0.0794
	Marine Trash IoU	0.8689	0.8496	+0.0194

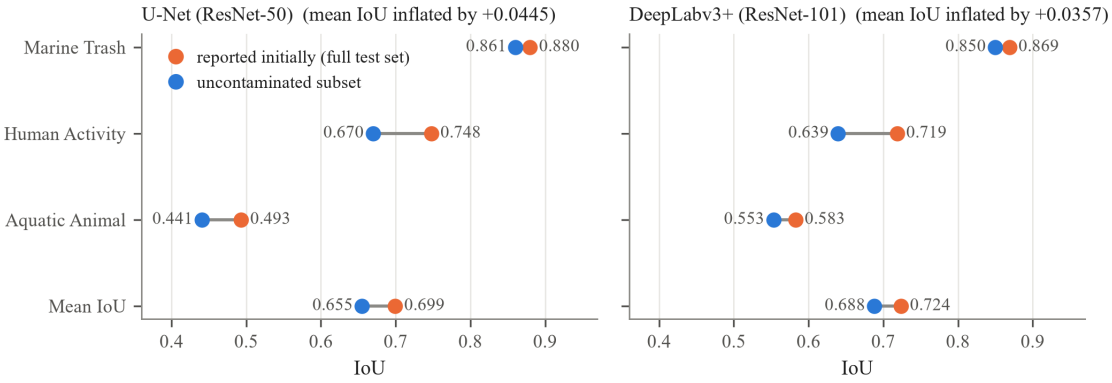


Figure 5.3: Mean IoU and per-class IoU of the initial U-Net and DeepLabv3+ on the reported test set and on its uncontaminated subset. The inflation is largest for Human Activity.

Residual leakage in the rebuilt splits. By design, both rebuilt splits pass the construction detector, and our re-implementation identifies one borderline pair in each, touching at most one test image. Mirrored and rotated copies are a different matter. The construction of Pipeline A did not connect them, and 109 of its 954 test images (11.4%) have a mirrored or rotated copy in training, the same rate as the original split (11.2%); 125 filename groups with a flip marker span its splits. Pipeline B has none. At looser similarity gates Pipeline A's test set is also less isolated (4.0% and 8.0% of test images have a look-alike in training against 0.9% and 3.4% for Pipeline B). We therefore investigated how much this might affect the Pipeline A results. Scoring every model separately on the roughly 845 test images without a mirrored counterpart (108 of the 109 affected images could be matched by filename) changes group means by at most 0.0025: the TAM-Net mean goes from 0.7468 to 0.7465, the SegFormer-B2 baseline from 0.7361 to 0.7355, the no-masking control from 0.7402 to 0.7403, Att-UNet from 0.6286 to 0.6261, CoGS on Att-UNet from 0.7000 to 0.7016, and CoGS on SegFormer-B2 from 0.7400 to 0.7387. The TAM-Net minus baseline difference is +0.0110 on this subset, compared to +0.0108 on the full test set. The pooled effect of the TAM-Net design over its two backbones

is +0.0120 on this subset (interval +0.0023 to +0.0219), against +0.0126 on the full test set. The 109 affected images score much lower, not higher, for every model (0.29 to 0.44 mIoU), so there is no sign that mirrored copies were memorised; but that subset is small and hard, so its absolute level is not interpretable. We conclude that this residual leakage does not change the Pipeline A conclusions, and we list it as a limitation.

5.1.4 Evaluation noise

Pipeline B trained SegFormer-B2 three times, changing only the random seed. Table 5.4 gives the results under the pre-registered convention (best-validation checkpoint) and under a final-epoch convention that we added afterwards and label exploratory. The three seeds span 0.0198 (best validation) or 0.0276 (final epoch) mIoU. The initially claimed advantage of the hybrid over DeepLabv3+ was 0.0175, which is smaller than either spread. This experiment measures variability and does not compare arms, so all three seeds are used: seeds 0 and 1 differ by only 0.0057, and the third seed extends the range to 0.0198, demonstrating the unreliability of a spread based on two runs.

Table 5.4: Test mIoU of three seeds of the same SegFormer-B2 configuration (Pipeline B).

Checkpoint	**Seed 0**	**Seed 1**	**Seed 2**	**Mean ± s.d.**	**Range**
Best validation (pre-registered)	0.6979	0.6922	0.6781	0.6894 ± 0.0102	0.0198
Final epoch (exploratory)	0.7055	0.7013	0.6779	0.6949 ± 0.0149	0.0276

Table 5.5 and Figure 5.4 apply a paired bootstrap to the three seed-only differences. Under the pre-registered convention no interval excludes zero, although one comes close (-0.0198, interval -0.0389 to +0.0011). Under the exploratory final-epoch convention two of the three intervals exclude zero. The second result is exploratory, and a reader should weigh it accordingly; but it shows a genuine limitation of the method. The image bootstrap assumes that the trained models are fixed and only resamples test images, thus it reflects uncertainty about which images were sampled, not uncertainty from training. It can vary greatly between two models that are exactly the same. The practical implication is the decision rule of Section 4.5: an interval that excludes zero is necessary but not sufficient, and replication over seeds is mandatory.

Table 5.5: Paired-bootstrap differences between seeds of the same configuration (Pipeline B, no TTA, 1,000 resamples).

Seed-only difference	**Best-validation: difference [95% CI]**	**Excludes 0?**	**Final epoch: difference [95% CI]**	**Excludes 0?**
seed 1 − seed 0	−0.0057 [−0.0251, +0.0149]	no	−0.0041 [−0.0208, +0.0137]	no
seed 2 − seed 0	−0.0198 [−0.0389, +0.0011]	no	−0.0276 [−0.0462, −0.0091]	yes
seed 2 − seed 1	−0.0141 [−0.0358, +0.0076]	no	−0.0235 [−0.0456, −0.0011]	yes

Test-time augmentation. Horizontal-flip TTA adds a further, systematic amount (Table 5.6). It is worth +0.0193 (interval +0.0121 to +0.0272) to the U-Net and +0.0084 to +0.0149 to the three SegFormer-B2 seeds in Pipeline B, with every

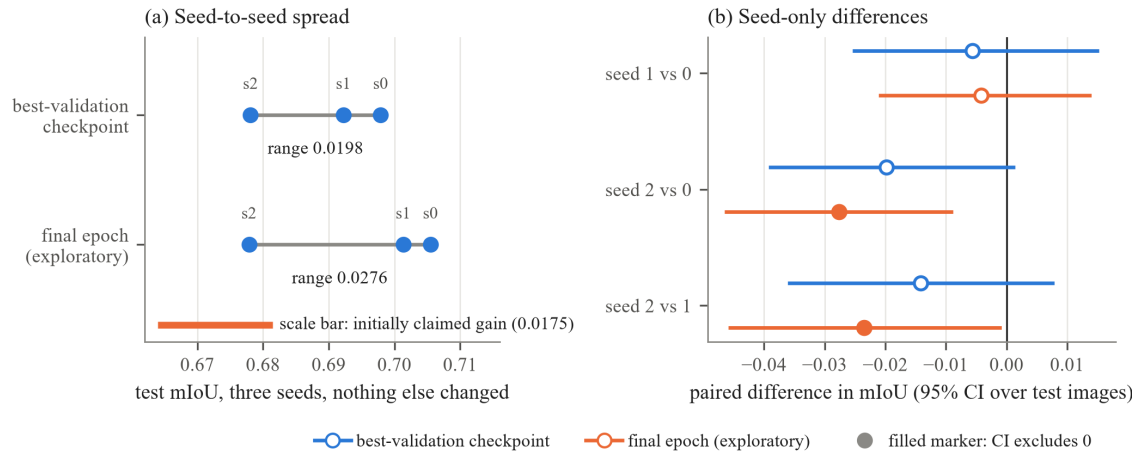


Figure 5.4: (a) The same model trained with three seeds, under the pre-registered and the exploratory checkpoint convention; the bar shows the size of the initially claimed gain on the same scale. (b) Paired bootstrap differences between seeds; filled markers are intervals that exclude zero.

interval excluding zero, and +0.003 to +0.009 to the SegFormer-family models of Pipeline A, which had already seen mirrored training images. The initial hybrid was scored with TTA on; the U-Net and DeepLabv3+ it was compared with were not. TTA alone is thus on the same order as its claimed margin. The hybrid also used a different loss (focal cross-entropy plus focal Tversky, against class-weighted cross-entropy) and a different epoch budget (up to 100 epochs, against 120 and 50 for its rivals).

Table 5.6: Effect of horizontal-flip TTA on test mIoU. Ensembles average class probabilities over the members.

Pipeline	**Model**	**Without TTA**	**With flip-TTA**	**Gain**
A	SegFormer-B2 baseline, seed 42	0.7395	0.7449	+0.0055
A	SegFormer-B2 baseline, seed 1	0.7327	0.7419	+0.0093
A	TAM-Net control, seed 42	0.7330	0.7389	+0.0059
A	TAM-Net, seed 42	0.7471	0.7518	+0.0047
A	TAM-Net, seed 1	0.7466	0.7498	+0.0032
A	2-model SegFormer-B2 ensemble	0.7432	0.7513	+0.0080
A	2-model TAM-Net ensemble (seeds 42, 1)	0.7531	0.7544	+0.0013
B	U-Net R50	0.6179	0.6373	+0.0193
B	SegFormer-B2, seed 0	0.6979	0.7072	+0.0093
B	SegFormer-B2, seed 1	0.6922	0.7006	+0.0084
B	SegFormer-B2, seed 2	0.6781	0.6930	+0.0149
B	3-model SegFormer-B2 ensemble	0.7134	0.7227	+0.0093

5.1.5 Matched-protocol results (Pipeline A)

Table 5.7 and Figure 5.5 give the main comparison on the 954-image test set of split A. Each row is a group of runs that share every setting except the seed; the number of runs is shown. Every multi-seed arm uses seeds 42 and 1; the third seeds that exist for five of the arms are analysed separately in Section 5.4.1. The full run list is Table A.1 in Appendix A.

Table 5.7: Test mIoU and per-class IoU on split A (954 images), mean over the runs shown in the Runs column (seeds 42 and 1) ± the sample standard deviation of the two. † trainable parameters only; the frozen CLIP or ConvNeXt tower is additional. The parameter count of the initial hybrid was not recorded.

Model	**Params (M)**	**Runs**	**mIoU**	**Aquatic**	**Artifacts**	**Human**	**Trash**
TAM-Net (SegFormer-B2 + detail path + teacher masking)	24.8	2	0.7468 ± 0.0003	0.610	0.769	0.775	0.833
TAM-Net control: same, masking disabled	24.8	2	0.7402 ± 0.0101	0.594	0.767	0.763	0.837
CoGS on SegFormer-B2	25.6†	2	0.7400 ± 0.0024	0.608	0.759	0.767	0.826
SegFormer-B2 baseline (lr 6×10^{-5})	24.7	2	0.7361 ± 0.0048	0.590	0.750	0.771	0.833
CoGS on DeepLabv3+	46.8†	2	0.7037 ± 0.0134	0.585	0.728	0.705	0.796
CoGS on Att-UNet (sub-concept prompts)	35.3†	2	0.7000 ± 0.0171	0.553	0.712	0.731	0.804
CoGS on Att-UNet (visual features only)	35.3†	2	0.6939 ± 0.0124	0.577	0.694	0.714	0.791
ImageNet-feature control on Att-UNet	35.3†	2	0.6771 ± 0.0073	0.595	0.660	0.671	0.783
TAM-Net on DeepLabv3+ (detail path + teacher masking)	45.8	2	0.6762 ± 0.0097	0.546	0.678	0.681	0.800
TAM-Net control on DeepLabv3+: same, masking disabled	45.8	1	0.6674	0.520	0.705	0.651	0.794
DeepLabv3+ (ResNet-101)	45.7	2	0.6619 ± 0.0112	0.521	0.675	0.658	0.793
Initial hybrid, retrained (10 epochs)	—	1	0.6435	0.525	0.646	0.626	0.777
U-Net (ResNet-50)	32.5	1	0.6416	0.527	0.621	0.643	0.775
PADS-Net (physics-based restoration)	34.6	1	0.6350	0.522	0.619	0.627	0.773
Att-UNet (ResNet-50), the initial baseline	34.4	2	0.6286 ± 0.0021	0.521	0.617	0.611	0.766

Four patterns stand out. **First**, the backbone matters most. SegFormer-B2 exceeds DeepLabv3+, U-Net, and Att-UNet by +0.082, +0.094, and +0.108 mIoU respectively (paired 95% intervals +0.059 to +0.106, +0.071 to +0.119, and +0.084 to +0.134), with the interval excluding zero on every class. The same holds independently in Pipeline B, where SegFormer-B2 exceeds U-Net by +0.080 (+0.055 to +0.103). **Second**, the initial hybrid, retrained under the common protocol, scores 0.6435 and sits with the CNN baselines (U-Net 0.6416, PADS-Net 0.6350, Att-UNet 0.6286 ± 0.0021), far below SegFormer-B2 (0.7361). **Third**, the top of the table is crowded: TAM-Net (0.7468), its no-masking control (0.7402), CoGS on SegFormer-B2 (0.7400), and the baseline (0.7361) lie within about one point, which is comparable to the seed-to-seed spread of the individual arms (standard deviations of two seeds, 0.0003 to 0.010, and a range of 0.020 to 0.028 across the three seeds of Section 5.1.4). TAM-Net is nevertheless the highest-scoring arm, and the same design attached to DeepLabv3+ (0.6762 over two seeds) also exceeds the plain

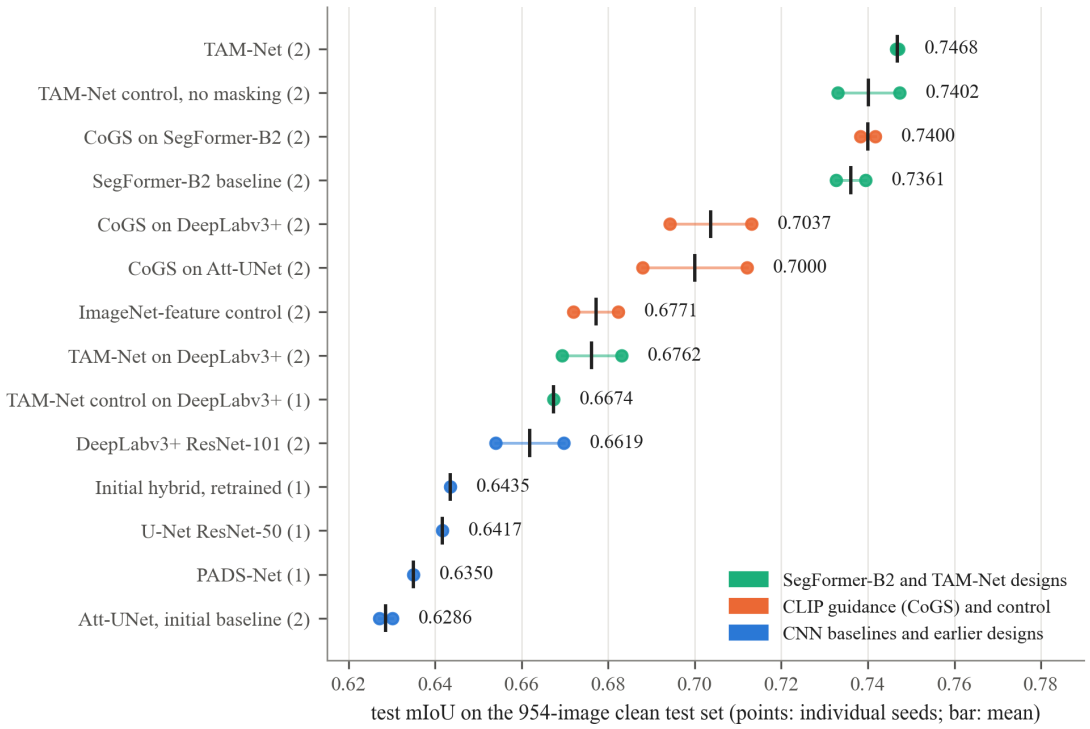


Figure 5.5: Test mIoU on split A. Points are individual seeds and the bar marks the mean. Green: the SegFormer-B2 family; orange: CLIP guidance and its control; blue: CNN baselines and earlier designs.

DeepLabv3+ (0.6619) and its own control (0.6674, one seed); Section 5.2.4 analyses the two backbones together. **Fourth**, Aquatic Animal is the weakest class in all 35 Pipeline A runs (and in all 23 Pipeline B runs), and Marine Trash is the strongest in every Pipeline A run. In Pipeline B, Artifacts and Metals is the strongest, which reflects the different test compositions.

5.1.6 Results of Pipeline B

Pipeline B's models are compared with each other and never with Pipeline A's, because the test sets differ. Table 5.8 gives the SegFormer-B2 baselines, their ensemble, and the U-Net. The three-seed ensemble, which costs three forward passes, scores 0.7134 without TTA and 0.7227 with it, and exceeds its own seed-0 member by +0.0155 (interval +0.0033 to +0.0279). Pipeline B's SegFormer-B2 used the shared learning rate of 3×10^{-4}. Pipeline A found 6×10^{-5} to be 0.051 mIoU better on the same architecture, so Pipeline B's SegFormer numbers are probably below what a tuned run would reach.

Table 5.8: Pipeline B baselines on split B (958 test images).

Model (Split B, 958 test images)	mIoU	95% CI	+ flip-TTA	Aquatic	Artifacts	Human / Trash
SegFormer-B2, seed 0	0.6979	[0.668, 0.726]	0.7072	0.609	0.785	0.727 / 0.671
SegFormer-B2, seed 1	0.6922	[0.660, 0.723]	0.7006	0.616	0.802	0.713 / 0.637
SegFormer-B2, seed 2	0.6781	[0.646, 0.709]	0.6930	0.592	0.785	0.694 / 0.641
3-seed SegFormer-B2 ensemble	0.7134	[0.683, 0.742]	0.7227	0.615	0.822	0.738 / 0.679
U-Net (ResNet-50), seed 0	0.6179	[0.590, 0.645]	0.6373	0.528	0.725	0.649 / 0.570

5.1.7 Findings that both pipelines reproduce

The two pipelines were built independently (Table 4.1), so a finding that appears in both is replicated in the strongest sense available in this thesis. Table 5.9 lists the findings that can be compared without comparing absolute scores across the pipelines, which we do not do: the size and direction of an effect, the ordering of the classes, the scale of the seed noise, and the state of the test sets.

The findings about the model family replicate: SegFormer-B2 exceeds the ResNet-50 U-Net by +0.094 in Pipeline A and by +0.080 in Pipeline B, with every per-class interval above zero in both. The class ordering at the bottom replicates as well (Aquatic Animal is the weakest class in every run of both pipelines), the seed-to-seed standard deviation is of the same scale in both (about 0.01), and both rebuilt test sets are nearly free of same-orientation near-duplicates of training images (1 of 954 and 0 of 958), against 41% to 48% of the original test images. The pipelines differ where they were designed to differ: only Pipeline B links mirrored copies, so only Pipeline A's test set still contains them (11.4%, which is listed among the limitations), and the strongest class differs because the test compositions differ.

5.1.8 External validation

To test generalisation we scored models on images from the public SUIM [33] and TrashCan [32] datasets that have no counterpart in our data (825 and 794 images), so no evaluation image was seen in training. Table 5.10 and Figure 5.6 show the result. On SUIM only Aquatic Animal and Artifacts and Metals could be scored (reefs and invertebrates and letter-box padding are ignored, Marine Trash is absent, and Human Activity has too few pixels), so the SUIM mIoU is the mean of two classes. On TrashCan, Artifacts and Metals is not scored, and the mean of three classes is dominated by Human Activity, which covers the robots and divers (IoU 0.70 to 0.76), while Aquatic Animal and Marine Trash reach only 0.13 to 0.22.

The CoGS variants, which use the Att-UNet backbone, lead DeepLabv3+ by +0.080 to +0.085 on SUIM and by +0.091 to +0.117 on TrashCan, consistent with their in-domain advantage over the same backbones. The SegFormer-B2 baseline (one run, seed 42) and TAM-Net (seeds 42 and 1) were scored later on the same images with the same protocol, and they change the picture for the strongest models. The SegFormer-B2 baseline reaches 0.737 on SUIM and 0.370 on TrashCan: level with the CoGS variants on SUIM (0.740 to 0.745) and above them on TrashCan (0.335 to 0.360), and ahead of DeepLabv3+ by +0.078 and +0.127. The external lead of CoGS

Table 5.9: Findings compared across the two pipelines. Absolute scores are not compared, because the test sets, losses, and schedules differ. Pipeline A: the two-seed SegFormer-B2 baseline against one U-Net run; Pipeline B: one run of each.

Finding	**Pipeline A (954 test images)**	**Pipeline B (958 test images)**	**Agree?**
SegFormer-B2 minus U-Net (ResNet-50), test mIoU [95% CI]	+0.0944 [+0.0707, +0.1193]	+0.0800 [+0.0551, +0.1027]	Yes: above zero in both
The same difference per class	All four intervals above zero	All four intervals above zero	Yes
Weakest class	Aquatic Animal in all 35 runs	Aquatic Animal in all 23 runs	Yes
Strongest class	Marine Trash and Plastic in all 35 runs	Artifacts and Metals in all 23 runs	No: the test compositions differ
Standard deviation of test mIoU across seeds of one configuration	0.0003 to 0.0171 (two seeds); 0.0062 to 0.0132 (three seeds)	0.0102 (three seeds; range 0.0198)	Yes: the same scale, about 0.01
Test images with a same-orientation near-duplicate in training (independent check)	1 (0.1%)	0 (0.0%)	Yes: practically none
Test images with a mirrored or rotated copy in training	109 (11.4%)	0 (0.0%)	No: only Pipeline B links mirrored copies
Original test images (347) with a training counterpart	Pipeline A's detector: 143 (41.2%); independent re-implementation: 141 (40.6%)	Pipeline B's detector, mirrored copies included: 165 (47.6%)	Yes: at least 41% in each

is therefore a lead over the older CNN baselines and not over a strong backbone, which agrees with the in-domain result that CoGS adds nothing on SegFormer-B2.

TAM-Net scores 0.795 and 0.774 on SUIM, +0.057 and +0.037 above the SegFormer-B2 baseline and above every CoGS variant. The no-masking control scores 0.768 and 0.792, so most of this gain is present without masking (Table 5.11). The whole design against the baseline is +0.0471 (interval +0.0207 to +0.0765, excluding zero), of which the detail path and the augmentation account for +0.0423 (+0.0182 to +0.0695) and the masking for +0.0048 (-0.0128 to +0.0235). On DeepLabv3+ the design gains +0.0099 (-0.0098 to +0.0298) and the masking -0.0091 (-0.0241 to +0.0052); pooled over both backbones the whole design gains +0.0285 (+0.0127 to +0.0463), the detail path and the augmentation +0.0307 (+0.0140 to +0.0487), and the masking -0.0021 (-0.0141 to +0.0102). The design therefore transfers to SUIM through its detail path and augmentation, and the masking adds nothing measurable

Table 5.10: External validation on images that do not overlap our data. Per-class IoU is given for the classes that were scored.

	SUIM (825 images)			TrashCan 1.0 (794 images)		
Model	**mIoU**	Aquatic	Artifacts	**mIoU**	Aquatic	Human / Trash
TAM-Net, seed 42	0.795	0.804	0.785	0.362	0.126	0.755 / 0.206
TAM-Net, seed 1	0.774	0.805	0.744	0.368	0.125	0.759 / 0.221
TAM-Net control (masking off), seed 42	0.768	0.797	0.739	0.344	0.129	0.730 / 0.174
TAM-Net control (masking off), seed 1	0.792	0.792	0.791	0.379	0.177	0.733 / 0.228
SegFormer-B2 baseline, seed 42	0.737	0.790	0.685	0.370	0.158	0.754 / 0.199
TAM-Net on DeepLabv3+, seed 42	0.683	0.716	0.650	0.254	0.099	0.596 / 0.066
TAM-Net on DeepLabv3+, seed 1	0.656	0.699	0.614	0.263	0.111	0.603 / 0.076
TAM-Net control on DeepLabv3+, seed 42	0.679	0.707	0.650	0.270	0.112	0.625 / 0.072
DeepLabv3+, seed 42	0.660	0.691	0.629	0.243	0.068	0.602 / 0.060
CoGS, sub-concepts, seed 42	0.745	0.796	0.694	0.341	0.145	0.703 / 0.176
CoGS, sub-concepts, seed 1	0.740	0.785	0.695	0.360	0.196	0.713 / 0.171
CoGS, visual features only	0.740	0.790	0.690	0.348	0.175	0.710 / 0.158
CoGS, class names	—	—	—	0.335	0.150	0.699 / 0.154
PADS-Net	0.624	0.674	0.574	0.225	0.040	0.564 / 0.070
U-Net	0.595	0.643	0.547	0.214	0.051	0.528 / 0.064
Att-UNet	0.594	0.638	0.550	0.232	0.061	0.566 / 0.069

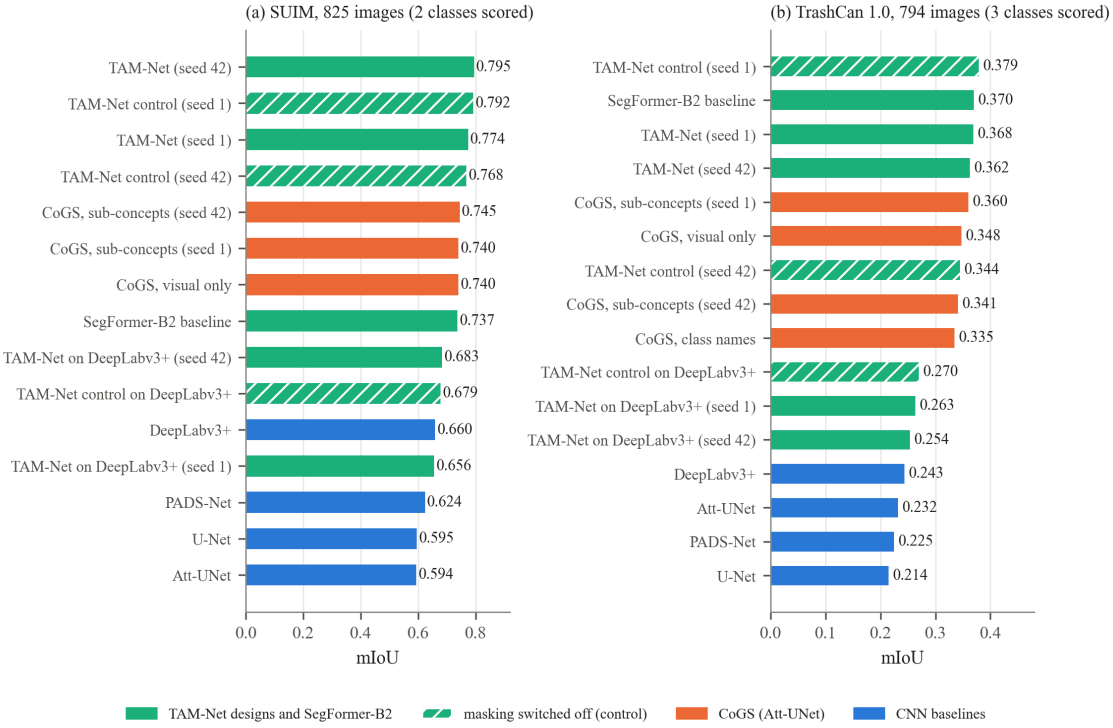


Figure 5.6: External validation. SegFormer-B2 and TAM-Net (green), CoGS variants (orange), and CNN baselines (blue).

there.

Table 5.11: Effects of the TAM-Net design on the 825 external SUIM images (two classes scored; 2,000 paired bootstrap resamples). The plain baselines were scored with seed 42 only, and the control on DeepLabv3+ has one seed.

Backbone	**Effect on SUIM**	**Difference**	**95% CI (images)**	**Ahead in**
SegFormer-B2	TAM-Net − plain model (whole design)	+0.0471	[+0.0207, +0.0765]	100.0%
	TAM-Net − control (masking alone)	+0.0048	[−0.0128, +0.0235]	69.2%
	control − plain model (detail path, flip/zoom)	+0.0423	[+0.0182, +0.0695]	100.0%
DeepLabv3+	TAM-Net − plain model (whole design)	+0.0099	[−0.0098, +0.0298]	83.2%
	TAM-Net − control (masking alone)	−0.0091	[−0.0241, +0.0052]	10.5%
	control − plain model (detail path, flip/zoom)	+0.0190	[−0.0048, +0.0413]	94.8%
Pooled over both backbones	TAM-Net − plain model (whole design)	+0.0285	[+0.0127, +0.0463]	100.0%
	TAM-Net − control (masking alone)	−0.0021	[−0.0141, +0.0102]	35.2%
	control − plain model (detail path, flip/zoom)	+0.0307	[+0.0140, +0.0487]	100.0%

On TrashCan the per-image counts were not kept, so there are no intervals. TAM-Net scores 0.362 and 0.368, its control 0.344 and 0.379, and the SegFormer-B2 baseline 0.370; on DeepLabv3+ the values are 0.254 and 0.263 for TAM-Net, 0.270 for the control, and 0.243 for the plain model. Nothing separates the SegFormer-B2 arms on TrashCan, where the Aquatic Animal IoU of TAM-Net (0.126 and 0.125) is below that of the baseline (0.158) and Marine Trash is slightly above (0.206 and 0.221 against 0.199). The plain baselines were scored with a single run, and the in-domain seed-to-seed standard deviation of the SegFormer-B2 baseline is 0.005, so differences of that size cannot be told from noise.

Three caveats bound what this evaluation shows. (i) Because our benchmark absorbed images from both sources, the models trained on imagery from them; the test is generalisation to unseen images of a *partly seen* source, not transfer to a new domain. (ii) CoGS was scored externally on the Att-UNet backbone only; CoGS on SegFormer-B2 and on DeepLabv3+ and the ImageNet-feature control were not run on either dataset, so we cannot say how CoGS on a strong backbone generalises. (iii) Absolute scores fall from about 0.70 to 0.75 in-domain to 0.34 to 0.37 on TrashCan, which shows a substantial domain shift for deep-sea footage.

5.1.9 Object size and detection

Composition. Table 4.7 shows that Aquatic Animals are small: the median is 609 pixels, and 60.5% of animals are smaller than 32×32 pixels, yet these hold only

3.4% of the class's pixels. The other classes are far larger (medians 6,495 to 16,615 pixels).

Detection by size. We bucketed Aquatic Animals by their share of the image content and counted an animal as detected when at least half of its pixels received the right class (Figure 5.7). On validation, detection is U-shaped: 42.3% of tiny animals (under 0.5% of the image, n=636) and 43.9% of huge ones (over 40%, n=19) are found, against 75–83% for small to large animals. At both ends the animal is mostly called *background* (58% and 55%) and rarely another class (about 4–5% and 0.3%). The tiny-animal failure is stable across seeds (42.3 ± 0.9%, 48.5 ± 0.9%, 67.8 ± 6.9% on test, n=656). The huge-animal dip is not: on test 67.8 6.9% of huge animals are found (n=30), and with 19 to 30 animals per bucket that end of the curve is fragile.

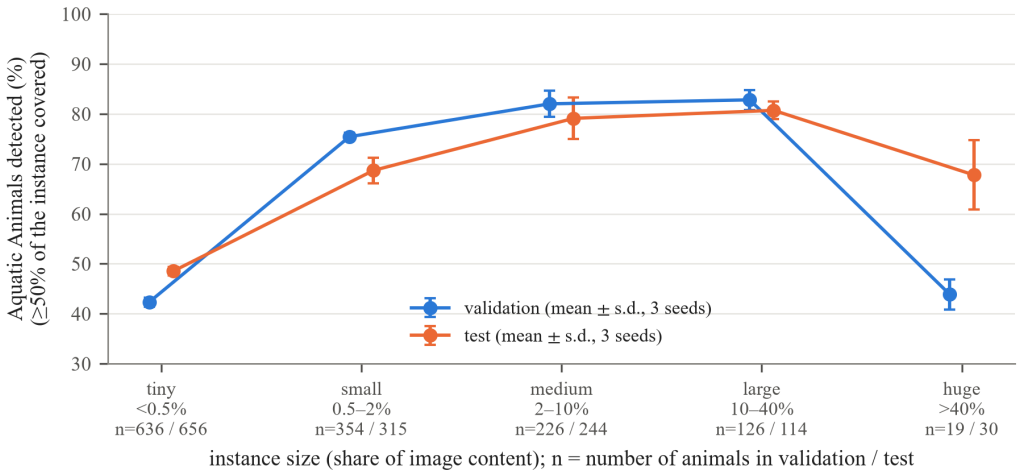


Figure 5.7: Share of Aquatic Animals detected by size bucket for the three SegFormer-B2 seeds (mean and standard deviation), on validation and on test. The tiny-animal shortfall is robust; the huge-animal dip is not.

5.1.10 Label audit

What happens in the audit. Some real objects were never labelled, so the training data tell the model to treat them as background. The labels alone cannot say how many, so the audit looks for regions where the label says background but another source says object, asks a person to judge those regions blindly, and corrects the person's answers for random guessing. It has five steps, and the paragraphs below give the numbers of each in this order.

1. *Candidate regions.* A region is a candidate when the label says background and either a model that never saw the image is confident that it contains an object (*model disagreement*: 1,082 candidates in the training set), or the same object is labelled in a near-identical video frame but not here (*video consistency*: the filtered check gave 40 candidates for review).

2. *Blind review.* The reviewers see each crop with the region outlined, in shuffled order, and do not know whether it is a candidate or a decoy. Round 1 used

385 crops from the training set (285 model candidates, 40 video candidates, 60 decoys); round 2 used 429 crops from 150 test images (379 candidates, 50 decoys).

3. *Decoy correction.* Decoys are random background regions with the same shapes as the candidates. The share of decoys that the reviewers still call an object (14.0% in round 1, 2.0% in round 2) measures guessing and is subtracted from every estimate.

4. *Estimate.* Estimated unlabelled objects of a class = mined candidates × decoy-adjusted precision. For the training set this gives about 770 unlabelled objects (a lower bound).

5. *Effect on scores.* The models are scored on the 150 re-reviewed test images with the original labels and with the confirmed objects added or ignored.

Figure 5.8 shows the same steps as a flowchart, with the estimate of Equation (4.7).

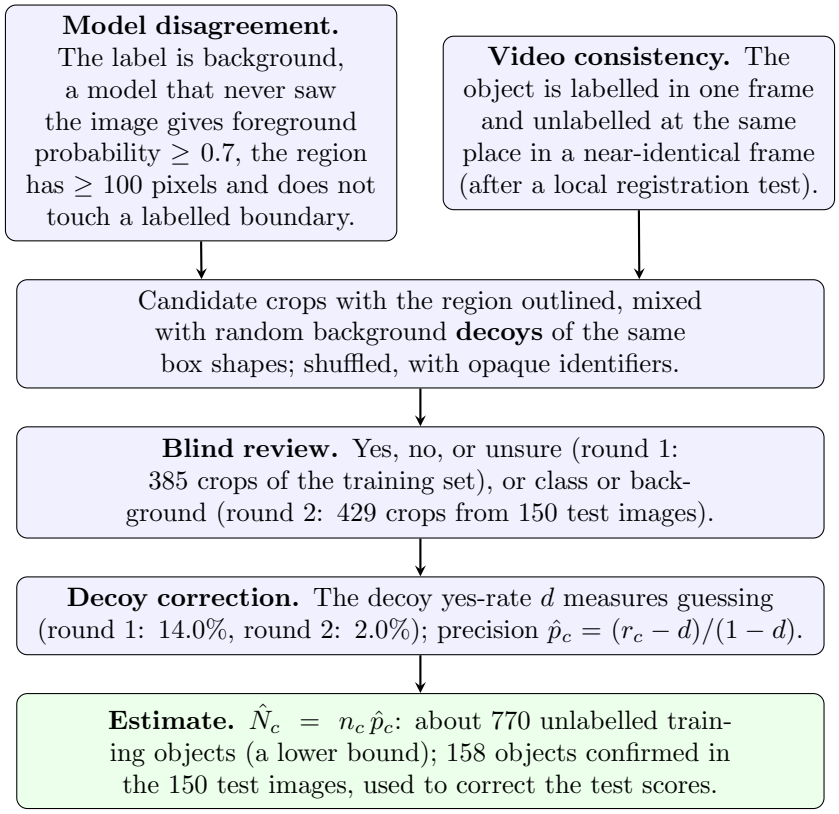


Figure 5.8: Workflow of the label audit. Candidates come from two sources, are mixed with decoys, are judged blind, and the yes-rate on decoys is used to correct the estimate.

Video consistency (a check that first failed). A first version of the video-consistency check, which used whole-frame similarity only, flagged 4,055 regions, 3,546 of them in Marine Trash, more than the 3,395 large Marine Trash instances in the whole training set, which is not credible. Whole-frame similarity does not mean that objects are in the same place; a slow pan keeps frames correlated while everything moves. Adding a local registration test (the instance's own patch must

still match at the same coordinates and carry real texture) rejected 2,712 flags because the scene had moved and 1,236 because the patch was flat, leaving 107 flags (55 after removing repeats of the same miss). About 97% of the apparent evidence had been an artefact of our own measuring instrument. What survives is informative for two classes only: Human Activity, 26 of 276 instances examined (9.4%), and Marine Trash, 10 of 1,203 (0.8%). Aquatic Animal (25 instances) and Artifacts and Metals (none) cannot be judged from video frames.

Model disagreement and blind review. Model disagreement produced 1,082 candidates in the training set. Table 5.12 and Figure 5.9 give the review. In round 1, 60.4% of model-disagreement candidates and 60.0% of video candidates were judged real objects, against 13.3% of random background decoys; counting only decisive answers, these are 75.1%, 77.4%, and 14.0%. Candidates are therefore about five times more likely to be real objects than random regions, and the decoy rate is subtracted from every estimate below. In round 2, on 150 test images and with candidates pooled from the three SegFormer-B2 seeds, 158 of 379 candidates (41.7%) were confirmed as an unlabelled object of a stated class, against 1 of 50 decoys (2.0%), roughly one unlabelled object per test image. The low decoy rate makes round 2 much more reliable than round 1.

Table 5.12: Blind-review answers. "Object share" counts unsure answers as not an object.

Crops shown	**n**	**Object**	**Not an object**	**Unsure**	**Object share**
Round 1: model-disagreement candidates	285	172	57	56	60.4%
Round 1: video-consistency candidates	40	24	7	9	60.0%
Round 1: decoys (random background)	60	8	49	3	13.3%
Round 2: model candidates (150 test images)	379	158	97	124	41.7%
Round 2: decoys (random background)	50	1	37	12	2.0%

Estimated missing objects. Table 5.13 converts the round-1 review into class-level estimates: about 770 unlabelled objects in the training set (467 Aquatic Animal, 120 Human Activity, 151 Marine Trash, 35 Artifacts and Metals), a rate of 3.8% to 7.8% of labelled instances per class, above the 3% threshold that we set in advance as the level worth acting on. This is a *lower bound*: only objects that some model detected could be reviewed. Three caveats apply. The Artifacts estimate rests on 30 reviewed crops, and the reviewers answered yes to 14% of random decoys in round 1. Two team members produced the answers, with one final answer recorded per crop, so agreement between them was not measured.

Table 5.13: Estimated unlabelled objects in the training set of split B. Precision is among decisive answers and corrected for the decoy yes-rate (14.0%).

Class	**Labelled instances**	**Mined candidates**	**Reviewed**	**Decoy-adjusted precision**	**Estimated unlabelled (rate)**
Aquatic Animal	6,349	601	136	77.8%	467 (6.9%)
Artifacts and Metals	892	133	30	26.3%	35 (3.8%)
Human Activity	1,425	150	29	79.9%	120 (7.8%)
Marine Trash and Plastic	3,547	198	34	76.1%	151 (4.1%)
All classes	12,213	1,082	—	—	**773**

Who is missed. Of the 110 Aquatic Animals confirmed as unlabelled in round 1, 85 (77%) are tiny (under 0.5% of the image content), 17 are small, and none fills the frame. The size that the models miss most is therefore also the size that annotators

skip most. A model trained on tiny animals that are labelled in some images and left as background in others learns to call them background.

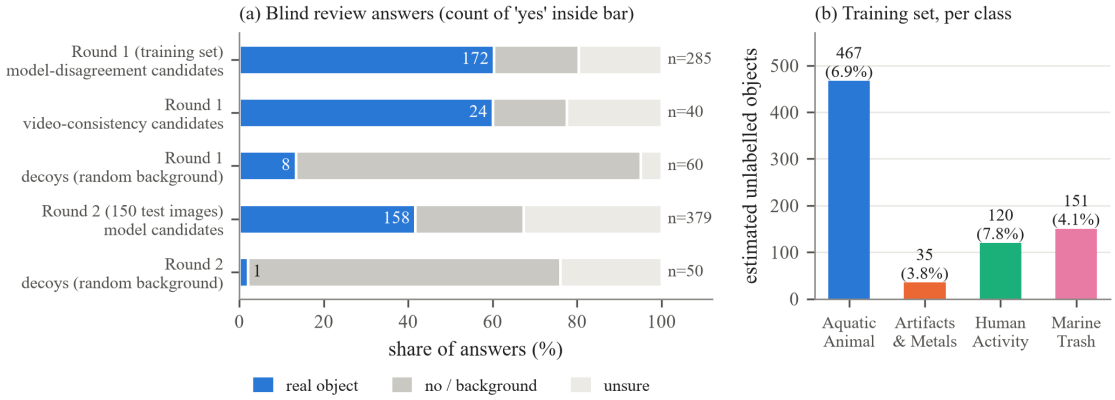


Figure 5.9: (a) Blind-review answers by group; the number inside each blue bar is the count of "real object" answers. (b) Estimated unlabelled objects per class in the training set.

Effect on scores. On the 150 re-reviewed test images we scored each model three ways (Table 5.14, Figure 5.10): with the original labels; *strict*, with the confirmed regions added as foreground; and *ignore*, with the confirmed regions excluded from scoring. Under *ignore* every model's score rises by +0.031 to +0.068, but this direction is guaranteed by construction, because removing pixels that were labelled background can only remove false positives; only its size is informative. The size is inflated for the SegFormer seeds, because the review candidates were pooled from them; the fairer estimate comes from the two models that played no part in choosing the regions (MAAS and its random control, +0.034 and +0.035). Under *strict*, every score falls by 0.035 to 0.057, because the added objects are ones that models do not currently find and the added region is the review box, not a tight polygon. The true value lies somewhere between the two readings. With 150 images the intervals are about ±0.08 wide, so the change in model ranking between conventions is within noise. The robust result is the count itself: 158 unlabelled objects in 150 test images, confirmed blind with a 2% decoy rate.

Table 5.14: Test mIoU on the 150 re-reviewed test images under three labelling conventions (Pipeline B).

Model (150 test images)	**Original labels**	**Strict**	**Ignore**	**Change (ignore)**
SegFormer-B2, seed 0	0.6477	0.6126	0.7156	+0.068
SegFormer-B2, seed 1	0.6324	0.5771	0.6635	+0.031
SegFormer-B2, seed 2	0.5991	0.5588	0.6462	+0.047
3-seed ensemble	0.6480	0.5916	0.6900	+0.042
MAAS (real labels)	0.6463	0.5911	0.6803	+0.034
Random control (real labels)	0.6439	0.5874	0.6788	+0.035

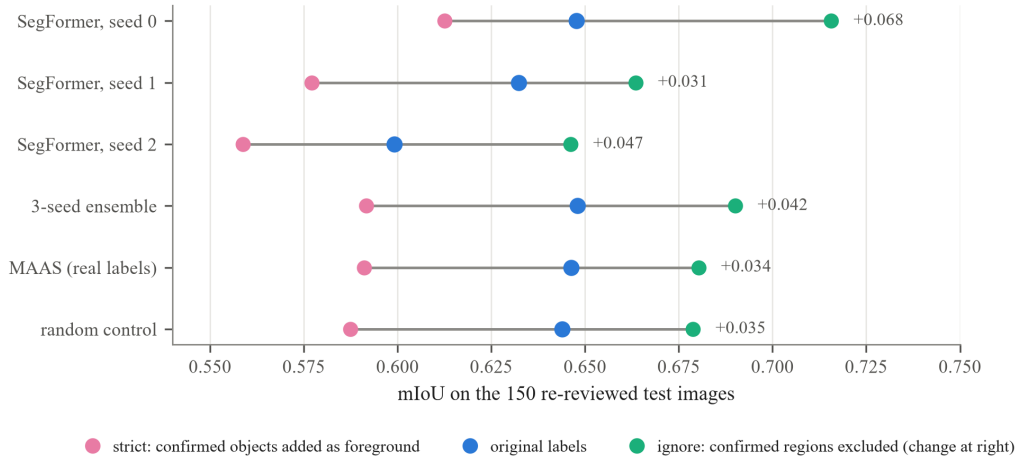


Figure 5.10: mIoU on the 150 re-reviewed test images with the original labels, with confirmed objects added (strict), and with confirmed regions ignored.

5.2 Analysis of Design Solutions

5.2.1 PADS-Net: physics-based restoration

PADS-Net scored 0.6350 on raw images and 0.6401 on CLAHE-enhanced images, against 0.6286 for the plain Att-UNet (mean of seeds 42 and 1), single-run differences of +0.006 and +0.012. These are inside the range of the Att-UNet seeds once the third seed is included (0.627 to 0.639, Table 5.28), which is why we do not read them as an effect. Enhancement itself does not help a plain model: Att-UNet on CLAHE-enhanced input scored 0.6263 against 0.6301 on raw input (one seed each, -0.0038). PADS-Net did not help on the class it was built for either (Aquatic Animal IoU 0.522 against 0.522 for Att-UNet).

The stored training log shows why the restoration is a null operation. At the last epoch the learned attenuation coefficients are $\beta^D \approx (8.16, 6.62, 5.37)$ and the backscatter coefficients $\beta^B \approx (4.19, 0.49, 0.41)$, with veiling light $B^\infty \approx (0.16, 0.91, 1.00)$, large values that look physical. But the coefficients only ever appear multiplied by the range z, so their scale is unidentifiable. We repeated the check on the stored weights of the best-validation epoch (epoch 46) by running the restoration module on the 954 test images (`verify_pads.py`): the range map has collapsed to a mean of $z = 0.0064$ with a within-image standard deviation of 0.0078 (the largest per-image mean is 0.040). With the coefficients of that checkpoint ($\beta^D \approx (7.6, 6.1, 5.0)$, $\beta^B \approx (3.9, 0.45, 0.36)$, $B^\infty \approx (0.20, 0.88, 1.00)$) the transmissions $e^{-\beta^D z}$ are 0.955, 0.963, and 0.970 and the backscatter terms 0.0045, 0.0022, and 0.0023, so the module reduces to multiplying by a near-constant between 1.03 and 1.06, subtracting a fixed offset, and passing the result through the fixed sigmoid soft clamp that the module always applies, which is the same curve for every pixel of every image and carries no information about the scene. The plausible-looking coefficients are an artefact of the scale ambiguity. The module learned to disable itself, because on this data colour degradation was not what limited the segmenter.

The small-object motivation is limited by the metric. Animals under 32×32 pixels hold 3.4% of the class's pixels (Table 4.7), so even perfect segmentation of all of them could raise Aquatic Animal IoU by at most a few points and mIoU by less than one. A validation-set oracle run during the study gave +0.011 for the class and +0.003 for mIoU; its prediction files were not among those transferred, so we could not repeat it. Mean IoU is nearly blind to the objects that the size analysis shows to be missed most, and instance-level detection should be reported alongside it.

5.2.2 CoGS: guidance from frozen CLIP features

The gain depends on the backbone. Table 5.15 shows the paired gain of CoGS over the same backbone without CLIP. On Att-UNet it is +0.0714 (interval +0.0531 to +0.0931), and every CoGS seed (0.688, 0.712) lies above every baseline seed (0.630, 0.627), which is the strongest separation two seeds per arm can give (exact permutation p=0.33, the smallest attainable). With the third seed of each arm (0.691 against 0.639) the gain is +0.0649 and the separation is still complete (p=0.10, Table 5.29). On DeepLabv3+ the gain is +0.0592 (+0.0367 to +0.0789) for seed 42 and +0.0244 for seed 1. On SegFormer-B2 it is +0.0039 (interval -0.0042 to +0.0123), no effect. The gain shrinks as the baseline gets stronger (0.629, 0.662, 0.736), which suggests that CLIP supplies knowledge that a weak backbone lacks and that SegFormer-B2 already has enough of.

Table 5.15: Paired gain of CoGS over the same backbone without CLIP (split A).

Backbone	**Without CLIP**	**With CoGS**	**Gain [95% CI over images]**
Att-UNet (2 seeds each)	0.6286	0.7000	+0.0714 [+0.0531, +0.0931]
DeepLabv3+, seed 42	0.6540	0.7131	+0.0592 [+0.0367, +0.0789]
DeepLabv3+, seed 1	0.6698	0.6942	+0.0244 (point estimate only)
SegFormer-B2 (2 seeds each)	0.7361	0.7400	+0.0039 [−0.0042, +0.0123]

The text adds little; the visual features do the work. Table 5.16 compares the variants on the Att-UNet backbone. Removing the text entirely (visual-only) gives 0.6939 against 0.7000 with 57 sub-concept prompts, a difference of +0.0062 in favour of the text whose interval includes zero (-0.0014 to +0.0142, ahead in 94% of resamples). The class-level picture is a trade-off: the text lowers Aquatic Animal IoU (-0.024, interval -0.040 to -0.008) and raises the other three classes (+0.013 to +0.019, each interval above zero; four classes tested and no correction). With the third seed of both arms the overall difference is -0.0001 (interval -0.0065 to +0.0062), so even its sign depends on which seeds are included. On the two external datasets the visual-only model lies within the seed spread of the sub-concept model (Table 5.10). The single class-names run scored 0.7071, inside the seed range of the other variants (0.685 to 0.712). The vocabulary was carefully built and frozen before training, and the conclusion is that it adds no reliable overall benefit once the network is trained. Before training it does help: with no training at all, scoring each pixel against the prompts gives 0.145 mIoU on the validation split with sub-concepts

and 0.140 with class names, and 67.2% of object pixels receive the correct class with sub-concepts against 35.0% with class names. Zero-shot CLIP alone is nevertheless not a usable segmenter.

Table 5.16: CoGS variants on the Att-UNet backbone (split A).

Variant (Att-UNet backbone)	**Seeds**	**Mean mIoU**	**Aquatic IoU**	**Difference to Att-UNet [95% CI]**
Att-UNet, no CLIP (initial baseline)	2	0.6286	0.521	—
CoGS, 57 sub-concept prompts	2	0.7000	0.553	+0.0714 [+0.0531, +0.0931]
CoGS, visual features only (no text)	2	0.6939	0.577	+0.0653 [+0.0476, +0.0850]
CoGS, four class names only	1	0.7071	0.613	+0.0785 (single run)
Same design, ImageNet-trained ConvNeXt-L features instead of CLIP	2	0.6771	0.595	+0.0485 (mean difference)

CLIP or any strong frozen backbone? Replacing CLIP with an ImageNet-22k ConvNeXt-L of the same size gives 0.6771 against 0.6939 for the visual-only CLIP variant, a further +0.0167 for CLIP (interval +0.0036 to +0.0296; exact p=0.33). The two-seed ranges barely separate (0.672 to 0.682 against 0.685 to 0.703) and overlap once the third seed of each arm is added (0.672 to 0.686 against 0.685 to 0.704), so this is a modest edge for language-aligned pre-training on top of a large gain (+0.0485) from any strong frozen network. The two controls differ in more than the pre-training data (the ImageNet features pass through a fixed random projection), so we do not claim more than this.

5.2.3 TAM-Net: teacher-guided masking of suspected missing labels

Main result and the effect of a third seed. With seeds 42 and 1 (Table 5.17, Figure 5.11), TAM-Net scored 0.7471 and 0.7466, a mean 0.0108 above the SegFormer-B2 baseline (interval -0.0013 to +0.0247, ahead in 95.5% of resamples, exact seed-level p=0.33, the smallest attainable), with an almost identical pair of scores (spread 0.0005). The numbers first reported for TAM-Net were based on these two seeds. A third TAM-Net seed, trained later on Kaggle with the identical recipe (the baseline and the control have no third seed), scored 0.7361. With it the TAM-Net mean is 0.7433 ± 0.0062, the difference from the baseline falls to +0.0072 (interval -0.0044 to +0.0199, ahead in 88.0% of resamples), and the seed spread is 0.0110 rather than 0.0005 (Section 5.4.1). The near-identical first two seeds therefore do not indicate unusual stability: with two seeds a spread cannot be estimated, and the third seed contradicts the impression. The paired comparisons of the main analysis are in Table 5.18.

Table 5.17: Seed-level test scores of the three arms (split A), seeds 42 and 1.

Arm	**Seed-level test mIoU**	**Mean**	**Std. dev.**
SegFormer-B2 baseline (seeds 42, 1)	0.7395, 0.7327	0.7361	0.0048
TAM-Net control (no masking) (seeds 42, 1)	0.7330, 0.7473	0.7402	0.0101
TAM-Net (seeds 42, 1)	0.7471, 0.7466	0.7468	0.0003

Table 5.18: Paired comparisons between the arms. "Seed-level exact p" is a two-sided permutation test on the seed means; the smallest attainable value is 0.33 for 2 against 2 seeds.

Comparison	**Difference**	**95% CI (images)**	**Ahead in**	**Seed-level exact p**
TAM-Net − baseline (total effect)	+0.0108	[−0.0013, +0.0247]	95.5%	0.33
TAM-Net − control (masking effect)	+0.0067	[−0.0043, +0.0177]	88.4%	1.00
control − baseline (detail path, flip/zoom)	+0.0041	[−0.0078, +0.0162]	74.1%	0.67

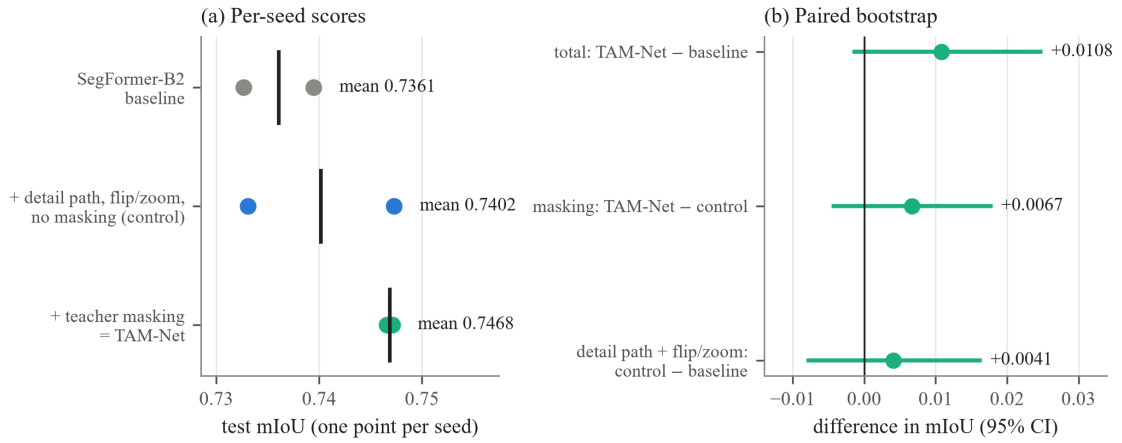


Figure 5.11: (a) Per-seed test mIoU of the baseline, the control, and TAM-Net. (b) Paired bootstrap differences between the arms. Every interval includes zero.

What the comparison isolates. The three arms differ stepwise. The control differs from the baseline in the stride-2 detail path, the on-the-fly flip and zoom-in augmentation, and the absence of the area-inverse pixel weighting that the baseline used; the three cannot be separated with these runs (+0.0041, interval -0.0078 to +0.0162). TAM-Net differs from the control only in the masking (and the reliability loss it enables): +0.0067, interval -0.0043 to +0.0177, ahead in 88% of resamples (+0.0031 with the third TAM-Net seed). Table 5.19 shows it per class; only Human Activity has an interval above zero (+0.0124, +0.0021 to +0.0230), and with four classes tested and no correction that is weak evidence on its own. The masked share is small (0.62% to 0.82% of pixels), which limits how much the mechanism could change.

Table 5.19: Per-class effect of masking (TAM-Net - control) and total effect (TAM-Net - baseline), seeds 42 and 1.

Class	Masking effect	95% CI	Total effect
Aquatic Animal	+0.0162	[−0.0172, +0.0507]	+0.0201
Artifacts and Metals	+0.0022	[−0.0223, +0.0278]	+0.0193
Human Activity	+0.0124	[+0.0021, +0.0230]	+0.0042
Marine Trash and Plastic	−0.0041	[−0.0178, +0.0093]	−0.0004

Validation to test. Table 5.20 lists, for each run, the validation score at the selected epoch and the test score. All four runs without masking lose 0.008 to 0.012 from validation to test (mean -0.0100). The two TAM-Net runs of the main analysis change by +0.0025 and -0.0012 (mean +0.0007), so they keep their validation score. The additional third seed changes by -0.0091, as much as the unmasked runs, so this is at most a suggestive tendency. Validation mIoU on a 980-image split was not a reliable guide to test rank: it placed the control at the level of TAM-Net.

Table 5.20: Validation score at the selected epoch against test score (split A).

Arm	Seed	Best epoch	Validation	Test	Change
SegFormer baseline	seed 42	43	0.7516	0.7395	−0.0121
SegFormer baseline	seed 1	49	0.7417	0.7326	−0.0091
TAM-Net control	seed 42	39	0.7440	0.7330	−0.0110
TAM-Net control	seed 1	55	0.7550	0.7472	−0.0077
TAM-Net	seed 42	53	0.7446	0.7471	+0.0025
TAM-Net	seed 1	46	0.7477	0.7466	−0.0012
TAM-Net	seed 2 (additional)	47	0.7452	0.7361	−0.0091

TTA and ensembling. With flip-TTA, TAM-Net seed 42 reaches 0.7518 and a two-model ensemble of seeds 42 and 1 reaches 0.7544, against 0.7513 for a two-model SegFormer-B2 ensemble given the same treatment (+0.0031, Table 5.6). These ensemble figures use the first two seeds only and predate the third; we do not extend them.

Thresholds. The per-class masking thresholds were derived from the per-class inconsistency rates that we later withdrew (Section 5.2.8). The mechanism and its measured effect against the control are unaffected, but the thresholds must be described as hand-set hyper-parameters.

5.2.4 TAM-Net on a second backbone and the pooled analysis

Why a second backbone. The SegFormer-B2 comparison rests on two seeds per arm and every interval in it includes zero, so it cannot tell whether the advantage of TAM-Net belongs to the design or to one backbone and one pair of seeds. After those results the same design was therefore attached to DeepLabv3+ (Section 4.2.5): TAM-Net on DeepLabv3+ with seeds 42 and 1, and a no-masking control with seed

42 (a control with seed 1 was not trained). Table 5.21 gives the six arms of the two backbones.

Table 5.21: Test mIoU on split A (954 images) of the plain model, the no-masking control, and TAM-Net on the two backbones. Values are computed from the per-image counts, so they can differ by 0.0001 in the fourth decimal from Appendix A.

Arm	**Seeds**	**Test mIoU per seed**	**Mean**	**Std. dev.**
SegFormer-B2 baseline	42, 1	0.7395, 0.7327	0.7361	0.0048
TAM-Net control (masking off)	42, 1	0.7330, 0.7473	0.7402	0.0101
TAM-Net	42, 1	0.7471, 0.7466	0.7468	0.0003
DeepLabv3+ (ResNet-101)	42, 1	0.6540, 0.6698	0.6619	0.0112
TAM-Net control on DeepLabv3+ (masking off)	42	0.6674	0.6674	single run
TAM-Net on DeepLabv3+	42, 1	0.6694, 0.6831	0.6762	0.0097

On DeepLabv3+ the masked runs score 0.6694 and 0.6831 (mean 0.6762), against 0.6540 and 0.6698 for the plain model (mean 0.6619) and 0.6674 for the control. The masked share of the pixels is 0.65% to 1.07% (0.74% to 0.78% at the last epoch), in the same range as on SegFormer-B2 (0.6% to 0.8%), and Aquatic Animal remains the weakest class (0.541 and 0.552).

Effects. Table 5.22 and Figure 5.12 give three effects for each backbone and pooled over both, with the same paired bootstrap as before (2,000 resamples of the 954 test images). The *whole design* (TAM-Net minus the plain model) gains +0.0108 on SegFormer-B2 and +0.0143 on DeepLabv3+. Each interval alone includes zero (-0.0013 to +0.0247 and -0.0019 to +0.0291), but the sign is the same in all four seed pairs (+0.0076, +0.0139, +0.0154, and +0.0133). Pooled over the two backbones the gain is +0.0126 (95% interval +0.0025 to +0.0225, ahead in 99.5% of the resamples). It therefore meets both conditions of the decision rule of Section 4.5: the interval excludes zero and the sign is the same in every seed pair. With the third TAM-Net seed of SegFormer-B2 (0.7361; there is no plain-model seed 2, so seed pairs cannot be formed for it) the pooled gain is +0.0108 (+0.0007 to +0.0205): still above zero, but only just.

The *masking alone* (TAM-Net minus the no-masking control) is +0.0067 on SegFormer-B2, +0.0088 on DeepLabv3+ (interval -0.0114 to +0.0276; the control has one seed), and +0.0077 pooled (-0.0032 to +0.0192, ahead in 90.8% of the resamples). It is positive in three of the four seed pairs (in seed 1 of SegFormer-B2 the control is 0.0007 above TAM-Net). The *detail path and the flip and zoom-in augmentation* together (control minus plain model) gain +0.0041 and +0.0056, and +0.0048 pooled (-0.0056 to +0.0159). The point estimates attribute about 60% of the pooled gain to the masking and 40% to the other two changes.

What this establishes. Taken together, the whole TAM-Net design (detail path, flip and zoom-in augmentation, and teacher-guided masking) improves on the plain model by about one point of mIoU on two different backbones, in every seed pair, and the pooled interval excludes zero. This is the strongest evidence in the study for a proposed component, and we treat the design as established. Three limits must be

Table 5.22: Effects of the TAM-Net design on the 954 test images (2,000 paired bootstrap resamples). "Same sign" counts the seed pairs (42 and 1) in which the difference is positive; on DeepLabv3+ both TAM-Net seeds are compared with the single control. The pooled effect is the average of the two backbone effects on the same resample; it was chosen after the DeepLabv3+ runs had finished.

Backbone	**Effect**	**Difference**	**95% CI (images)**	**Ahead in**	**Same sign, every seed pair**
SegFormer-B2	TAM-Net − plain model (whole design)	+0.0108	[−0.0013, +0.0247]	95.5%	yes (2/2)
	TAM-Net − control (masking alone)	+0.0067	[−0.0043, +0.0177]	88.4%	no (1/2)
	control − plain model (detail path, flip/zoom)	+0.0041	[−0.0078, +0.0162]	74.1%	no (1/2)
DeepLabv3+	TAM-Net − plain model (whole design)	+0.0143	[−0.0019, +0.0291]	95.8%	yes (2/2)
	TAM-Net − control (masking alone)	+0.0088	[−0.0114, +0.0276]	81.3%	yes (2/2)
	control − plain model (detail path, flip/zoom)	+0.0056	[−0.0121, +0.0235]	73.0%	no (1/2)
Pooled over both backbones	TAM-Net − plain model (whole design)	+0.0126	[+0.0025, +0.0225]	99.5%	yes (4/4)
	TAM-Net − control (masking alone)	+0.0077	[−0.0032, +0.0192]	90.8%	no (3/4)
	control − plain model (detail path, flip/zoom)	+0.0048	[−0.0056, +0.0159]	80.7%	no (2/4)

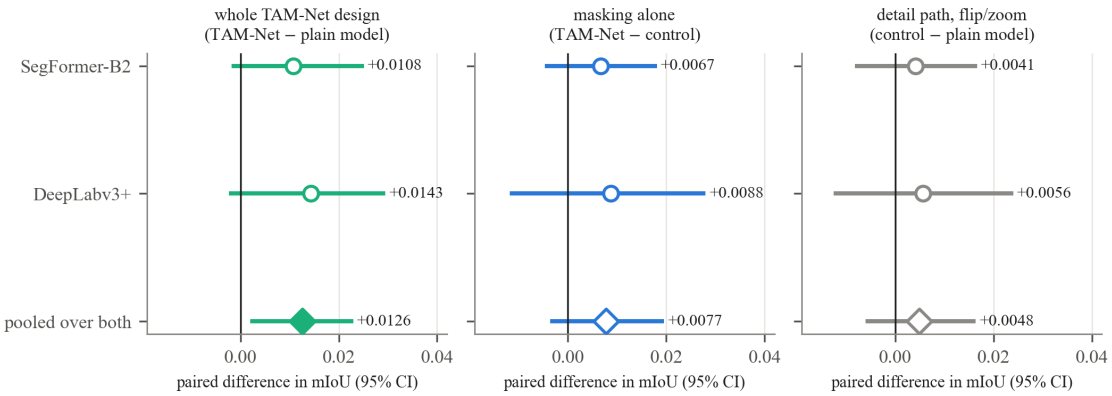


Figure 5.12: Paired effects on the 954 test images with 95% percentile intervals over images. Left: the whole design against the plain model; centre: masking alone against the no-masking control; right: detail path and flip and zoom-in augmentation against the plain model. Circles: one backbone; diamonds: pooled over both; filled markers: the interval excludes zero.

stated. The pooling over the two backbones was decided after the DeepLabv3+ runs had finished, so it is exploratory and not pre-registered, and the pooled interval is the only interval in this thesis that was chosen after the results were known. The control on DeepLabv3+ has a single seed, so the masking-alone effect on that backbone carries the seed noise of a single control run (0.014 between the two SegFormer-B2 control seeds).

Data and run notes. Five of the 10,446 training images (copies of one source photograph) could not be read on the machine that trained the three DeepLabv3+ runs; whenever one of them was drawn, the data loader used a random other training image (at most 0.05% of the draws). The plain DeepLabv3+ runs and all SegFormer-B2 runs used the complete set. Each of the three runs was resumed from its checkpoint after an interruption (the control twice), and together they took 13.5 GPU-hours.

5.2.5 MAAS: a controlled test of missing-annotation-aware training

MAAS was tested with a pre-registered criterion. To create a known problem, 30% of the training polygon instances were deleted (stratified by class) and every model was scored on the original, uncorrupted test labels. The deletion costs -0.1113 mIoU (0.6979 to 0.5866). Success was defined in advance as beating the plain model, the random control, *and* the threshold control, with paired intervals excluding zero. Table 5.23 gives the scores, Table 5.24 the paired differences, and Figure 5.13 summarises both.

Table 5.23: The 30%-deleted-label experiment (Pipeline B, seed 0). "Damage repaired" is the share of the 0.1113 loss recovered.

Model	**Test mIoU**	**+ flip-TTA**	**Damage repaired**	**Paired Δ vs. plain [95% CI]**
Clean-label ceiling (plain SegFormer-B2, seed 0)	0.6979	0.7072	—	—
MAAS v2	0.6145	0.6239	+25.1%	+0.0279 [+0.0050, +0.0522]
Random control v2 (same pixel budget)	0.6038	0.6135	+15.5%	+0.0172 [−0.0004, +0.0358]
Threshold control v2	0.5886	0.5935	+1.8%	+0.0020 [−0.0007, +0.0049]
Plain training on the 30%-deleted labels	0.5866	0.5897	—	—
MAAS v1	0.5702	0.5771	−14.7%	−0.0164 [−0.0378, +0.0055]

Table 5.24: Paired differences in the MAAS experiments (2,000 resamples).

Comparison (30% deleted labels)	**Difference**	**95% CI**	**CI excludes 0?**
MAAS v2 − plain training	+0.0279	[+0.0050, +0.0522]	yes
MAAS v2 − random control (pre-registered)	+0.0107	[−0.0110, +0.0323]	no
MAAS v2 − threshold control (pre-registered)	+0.0259	[+0.0045, +0.0487]	yes
random control − plain training	+0.0172	[−0.0004, +0.0358]	no
MAAS − plain SegFormer, real labels	−0.0066	[−0.0217, +0.0089]	no
random control − plain SegFormer, real labels	−0.0039	[−0.0191, +0.0119]	no
MAAS − random control, real labels	−0.0026	[−0.0194, +0.0152]	no

MAAS fails its own criterion. Version 2 beats plain training (+0.0279, interval +0.0050 to +0.0522) and the threshold control (+0.0259), but not the random

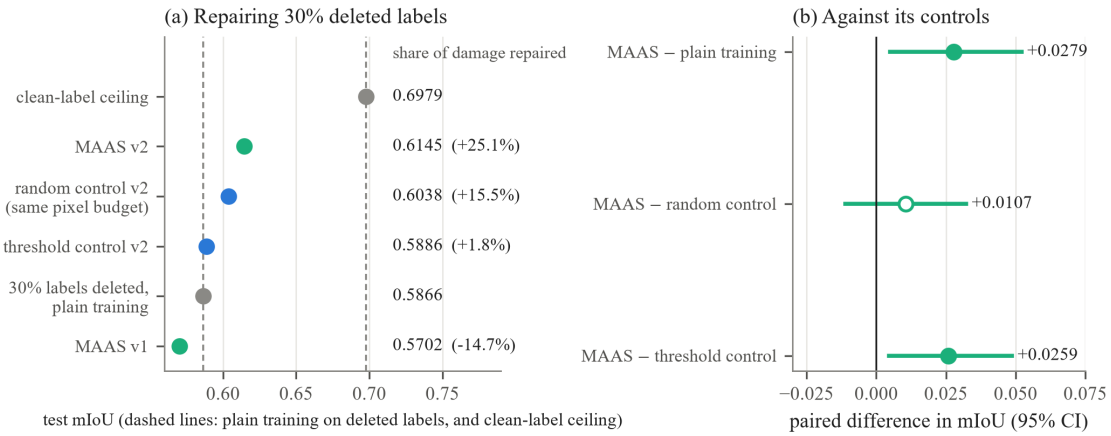


Figure 5.13: (a) Test mIoU when 30% of training labels are deleted. (b) Paired differences of MAAS against its controls.

control (+0.0107, interval -0.0110 to +0.0323). Of MAAS's 25.1% repair, random down-weighting of the same pixel budget already achieves 15.5%, so most of the benefit comes from down-weighting *some* background under label noise, not from choosing the right pixels. The detector itself works: version 2's suspected pixels fall inside truly deleted objects with a precision of 0.601 against a chance rate of 0.045 (13.4 times chance; version 1 reached 0.843, or 18.8 times), while down-weighting only 0.55% of background pixels. Because the deleted objects are known, three quantities can be computed exactly from the training log. *Precision* is the share of the pixels that MAAS down-weights which lie inside a deleted object (3,538,855 of 5,888,958 pixels in the last epoch of version 2). The *chance rate* is the precision that a random choice of pixels would reach: MAAS may only act on pixels labelled as background, and after the deletion 1,064,043,837 of the pixels of the 4,467 training images carry the background label, of which 47,753,685 actually belong to a deleted object, so a randomly chosen background pixel is a deleted-object pixel with probability 47,753,685/1,064,043,837=0.0449. The ratio of the two is 0.601/0.0449 = 13.4. *Recall* is the share of all deleted pixels that MAAS reaches: 3,538,855/47,753,685 = 7.4%. That recall is low, which caps what it can repair (5.7% for version 1). Version 1 was worse than plain training (-0.0164) because a hard-coded boundary filter rejected 79% of its candidate regions (6,272 rejected against 1,690 kept in the last epoch); adding the boundary width to the validation search, which chose zero, produced version 2 (+0.0443 over version 1).

On the real labels the effect is not established. On the uncorrupted labels MAAS scores 0.6914 and its random control 0.6940 against 0.6979 for the plain model: MAAS - plain is -0.0066 (interval -0.0217 to +0.0089). Real missing labels are rarer and harder than synthetic deletions, and the effect is not established. A protocol error is worth recording: an early report used the best-validation checkpoint for the corrupted baseline, which is epoch 4 of 40, before the corrupted labels had done their damage, so both checkpoint conventions are now reported.

5.2.6 AquaScale: scale-aware design

AquaScale was built for the U-shaped detection profile (Section 5.1.9). Table 5.25 shows that no variant beats the baseline (0.6979): five variants score 0.6838 to 0.6970. Architecture alone gives +0.5 percentage points on tiny-animal detection (interval -2.7 to +4.0), that is, no measurable gain. Zoom augmentation only moves accuracy between sizes: version 1 raises huge-animal detection on validation (47.4% to 68.4%, interval of the change +5.3 to +42.1 points) but lowers tiny-animal detection (43.4% to 32.7%, -10.7 points, interval -16.4 to -5.8); version 2 keeps the tiny animals but loses the huge-animal gain and lowers medium ones (-6.2 points, interval -12.2 to -0.9). None passed the seed-0 screen, so none was seed-verified. This is a negative result with a diagnosis: the scale-aware architecture could not move a metric in which the small objects contribute few pixels, and zoom only redistributes performance across sizes.

Table 5.25: AquaScale variants against the SegFormer-B2 baseline (Pipeline B, seed 0). "Detected" is the share of animals in that size bucket with at least half their pixels correct.

Variant (seed 0)	Params (M)	Test mIoU	+ TTA	Val. tiny detected (%)	Val. huge detected (%)	Test tiny detected (%)
SegFormer-B2 baseline	24.72	0.6979	0.7072	43.4	47.4	49.2
architecture only (scale attention + detail branch)	24.95	0.6970	0.7021	43.9	36.8	51.7
zoom only, all sizes (v1)	24.72	0.6858	0.6991	32.7	68.4	41.8
architecture + zoom (v1)	24.95	0.6912	0.7017	36.2	63.2	41.3
zoom only, large sizes (v2)	24.72	0.6864	0.6995	43.4	52.6	47.7
architecture + zoom (v2)	24.95	0.6838	0.6959	39.0	36.8	42.5

5.2.7 The initial hybrid, retrained

We ported the initial architecture unchanged (a shared ResNet-50 encoder, an attention-gated decoder to full resolution, a CenterNet-style detection head at stride 8, detection-guided fusion, and two deep-supervision heads) and trained it under the Pipeline A protocol. Because the full-resolution decoder and four loss terms took about ten minutes per epoch on a Kaggle T4, 60 epochs would have exceeded the 12-hour session limit, so it was trained for 10 epochs with the schedule sized to finish. It reached 0.6752 on validation at epoch 9 and **0.6435** on test (Aquatic Animal 0.525, Artifacts 0.646, Human Activity 0.626, Marine Trash 0.777), in the band of the CNN baselines and 0.093 below SegFormer-B2. This probably understates the architecture, and we cannot exclude that 60 epochs would score higher. The initial benchmark's own log points the other way: the model early-stopped at epoch 72 with a best validation mIoU of 0.6369 while its reported test mIoU was 0.7412, a gap of 0.104 in the favourable direction. The initial validation set was contaminated as well (39.6%), so we attribute that gap to no single cause. Under the common protocol, the claim that the hybrid was the best model is not supported.

5.2.8 A measurement we withdrew

During Pipeline A we measured per-class annotation-inconsistency rates by grouping images of the "same photograph" by filename stem and checking whether the same

classes were annotated in every copy. The rates were 41.7% (Aquatic Animal), 65.9% (Human Activity), 85.7% (Artifacts and Metals), and 7.7% (Marine Trash), with a negative control at 0.1%. **These numbers are invalid.** Generic stems such as “12”, “10”, and “13” occur in several source datasets, so many groups contained different photographs. The file that stores the verified groups shows that the flagship example (“photo 12” annotated three different ways) is three different photographs (image correlation 0.14; the groups for stems “10” and “13” have correlations 0.06 and -0.09). The augmentation-copy control could not catch this failure mode. After checking every one of the 227 groups for pixel identity, only 110 were the same photograph and 117 were name collisions. Among the verified groups, disagreement appears only for Aquatic Animal (4 of 22 groups, 18.2%), and none for Human Activity (0 of 14) or Marine Trash (0 of 80); Artifacts and Metals has no verified group. We present this only as limited evidence consistent with small, easily missed animals, not as a measured noise rate. The invalid rates had been used to set TAM-Net’s thresholds, which is why they are now described as hand-set.

5.2.9 Assessment of the alternative designs

Table 5.26 sets the designs side by side on the criteria of this section: accuracy (the effect and how firmly it is established), efficiency (extra parameters and what must run at inference), cost (training time per run), and feasibility, strengths, and weaknesses.

Table 5.26: Assessment of the alternative designs. Training time is the median epoch time times the number of epochs (Pipeline A, hours) or the logged training time (Pipeline B, minutes); the two pipelines are not comparable with each other.

Design	**Accuracy: effect and status**	**Efficiency: extra parameters and inference**	**Cost: training time per run**	**Feasibility, strengths, weaknesses**
SegFormer-B2 (tuned baseline)	+0.08 to +0.11 over the ResNet baselines in both pipelines, established; the learning rate 6×10^{-5} instead of 3×10^{-4} is worth +0.051 (seed 42)	24.7 M parameters; nothing extra	2.5 h (Pipeline A); 58 min (Pipeline B)	Strong, cheap and simple; needs a tuned learning rate

Design	**Accuracy: effect and status**	**Efficiency: extra parameters and inference**	**Cost: training time per run**	**Feasibility, strengths, weaknesses**
CoGS	+0.071 on Att-UNet, established; +0.004 on SegFormer-B2, not supported; the text prompts add nothing reliable	+0.9 to 1.1 M parameters; a large frozen CLIP tower must run at inference	2.9 h on Att-UNet (baseline 2.4 h); 3.4 h on SegFormer-B2	Helps weak backbones and needs no vocabulary; the tower is a deployment burden with no gain on a strong backbone
PADS-Net	+0.006 (raw) and +0.012 (CLAHE) over Att-UNet, inside the seed range; not established	+0.2 M parameters; a restoration module at inference	2.7 h (Att-UNet 2.4 h)	Physically motivated, but the restoration learned to switch itself off ($z \approx 0.006$)
TAM-Net	Best-scoring arm (0.7468). The whole design gains +0.0126 pooled over two backbones, established (pooling chosen after the results); +0.0108 on SegFormer-B2 alone and the masking alone (+0.0077 pooled) are not established	+82,598 parameters (0.33%); the stride-2 detail path stays, the teacher and the masking are training-only	3.6 to 4.2 h against 2.5 h for the baseline (control 3.0 h)	Best score, the same sign in every seed pair, small inference cost; about 1.5 to 1.7 times the training time, hand-set thresholds
MAAS	+0.0279 over plain training under 30% label deletion but only +0.0107 over its random control (interval includes 0); no effect on the real labels; not established	None	73 to 82 min (plain 56 to 58 min)	Free at inference and its detector works (13 times chance precision), but most of the benefit comes from down-weighting background in general

Design	Accuracy: effect and status	Efficiency: extra parameters and inference	Cost: training time per run	Feasibility, strengths, weaknesses
AquaScale	No variant beats the baseline (0.6838 to 0.6970 against 0.6979)	+0.23 M parameters; scale attention and a detail branch	59 to 68 min (baseline 58 min)	Targets the U-shaped size profile, but zoom only moves accuracy between sizes

5.3 Final Design Adjustments

The evaluation led to the following changes to the design, all chosen on validation or on measured evidence.

- **Benchmark.** The original 347-image test set is retired as a basis for claims. Two rebuilt splits replace it, and mirrored copies are linked in Pipeline B's construction. Initial scores are reported only as history.
- **Backbone and learning rate.** SegFormer-B2 replaces the ResNet-based models as the reference, and its learning rate islowered from 3×10^{-4} to 6×10^{-5}(validation 0.7155 to 0.7516; test 0.6883 to 0.7395 at seed 42).
- **Enhancement.** The colour-enhancement stage is removed from the main pipeline: it is worth -0.004 mIoU to a plain model, within noise, and the physics-based version learned to switch itself off.
- **Training.** On-the-fly horizontal flip and zoom-in augmentation are kept for the SegFormer-family runs of Pipeline A, because flip-TTA still adds a small, consistent gain (Table 5.6), which shows that mirrored views remain useful to the models, and TTA is always reported.
- **CLIP guidance.** If CLIP guidance is used at all, the visual-only form is recommended: it is simpler than the text-based form and scores within noise of it (0.6939 against 0.7000). It is not recommended on top of SegFormer-B2.
- **The TAM-Net design.** TAM-Net (stride-2 detail path, flip and zoom-in augmentation, and teacher-guided masking) is adopted for the final system: it is the best-scoring arm on SegFormer-B2 and improves on the plain model on both backbones (Section 5.2.4). The masking alone is small and not established, so it is kept as one component of the design and is not presented as the cause of the gain.
- **Retired ideas.** PADS-Net, AquaScale, and MAAS are dropped from the final system.

- **Reporting.** Every future comparison uses at least four seeds per arm, since with two seeds no comparison can reach an exact permutation p below 0.33 and with three seeds below 0.10 (Section 5.4).

The resulting recommended configuration is TAM-Net on a SegFormer-B2 with an ImageNet-pretrained encoder, learning rate 6×10^{-5}, 60 epochs with OneCycle, focal cross-entropy plus focal Tversky loss, the stride-2 detail path, on-the-fly flip and zoom-in augmentation, teacher-guided masking after epoch 15, best-validation checkpoint, flip-TTA at inference, and, when compute allows, an ensemble of two to three seeds. The deployable package of Pipeline B (a SegFormer-B2, single or three-model ensemble) was trained at the untuned rate and would be expected to improve if retrained this way; that expectation is an inference, not a measurement.

5.4 Statistical Analysis

Techniques. Confidence intervals are 95% percentile bootstrap intervals over test images (1,000 to 2,000 resamples, fixed seeds). Differences use a paired bootstrap with the same resampling for both models. For groups of seeds, the per-resample score of a group is the mean of its members. For seed-level inference we use an exact permutation test on the seed means, which needs no distributional assumption.

What the bootstrap does not cover. The image bootstrap captures test-set sampling variation only. Section 5.1.4 shows that two identically configured models can differ with an interval excluding zero (final-epoch convention: seed 2 minus seed 0, -0.0276, interval -0.0462 to -0.0091). Intervals are therefore reported together with seed-level information, and no single-seed result is called established.

Power. With n1 and n2 seeds in the two groups, the smallest attainable two-sided exact permutation p is

$$p_{\min} = \frac{2}{\binom{n_1+n_2}{n_1}}. \tag{5.1}$$

For two seeds against two it is 0.33, for three against two 0.20, and for three against three 0.10. Every comparison of the main analysis is two against two (or two against one), so none can go below 0.33 and none can reach $p < 0.05$ at seed level; the strongest ones, such as CoGS on Att-UNet against Att-UNet, attain that minimum because both CoGS seeds exceed both baseline seeds. Even with all third seeds included (Section 5.4.1) the smallest value is 0.10. This is a limit of the replication budget: reaching 0.05 needs at least four seeds per arm (four against four gives a minimum of 0.029; three against four gives 0.057).

Multiple comparisons. Where we look at four classes at once (Table 5.19) no correction is applied, and a single class interval excluding zero is treated as weak evidence.

Pooling over backbones. The TAM-Net design was tested on two backbones (Section 5.2.4). The effect on each backbone is the difference between the seed means of two arms, and the pooled effect is the average of the two backbone effects, computed on the same bootstrap resample of test images, so that the pooled interval

reflects the image sampling of both. Pooling is a way to use more runs in one comparison. It was chosen after the DeepLabv3+ runs had finished and is therefore exploratory; it is applied to one family of three effects only (the whole design, the masking alone, and the detail path with the augmentation), and the decision rule is applied to it as to every other comparison.

Table 5.27: What can and cannot be claimed.

Claim	**Status**	**Basis**
41% of the original test images have a near-duplicate in training	Established	Two detectors, both pipelines; 141 and 143 of 347
Initial scores were inflated by 3.6 to 4.5 points	Established (size approximate)	Re-scoring on 204 clean images; subset not random
The initial hybrid was not shown to be best	Established	Retrain 0.6435 against 0.7361; unmatched protocol; seed noise 0.020 to 0.028
SegFormer-B2 is better than the ResNet baselines	Established	+0.08 to +0.11, all classes, both pipelines
CLIP guidance helps weak backbones	Established for Att-UNet; consistent for DeepLabv3+	+0.071 (2 seeds each, complete separation; +0.065 with three); 2 of 2 seeds
CLIP guidance helps SegFormer-B2	Not supported	+0.0039, interval includes 0
Text prompts add value beyond visual features	Not supported	+0.0062, interval -0.0014 to +0.0142 (-0.0001 with three seeds)
The TAM-Net design improves on the plain model (SegFormer-B2 and DeepLabv3+ pooled)	Established (pooled; chosen after the results)	+0.0126, interval +0.0025 to +0.0225; same sign in all four seed pairs; +0.0108 (+0.0007 to +0.0205) with a third SegFormer-B2 seed
TAM-Net beats a tuned SegFormer-B2 on that backbone alone	Not established	+0.0108, interval -0.0013 to +0.0247, two seeds (+0.0072 with a third TAM-Net seed)
Teacher masking is what helps	Not established	+0.0067 on SegFormer-B2 (interval -0.0043 to +0.0177), +0.0088 on DeepLabv3+ (-0.0114 to +0.0276), +0.0077 pooled (-0.0032 to +0.0192)
TAM-Net is more stable across seeds	Not supported	Spread 0.0005 with two seeds, 0.011 with three (SegFormer-B2); 0.0137 on DeepLabv3+

Claim	Status	Basis
The gain of the TAM-Net design transfers to external images	Supported on SUIM, not on TrashCan	+0.0471 (+0.0207 to +0.0765) over the baseline on SUIM, from the detail path and augmentation (masking +0.0048); no difference on TrashCan; baselines with one run
MAAS beats its matched random control	Not established (fails pre-registered criterion)	+0.0107, interval -0.0110 to +0.0323
PADS-Net, AquaScale improve over baselines	No	Within seed noise / below baseline
About 770 training objects are unlabelled	Supported (lower bound; two reviewers)	Blind review with decoys
Correcting test labels raises scores by 0.03 to 0.07	Direction by construction; size indicative	150 images, two reviewers

5.4.1 Sensitivity to the third seeds

Five arms of Pipeline A were trained with a third seed (seed 2): the Att-UNet baseline, both CoGS variants on Att-UNet, the ImageNet-feature control, and TAM-Net. The other multi-seed arms have two seeds (U-Net, PADS-Net, and the hybrid have one run each), and mixing three-seed and two-seed groups would make the arms unequal, so the main analysis uses seeds 42 and 1 for every arm. The rule is based on availability and not on the scores, but it was applied after all runs had finished; the third seeds are therefore reported here in full, so that the choice of two seeds cannot hide any result. Table 5.28 lists the arm means and Table 5.29 repeats the key comparisons with every trained seed (the SegFormer baseline and the TAM-Net control stay at two seeds).

Table 5.28: Third seeds against the two-seed means used in the main analysis (test mIoU, split A).

Arm	Seeds 42 and 1 (mean)	Seed 2	All three seeds (mean ± s.d.)
Att-UNet	0.6286	0.6391	0.6321 ± 0.0062
CoGS on Att-UNet, sub-concepts	0.7000	0.6908	0.6970 ± 0.0132
CoGS on Att-UNet, visual only	0.6939	0.7035	0.6971 ± 0.0103
ImageNet-feature control	0.6771	0.6857	0.6800 ± 0.0072
TAM-Net	0.7468	0.7361	0.7433 ± 0.0062

Table 5.29: Key comparisons with two seeds per arm (main analysis) and with every trained seed. Exact p is the seed-level permutation test.

Comparison	**Two seeds per arm: difference [95% CI]; exact p**	**All trained seeds: difference [95% CI]; exact p**
TAM-Net − baseline (baseline: 2 seeds)	+0.0108 [−0.0013, +0.0247]; 0.33	+0.0072 [−0.0044, +0.0199]; 0.30
TAM-Net − control (control: 2 seeds)	+0.0067 [−0.0043, +0.0177]; 1.00	+0.0031 [−0.0071, +0.0128]; 0.70
CoGS (sub-concepts) − Att-UNet	+0.0714 [+0.0531, +0.0931]; 0.33	+0.0649 [+0.0470, +0.0856]; 0.10
CoGS (visual only) − Att-UNet	+0.0653 [+0.0476, +0.0850]; 0.33	+0.0650 [+0.0476, +0.0845]; 0.10
CoGS (visual only) − ImageNet control	+0.0167 [+0.0036, +0.0296]; 0.33	+0.0171 [+0.0050, +0.0292]; 0.20
SegFormer-B2 baseline − Att-UNet	+0.1075 [+0.0837, +0.1339]; 0.33	+0.1040 [+0.0806, +0.1290]; 0.10

Two points follow. First, for the Att-UNet family the conclusions are unchanged: the gain of CLIP guidance is +0.071 with two seeds and +0.065 with three, the backbone gap is +0.108 against +0.104, and the CLIP edge over the ImageNet control is +0.0167 against +0.0171; the third seeds only lower the attainable p from 0.33 to 0.10 or 0.20. Second, TAM-Net is the one arm where the third seed changes the reading. Its score (0.7361) is 0.011 below the first two seeds, which lowers the gain over the baseline from +0.0108 to +0.0072 and the masking effect from +0.0067 to +0.0031, and it removes the impression of stability given by the first two seeds. Neither version is statistically established, and the smaller figures are the more cautious estimate of the effect.

5.5 Comparisons and Relationships

Against the initial benchmark. The initial ranking, with the hybrid first and DeepLabv3+ and YOLOv8m-seg close behind, does not survive matched conditions. Under one protocol the SegFormer-B2 arms come first (TAM-Net 0.7468, the baseline 0.7361), then DeepLabv3+, then the hybrid, U-Net, PADS-Net, and Att-UNet within about 1.5 points of each other. The initial numbers themselves were measured correctly; the test set was too easy and the comparison was not matched.

Against the literature. We do not compare our absolute scores with published numbers on other underwater datasets, because label sets, classes, splits, and metrics differ; the SUIM and TrashCan results above are on different classes from ours and are not the datasets' own benchmarks. The closest published figure, 82.12 mIoU on SUIM for UWSegFormer [72] (as reported in its abstract), uses SUIM's own label set and protocol and cannot be set beside our two-class external scores (0.59 to 0.80). What our results do agree with is the general pattern from the reliability literature: hidden overlap between training and test sets inflates scores [24], [28], test-label errors matter [40], and seed variance can exceed a claimed gain [37].

Relationships between quantities. (i) *Benefit of CLIP against baseline strength*: the gain falls as the baseline improves (Att-UNet 0.629: +0.071; DeepLabv3+ 0.662: +0.024 to +0.059; SegFormer-B2 0.736: +0.004). (ii) *Detection against size*: a plateau of 75–83% for small to large animals with shortfalls at both ends. (iii) *What models miss against what annotators skip*: tiny animals are both the least detected and, among confirmed unlabelled animals, the most common (77%). (iv)

Label completeness against score: correcting the test labels moves scores by 0.03 to 0.07, comparable to the differences between methods; part of the remaining gap is therefore a label ceiling, not a model ceiling. (v) *Learning rate against architecture*: for SegFormer-B2, tuning the learning rate was worth 0.051 (seed 42), more than any design idea except the backbone change. Figure 5.14 shows predictions of the baseline and TAM-Net.

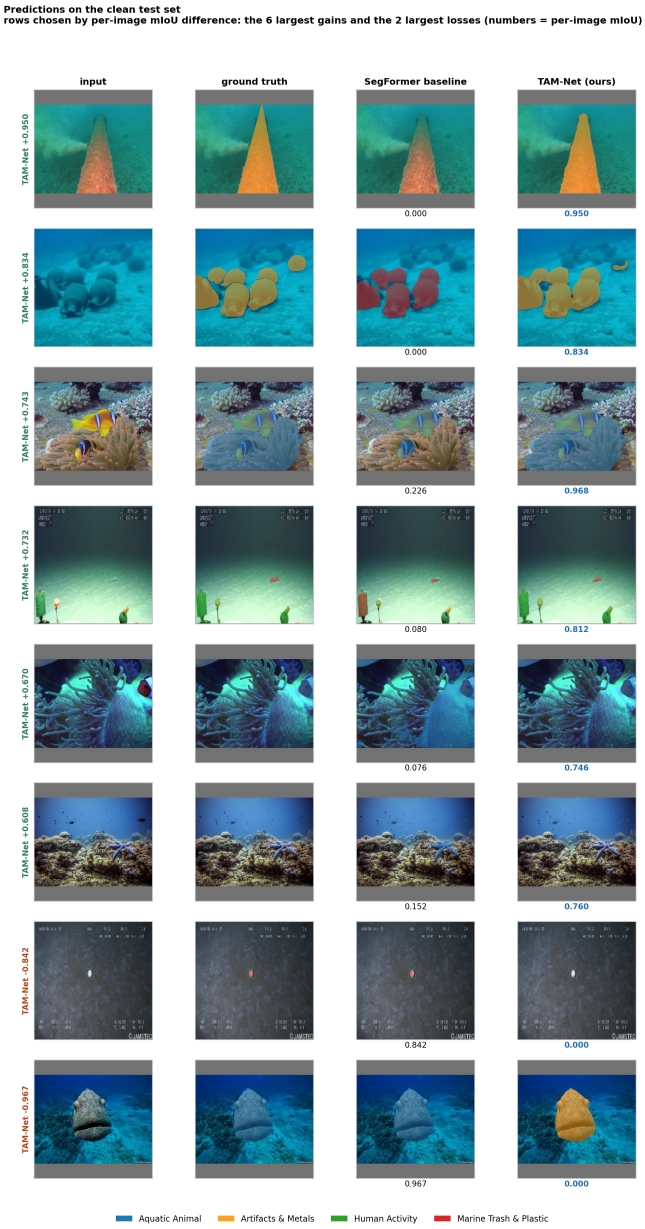


Figure 5.14: Predictions on split-A test images (SegFormer-B2 baseline against TAM-Net, seed 42; numbers are per-image mIoU). The rows are chosen by per-image difference: the six largest gains and the two largest losses of TAM-Net, so they illustrate failure modes and are not representative. Over the 869 of 954 test images that contain at least one object (a per-image mIoU is undefined for the other 85), TAM-Net scores higher on 409, lower on 452, and ties on 8, with a mean per-image difference of +0.005.

5.6 Discussions

Interpretation. The dominant lesson is about measurement. After the backbone (+0.08 to +0.11) and the learning rate (+0.051, one seed), the largest effects on a reported score were introduced by the evaluation and not by any model idea: the benchmark (a 0.036 to 0.045 inflation), seed noise (0.020 to 0.028), and test-time augmentation (up to +0.019). Each of them is larger than the effect of teacher-guided masking (+0.003 to +0.009) or of any of the failed ideas. Only after these were controlled could the smaller effects be judged, and most of them turned out to be within noise. The three ideas that targeted the small-object and label problems from the model side, restoration, scale awareness, and missing-annotation weighting, did not beat their controls, and the audit explains why: the missing objects are mostly tiny, sparse in pixels, and only partly recoverable by any pixel-level loss adjustment.

Answers to the research questions. *RQ1:* 40.6% to 41.2% of the original test images overlapped the training set, partly because 48.4% of SUIM and 22.8% of TrashCan were absorbed without deduplication, and initial scores were inflated by 3.6 to 4.5 points. *RQ2:* seed noise reaches 0.020 to 0.028 mIoU and TTA up to 0.019, larger than the initially claimed 0.0175. *RQ3:* about 770 objects are unlabelled in training (a lower bound) and about one per test image, and correcting them shifts scores by 0.03 to 0.07. *RQ4:* the backbone helps decisively; CLIP guidance helps weak backbones but not SegFormer-B2; the TAM-Net design (detail path, augmentation, and teacher-guided masking) is the best-scoring model and improves on the plain model on two backbones (pooled +0.0126, 95% CI +0.0025 to +0.0225); restoration, scale-aware design, and MAAS do not help. *RQ5:* CoGS keeps its lead over the older CNN baselines on SUIM and TrashCan but not over SegFormer-B2, which scores as well as or better than CoGS on both; the TAM-Net design transfers to SUIM (+0.047 over the SegFormer-B2 baseline, interval +0.021 to +0.077, coming from the detail path and the augmentation and not from the masking) but not to TrashCan (0.362 and 0.368 against 0.370, without intervals); the baselines were scored with one run; and the sources are partly seen.

Implications. For anyone building a segmentation dataset from redistributed public sources, deduplicate against the sources and against mirrored copies before splitting; report seeds and intervals; and check what fraction of a class's objects an annotator can reasonably be expected to label. For deployment, TAM-Net on a tuned SegFormer-B2 with flip-TTA is the recommended choice (its extra cost at inference is the 82,598-parameter detail path), and a tuned SegFormer-B2 with flip-TTA the safe fallback, and its expected accuracy on a new site should be taken from external-domain figures, not from the in-domain ones; the external figures we have for SegFormer-B2 (one run, without TTA) are 0.737 on SUIM and 0.370 on the deep-sea TrashCan set, and 0.34 to 0.37 mean IoU is the level reached by the best models there.

Limitations and threats to validity. (1) *Two pipelines, two test sets.* Scores cannot be compared across them, and only Pipeline A hosts the model comparison. (2) *Residual leakage in split A* through mirrored copies (11.4% of test images), whose effect we bounded but did not eliminate. (3) *Baseline mismatch.* The baseline

SegFormer-B2 uses an area-inverse pixel weighting that TAM-Net, its control, and CoGS on SegFormer-B2 do not, so the baseline–control step bundles three changes, and the CoGS-on-SegFormer comparison also differs from its baseline in this weighting (CoGS on Att-UNet and on DeepLabv3+ share the weighting of their baselines). The same holds on DeepLabv3+, where the control has a single seed. (4) *Few seeds.* Two seeds per arm cap the attainable exact p at 0.33. The pooled analysis of the TAM-Net design (Section 5.2.4) uses more runs in one comparison, but it was chosen after the DeepLabv3+ runs had finished. Five arms have a third seed, which is reported separately (Section 5.4.1) because the other arms do not; the choice of seeds 42 and 1 for the main analysis was made after all runs had finished, on availability and not on score. (5) *Pipeline B's SegFormer used an untuned learning rate*, so the MAAS and AquaScale baselines are probably weak. (6) *Label audit* rests on two reviewers whose agreement was not measured, round 1 carried a 14% decoy rate, and the estimates are lower bounds; the original labels were also produced by several annotators, whose inconsistency the audit measures only for missing objects, not for boundaries or class choices. (7) *External datasets* share sources with our training data; the plain baselines were scored with one run; TrashCan has no per-image counts and therefore no intervals; CoGS was scored on Att-UNet only; and SUIM's mean covers two classes. (8) *Pipeline A's test masks omit box-only and degenerate rows* (30% of Marine Trash test instances are dropped), although their pixel area is small. (9) *The initial hybrid was trained for 10 epochs*, so its 0.6435 probably understates what a full schedule would give. (10) *Video frames* sit almost entirely in the training splits, so the test sets measure the still-image domain. (11) *Two stored diagnostics*, namely the model-cost table (latency) and the small-object oracle experiment, were not among the transferred files and were not repeated; where we report them we say so. (12) *Cost and latency* were not measured for CoGS. (13) *Unreadable training images.* Five of the 10,446 training images could not be read on the machine that trained the three TAM-Net runs on DeepLabv3+, and the data loader replaced them by random other images (at most 0.05% of the draws).

Chapter 6

Conclusion

6.1 Summary of Findings

This thesis set out to improve four-class underwater semantic segmentation and found that the first task was to find out how far the existing results could be trusted. The main findings, each recomputed from stored result files, are these.

1. **The proposed TAM-Net design is the best-scoring model, and its benefit is established** TAM-Net reaches 0.7468 over two seeds on SegFormer-B2, +0.0108 (interval −0.0013 to +0.0247) above the baseline and +0.0067 (interval −0.0043 to +0.0177) above its own no-masking control; Attached to DeepLabv3+ the same design gains +0.0143, and pooled over the two backbones the whole design improves on the plain model by +0.0126 (95% interval +0.0025 to +0.0225, same sign in all four seed pairs; +0.0108 with the third seed). The pooling was chosen after the results, and the masking alone (+0.0077, interval −0.0032 to +0.0192) is not established.

2. **The original benchmark was contaminated.** 41.2% of the original test images (143 of 347) had a near-duplicate in training; an independent re-implementation found 40.6%, and the original validation set was contaminated at a similar rate (39.6%). The overlap came largely from absorbing public datasets without deduplication: 48.4% of SUIM and 22.8% of TrashCan are in our data.

3. **The initial result does not stand.** Re-scored on the uncontaminated subset, the initial U-Net and DeepLabv3+ lose 0.0445 and 0.0357 mIoU. The claimed 0.0175 advantage of the hybrid is smaller than the seed-to-seed spread (0.020 to 0.028) and of the same order as the effect of flip-TTA (+0.008 to +0.019), which the hybrid alone received, and under a common protocol the retrained hybrid scores 0.6435 against 0.7361 for a tuned SegFormer-B2.

4. **Evaluation needs replication.** An image-bootstrap interval that excludes zero can occur between two identically configured runs. We therefore require agreement across seeds as well, and with two seeds per arm no comparison can reach an exact permutation p below 0.33.

5. **Labels are incomplete.** A blind review with decoys estimates about 770 unlabelled objects in the training set (a lower bound) and confirmed 158 unlabelled objects in 150 test images (41.7% of candidates against a 2.0% decoy rate). Most unlabelled animals are tiny (77%), which is also the size the models miss most.

6. **What improves segmentation.** The backbone dominates: SegFormer-B2 exceeds ResNet-based baselines by 0.08 to 0.11 mIoU overall, with the interval excluding zero on every class in both pipelines, and tuning its learning rate was worth a further 0.051 (seed 42). Frozen CLIP features raise a weak backbone by +0.071 (95% interval +0.053 to +0.093 on Att-UNet), but not SegFormer-B2 (+0.004), and the text prompts add no reliable benefit beyond the visual features.

7. **Three ideas failed, each with a diagnosed cause.** Physics-based restoration (PADS-Net) learned to switch itself off and colour enhancement is worth −0.004 to a plain model; the scale-aware design (AquaScale) did not beat the baseline, and the size analysis points to the cause: small objects hold few pixels and zoom only redistributes accuracy across sizes; missing-annotation-aware training (MAAS) beat the plain model but not its matched random control, and showed no established effect on the real labels.

8. **Generalisation.** On images from two public datasets that do not overlap ours, the CLIP-guided models keep their lead over the older CNN baselines but not over SegFormer-B2, which scores as well as or better than they do; the TAM-Net design is ahead of SegFormer-B2 on SUIM (+0.0471, interval +0.0207 to +0.0765) through its detail path and augmentation, with no contribution from the masking, and level with it on TrashCan; the baselines were scored with one run, the sources are partly seen, and scores on deep-sea footage fall to about 0.34 to 0.37.

6.2 Contributions to the Field

The contributions are a measured account of the benchmark, a method for measuring, and a design that is the best-performing model of the comparison.

- **An audited benchmark and how to audit one.** A near-duplicate detector that uses a perceptual hash, a correlation check, and mirrored and rotated versions; two independently created clean splits; a worked example of contamination in an underwater segmentation dataset from redistributed public sources, including the provenance of the contamination and its impact on published scores.

- **A protocol that survives its own controls.** Explicit statements of the claims (Table 5.27) together with Dataset-global IoU, paired bootstrap, seed replication using accurate permutation tests, matched controls, validation-only selection, Along with the evidence that a single-run comparison and an interval

excluding zero are each insufficient here, and a replication of the main findings in two independently built pipelines (Table 5.9).

- **A decoy-calibrated audit of missing labels.** A review design that uses random background regions to measure the reviewers' yes-rate, and then corrects all reported rates by this yes-rate, giving lower-bound estimates of the number of missing objects at the class level and a size profile of who is missed.
- **A controlled comparison.** A strong and reproducible baseline configuration (SegFormer-B2 at a tuned learning rate), and the observation that CLIP guidance is beneficial for weak but not strong backbones and that the language component is not essential to achieve the gain.
- **A proposed design, TAM-Net.** A stride-2 detail path, flip and zoom-in augmentation, and teacher-guided masking of suspected missing labels make up the best-scoring model of the study (0.7468) and improve on the plain model on two backbones (pooled +0.0126, 95% interval +0.0025 to +0.0225; established when pooled, with the masking alone not established). The only thing that is still in inference is the small detail path.
- **Diagnosed negative results.** Each failure is documented, so that others don't repeat it.
- **Reproducible artefacts.** Scripts, both splits, per-image confusion counts, checkpoints, the review pages and answers, and a runnable model, each number in this thesis being traceable to a stored file (Appendix B).

What we do not claim. We limit our claims to only the datasets we compared the models in, and make no claim about state-of-the-art performance on other benchmarks, which we did not run. The components of TAM-Net and CoGS have precedent (mean teachers [12], background recalibration [36], zero-initialised fusion [62], CLIP-guided segmentation [47], [52]); we present TAM-Net as an adaptation of them whose combined benefit, about one point of mIoU on two backbones, is what this comparison measured. We also make no concrete claim that the dataset is completely leakage free in its current state, only near-duplicates and same video frames at the stated thresholds. Furthermore, per-class annotation noise rate was not measured and therefore, we make no claim about how much each class was affected by unreliable annotations.

6.3 Recommendations for Future Work

- **More seeds and a cleaner ablation.** Run each model with four different random seeds. Use the same criteria for the baseline as TAM-Net. Run no-masking-experiment for all backbone (at least 2 seeds). This will help us to understand the effects of detail path, augmentation and masking separately.
- **One benchmark.** Test all the models on the same test set.We can either run Pipeline A on Pipeline B's split or we can create a new split for Pipeline

A. This will remove the remaining 11% mirrored-copy leakage. This will also make the model comparison more consistent.

- **Tune before you compare.** Repeat the MAAS and AquaScale experiments using a proper tuned SegFormer-B2 baseline.

- **Repair the labels.** Record all reviewers' decisions on training images. Measure how much they agree, correct the wrong labels and train the models again. This will help us to know whether levels are contributing to bad results. It can also help find small objects which annotators may miss. With corrected labels, TAM-Net's thresholds can be set using actual measurements of annotation quality instead of choosing manually.

- **Report instance-level metrics.** Add detection results based on object size to every table. Because mean IoU does not show small-object performance high.

- **External and stronger baselines.** Repeat the external evaluation on SUIM and TrashCan with many seeds and bootstrap intervals. Add CoGS and the ImageNet-feature control with strong backbones.Test the model on complete unseen test sets and measure the inference time and computation costs of CoGS.

- **Open long-tailed recognition.** The open-set and long-tail direction idea is still an area of study. This includes detecting unknown objects and rejecting the unknown classes. Our current benchmark provides a reliable starting point for this study.

- **Record image provenance.** Record all the original images. and if the 33 files of unknown origin (Chapter 3) are found synthetic, then remove them and repeat the affected experiments again.

- **Video and deep-sea data.** Add test sets from video at different times. Study domain adaptation for deep-sea underwater images.

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Appendix A

Complete Run Tables

This appendix lists every trained and evaluated run of the controlled study. Scores are test mIoU and per-class IoU without test-time augmentation unless a column says otherwise. Runs of Pipeline A and Pipeline B were evaluated on different test sets. So we did not compare with each other. All values are read from the stored evaluation and result files. The analyses in Chapter 5 use the per-image counts of the same runs. It differs from the values in the evaluation files by at most 0.00005; for five runs (Att-UNet seed 2, U-Net, SegFormer-B2 seed 1, the TAM-Net control seed 1, and TAM-Net on DeepLabv3+ seed 42) this shows as a difference of 0.0001 in the fourth decimal between this appendix and the text.

A.1 Pipeline A

Table A.1 lists the 35 runs of Pipeline A: 33 trained on the workstation (three of them TAM-Net runs on DeepLabv3+, Section 5.2.4), the retrained initial hybrid, and TAM-Net seed 2, both of which were trained on Kaggle. The best epoch and validation score of TAM-Net seed 2 were read from its checkpoint metadata. Five runs are marked "additional": these are the third seeds (seed 2) of five arms, which enter only the sensitivity analysis of Section 5.4.1; every other multi-seed arm, and the main analysis, use seeds 42 and 1.

Table A.1: Every trained and evaluated run of Pipeline A on the 954-image test set of split A, sorted by test mIoU. Ep.: epoch chosen on validation (1-based); Val.: validation mIoU at that epoch; Test: test mIoU; Aq., Art., Hum., Tr.: per-class test IoU (Aquatic Animal, Artifacts and Metals, Human Activity, Marine Trash and Plastic). Rows marked "additional" are third seeds, used only in the sensitivity analysis.

Run	Ep.	Val.	Test	Aq.	Art.	Hum.	Tr.
TAM-Net control (no masking), seed 1	55	0.7550	0.7472	0.605	0.776	0.762	0.845
TAM-Net, seed 42	53	0.7446	0.7471	0.601	0.775	0.775	0.836
TAM-Net, seed 1	46	0.7477	0.7466	0.619	0.764	0.775	0.829
CoGS SegFormer-B2, seed 1	55	0.7482	0.7417	0.619	0.756	0.768	0.823
SegFormer-B2, lr 6e-5, seed 42	43	0.7516	0.7395	0.622	0.753	0.753	0.830
CoGS SegFormer-B2, seed 42	54	0.7406	0.7383	0.597	0.761	0.766	0.828

Run	Ep.	Val.	Test	Aq.	Art.	Hum.	Tr.
TAM-Net, seed 2 (additional; Kaggle)	47	0.7452	0.7361	0.584	0.772	0.767	0.822
TAM-Net control (no masking), seed 42	39	0.7440	0.7330	0.582	0.758	0.763	0.828
SegFormer-B2, lr 6e-5, seed 1	49	0.7417	0.7326	0.558	0.747	0.789	0.836
CoGS DeepLabv3+, seed 42	18	0.7195	0.7131	0.618	0.732	0.714	0.788
CoGS Att-UNet, sub-concepts, seed 1	40	0.7296	0.7121	0.582	0.712	0.740	0.814
CoGS Att-UNet, class names, seed 42	30	0.7244	0.7071	0.613	0.701	0.728	0.787
CoGS Att-UNet, visual only, seed 2 (additional)	32	0.7283	0.7035	0.590	0.706	0.725	0.793
CoGS Att-UNet, visual only, seed 42	34	0.7241	0.7026	0.606	0.701	0.710	0.793
CoGS DeepLabv3+, seed 1	35	0.7242	0.6942	0.552	0.724	0.697	0.804
CoGS Att-UNet, sub-concepts, seed 2 (additional)	13	0.7267	0.6908	0.573	0.684	0.711	0.795
SegFormer-B2, lr 3e-4, seed 42	39	0.7155	0.6883	0.540	0.729	0.685	0.800
CoGS Att-UNet, sub-concepts, seed 42	19	0.7254	0.6880	0.524	0.712	0.721	0.795
ImageNet-feature control, seed 2 (additional)	52	0.6998	0.6857	0.605	0.677	0.675	0.786
CoGS Att-UNet, visual only, seed 1	14	0.7262	0.6851	0.548	0.686	0.717	0.789
TAM-Net on DeepLabv3+, seed 1	49	0.7138	0.6831	0.552	0.669	0.714	0.798
ImageNet-feature control, seed 42	52	0.7010	0.6823	0.609	0.670	0.670	0.780
ImageNet-feature control, seed 1	33	0.6945	0.6720	0.580	0.650	0.672	0.786
DeepLabv3+ R101, seed 1	48	0.7006	0.6698	0.539	0.678	0.667	0.795
TAM-Net on DeepLabv3+, seed 42	45	0.7141	0.6693	0.541	0.686	0.648	0.802
TAM-Net control on DeepLabv3+ (no masking), seed 42	48	0.7139	0.6674	0.520	0.705	0.651	0.794
DeepLabv3+ R101, seed 42	34	0.7021	0.6540	0.503	0.672	0.649	0.792
Initial hybrid, retrained (Kaggle, 10 epochs)	9	0.6752	0.6435	0.525	0.646	0.626	0.777
U-Net R50	57	0.6683	0.6416	0.527	0.621	0.643	0.775
PADS-Net (CLAHE input)	35	0.6767	0.6401	0.509	0.629	0.637	0.786
Att-UNet, seed 2 (additional)	43	0.6753	0.6390	0.537	0.628	0.626	0.766
PADS-Net (raw input)	47	0.6817	0.6350	0.522	0.619	0.627	0.773
Att-UNet, seed 42	54	0.6685	0.6301	0.532	0.620	0.606	0.762
Att-UNet, seed 1	41	0.6848	0.6271	0.511	0.613	0.616	0.769
Att-UNet, CLAHE-enhanced input	49	0.6740	0.6263	0.514	0.630	0.591	0.770

A.2 Pipeline B

Table A.2 lists the 23 result directories of Pipeline B. Nineteen are trained models; the ensemble and the three threshold controls are computed from trained models without further training.

Table A.2: Every run of Pipeline B on the 958-image test set of split B, sorted by test mIoU without TTA. "Ep." is the epoch chosen on validation and "Val." the validation mIoU there; "Test" is the test mIoU and "+TTA" the same model with horizontal-flip test-time augmentation; the last four columns are per-class IoU (Aq.: Aquatic Animal, Art.: Artifacts and Metals, Hum.: Human Activity, Tr.: Marine Trash and Plastic). Threshold controls and the ensemble involve no training and therefore have no best epoch.

Run	Ep.	Val.	Test	+TTA	Aq.	Art.	Hum.	Tr.
3-seed SegFormer-B2 ensemble	—	—	0.7134	0.7227	0.615	0.822	0.738	0.679

Run	Ep.	Val.	Test	+TTA	Aq.	Art.	Hum.	Tr.
Threshold control of the baseline	—	—	0.6981	0.7037	0.601	0.796	0.726	0.670
SegFormer-B2 baseline, seed 0	29	0.7089	0.6979	0.7072	0.609	0.785	0.727	0.671
AquaScale: architecture only	37	0.7011	0.6970	0.7021	0.599	0.796	0.738	0.655
3-fold audit model, fold 1 (20 ep.)	18	0.6918	0.6961	0.7052	0.613	0.795	0.733	0.643
Random control on the real labels	39	0.7048	0.6940	0.7048	0.614	0.776	0.715	0.671
SegFormer-B2 baseline, seed 1	32	0.6914	0.6922	0.7006	0.616	0.802	0.713	0.637
MAAS on the real labels	37	0.7012	0.6914	0.7001	0.607	0.772	0.721	0.665
AquaScale: full (v1)	37	0.7017	0.6912	0.7017	0.600	0.782	0.705	0.678
AquaScale: zoom only (v2)	32	0.6882	0.6864	0.6995	0.603	0.765	0.716	0.661
AquaScale: zoom only (v1)	35	0.7064	0.6858	0.6991	0.605	0.802	0.697	0.640
AquaScale: full (v2)	27	0.6910	0.6838	0.6959	0.617	0.773	0.692	0.653
3-fold audit model, fold 0 (20 ep.)	15	0.6983	0.6805	0.6875	0.595	0.760	0.707	0.660
SegFormer-B2 baseline, seed 2	37	0.6941	0.6781	0.6930	0.592	0.785	0.694	0.641
3-fold audit model, fold 2 (20 ep.)	17	0.6837	0.6770	0.6820	0.584	0.764	0.705	0.656
U-Net R50 baseline	38	0.6395	0.6179	0.6373	0.528	0.725	0.649	0.570
30% deleted, MAAS v2	11	0.6187	0.6145	0.6239	0.557	0.720	0.621	0.560
30% deleted, random control v2	22	0.6033	0.6038	0.6135	0.513	0.702	0.633	0.567
30% deleted, threshold control v2	—	—	0.5886	0.5935	0.495	0.700	0.603	0.557
30% labels deleted, plain training	4	0.5774	0.5866	0.5897	0.486	0.706	0.604	0.551
30% deleted, random control v1	4	0.5948	0.5843	0.5895	0.508	0.661	0.642	0.526
30% deleted, threshold control v1	—	—	0.5705	0.5707	0.458	0.705	0.597	0.522
30% deleted, MAAS v1	21	0.5830	0.5702	0.5771	0.463	0.697	0.624	0.497

Appendix B

Reproducibility and Verification

The purpose of this appendix is to let a reader check the numbers in this thesis. Nothing in the results chapter was taken from a summary document without being recomputed from the stored raw files; where a number in a working document disagreed with the raw files, the raw files were used.

B.1 Where each number comes from

Table B.1 lists, for each group of results, the stored files from which the numbers were taken.

B.2 Independent checks

The checks below were run for this thesis with scripts that are kept in the `reproducibility` folder of the source tree. Absolute paths in those scripts refer to the machine on which they were run.

- **Scores from raw counts.** For all 31 Pipeline A runs that have per-image counts, mIoU recomputed from the counts agrees with the stored evaluation file to within 5×10^{-5} (exactly for the 14 CoGS and ImageNet-control runs). For all 23 Pipeline B runs, mIoU recomputed from the counts equals the stored value at four decimals, with and without TTA.
- **Contamination.** A separate implementation of the near-duplicate detector (PIL instead of OpenCV) found 141 of 347 original test images (the stored list has 143; all 141 are in it), 132 with a shared filename stem (identical to the stored count), and 251 of 634 in the validation set.
- **Splits.** Per-split image and per-class annotation counts of the original export and both rebuilt splits were recounted from the 6,382 label files (19,114 rows, 1,332 box-only, 167 background-only images), and the agreement between the two rebuilt splits was measured (52.2% against 53.3% expected by chance from the marginals).

Table B.1: Raw files behind the results.

Result	**Stored source**
Pipeline A scores	`eval_*.json` (one per run), `perimg_*.npz` (a 5×5 confusion matrix for each of the 954 test images), `run_histories/*.json` (per-epoch loss, validation, masked share)
Pipeline A splits and audit	`splits_clean.json`, `contaminated_test.json`, `inflation.json`, `suim_overlap.json`, `trashcan_overlap.json`, `suim_crossdata.json`, `trashcan_crossdata.json`, `label_noise_verified.json`
Retrained hybrid and TAM-Net seed 2	`eval_clean_hybrid_raw_best.json` and its history; `tamnet_s2` checkpoint metadata and `eval_tamnet_s2_raw_best.json`, `tamnet_masking_bootstrap_3seed.json`
TAM-Net on DeepLabv3+ and the pooled analysis	`eval_tamnet_dl*_raw_best.json`, `perimg_tamnet_dl*.npz`, `perimg_suim_*.npz`, `tamnet_dl_masking_bootstrap.json`, `run_histories/tamnet_dl*_history.json`, and `reproducibility/dl_pooled.py` (writes `dl_pooled.json` and the tables of the pooled analysis)
Agreement between the pipelines (Table 5.9)	`m1_numbers.json`, `pipeline_b_backbone.json` (written by `verify_m2.py`), the per-image counts of both pipelines, and `make_tables.py`
Dataset composition (Figure 4.7)	the image and label files of the export, and `reproducibility/make_source_figure.py`
PADS-Net range map	`padsnet_full_raw_best.pth` and `reproducibility/verify_pads.py` (writes `pads_check.json`)
Assessment of the designs	`run_histories/*.json` (median epoch time times the number of epochs), `model_weights/RUN/result.json` (`train_minutes`); table written by `make_tables.py`
Pipeline B scores	`model_weights/RUN/result.json`, `test_counts_tta*.npz`, `final_test_counts_tta0.npz`, `log.jsonl`
Pipeline B analysis	`metrics/analysis/*.json`: noise floor, audit estimate, video and disagreement reports, scale analysis, corrected test scores; `metrics/AquaScale/`
Human review	`human_review/round1_train_audit/answers.csv`, `round2_test_correction/answers.csv` and the hidden keys `review_key.json`, `test_review_key.json`
Initial benchmark	`model/*/results/*_results.json`, the YOLOv8 and hybrid notebook outputs, `comparison_outputs/00_master_summary.csv`

- **Mirrored copies.** A flip-aware check (horizontal flip, vertical flip, 180° rotation) found 39 (11.2%), 109 (11.4%), and 0 test images with a mirrored copy in training for the original split, split A, and split B; the Pipeline A scores were recomputed without the affected images.
- **Bootstrap.** The paired bootstraps of the TAM-Net comparison, the CoGS comparisons, the seed pairs, the MAAS experiments, and the TTA effects were recomputed from the per-image counts with independent random seeds; intervals agree with the stored ones to within resampling error (for example, TAM-Net (two seeds) minus baseline: stored -0.0014 to $+0.0238$, recomputed -0.0013 to $+0.0247$).
- **Review.** Both review rounds were recomputed from the answer sheets and the hidden keys (decoy rates 13.3% and 2.0%; 158 of 379 candidates in round 2; 85 of 110 confirmed animals tiny), and the MAAS recovery precision and recall were recomputed from the training logs (precision 0.843 for version 1 and 0.601 for version 2; recall 5.7% and 7.4%).
- **Sizes.** Instance sizes were recomputed from the polygon labels of all images, and the per-size detection tables from the stored analysis files.
- **Model sizes.** Parameter counts were checked by counting tensor shapes inside the stored checkpoints (SegFormer-B2 24,723,397; TAM-Net 24,805,995, of which the detail path 82,341 and the reliability head 257) and against the counts printed at the start of each training log.
- **Qualitative figure.** The per-image scores behind Figure 5.14 agree with the stored per-image confusion counts to within 0.0086 for all 869 images, ruling out the mis-initialised-weights error that an earlier version of that figure had.
- **Initial benchmark.** The initial result files were read and the YOLOv8 test metrics were read from the training notebook, since no results file was stored for that model.

B.3 Items that could not be checked

Some quantities from working documents were not among the transferred files, and are therefore quoted with that caveat in the text or omitted: the model-cost table (parameters were verified, latency was not), the small-object oracle experiment, and the exact number of byte-identical SUIM files. The PADS-Net range-map statistics were re-derived from the stored checkpoint (`verify_pads.py`). The GPU model of the Pipeline A workstation (an RTX 4080 SUPER) is not recorded in the stored logs; it is stated by the authors.

B.4 Rebuilding

To rerun a result, install the environment of Table 3.1, place the Roboflow export at the path expected by the scripts, and run the steps in the order of their

numeric prefixes (split construction, cache and mask materialisation, training, evaluation, reports). The figures and the numeric tables of this thesis are generated by scripts in the `reproducibility` folder that read only the stored result files: `m1_numbers.py`, `dl_pooled.py`, and `verify_m2.py` write the intermediate JSON files, and `make_tables.py`, `make_paper_figures.py`, and `make_source_figure.py` then regenerate every result table body and figure.